EBU5303

Multimedia Fundamentals

Digitisation

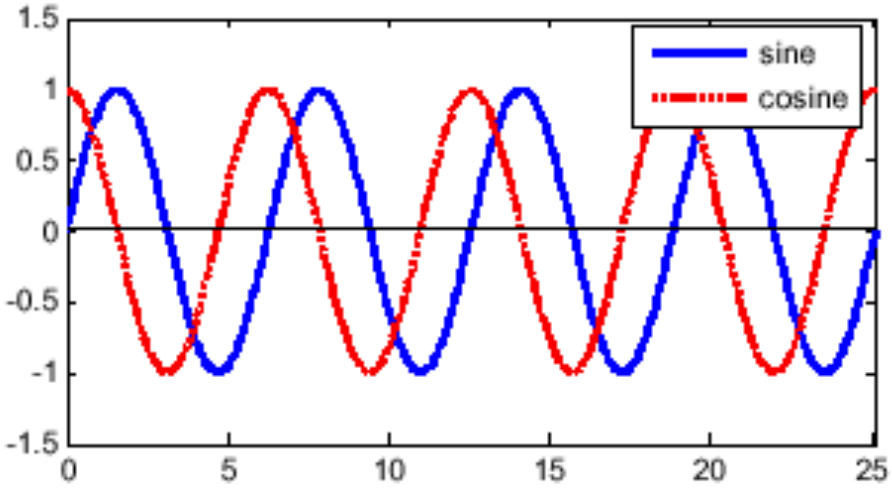
Agenda

- Sound and images are analogue phenomena that can be represented by complex waveforms
- They must be digitised to be handled by computers: sampling and quantisation
- Sampling and quantisation rates determine the size of the digitised data

取样 量子化

two process to digitize an image

Sinusoidal functions

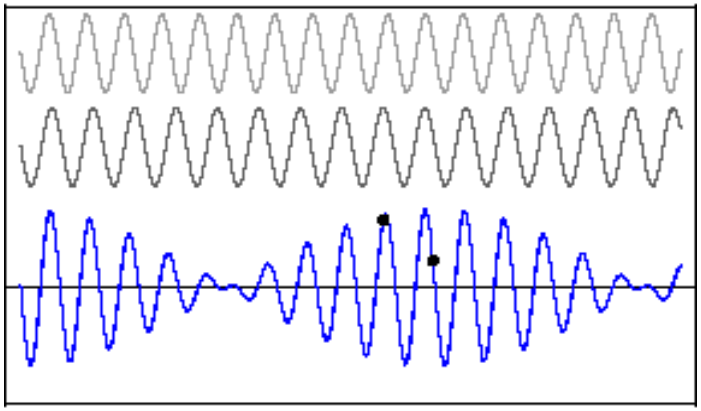


- cycle
- wavelength (m) 波长
- f = frequency (Hz)
- T = period (s) = 1/f
- A = amplitude
- ω = angular frequency (rad/s) = $2\pi f$ 角频率
- ϕ = phase

$$s(t) = A \sin(\omega t + \phi)$$

Complex Waveforms

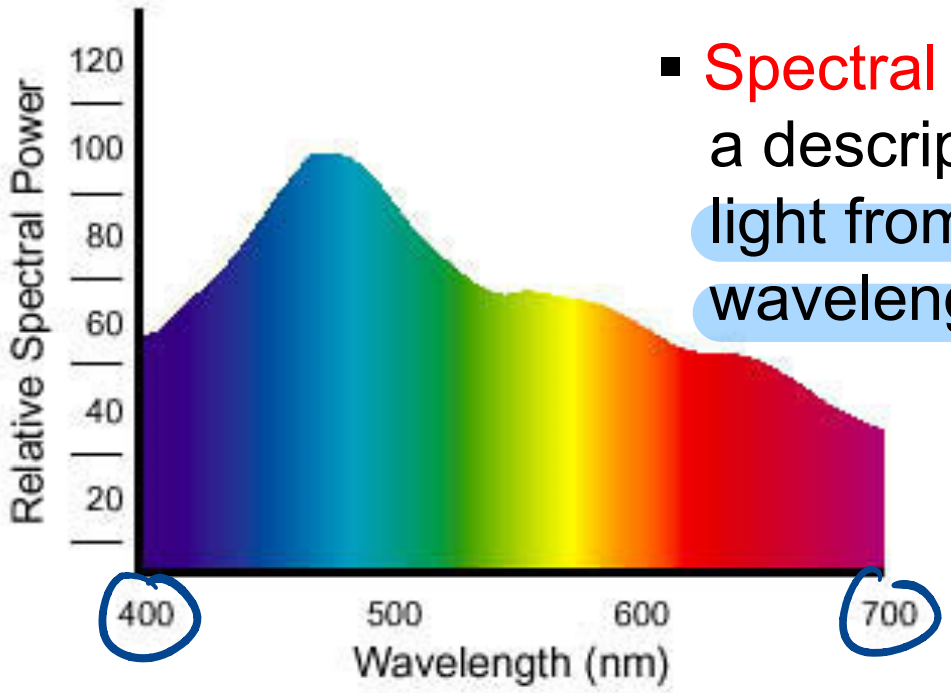
- Most waveforms are complex, i.e. their shape is the result of adding 2 or more waveforms.
- Each complex waveform may be described as the sum of a number of simple sine waves, each with a particular amplitude, frequency (or wavelength) and phase.



complex waveform

Example: Visible Light

- The wavelengths of visible light lie roughly between 400 nm and 700 nm.



- Spectral Power Distribution (SPD): a description of how the intensity of light from some source varies with wavelength.

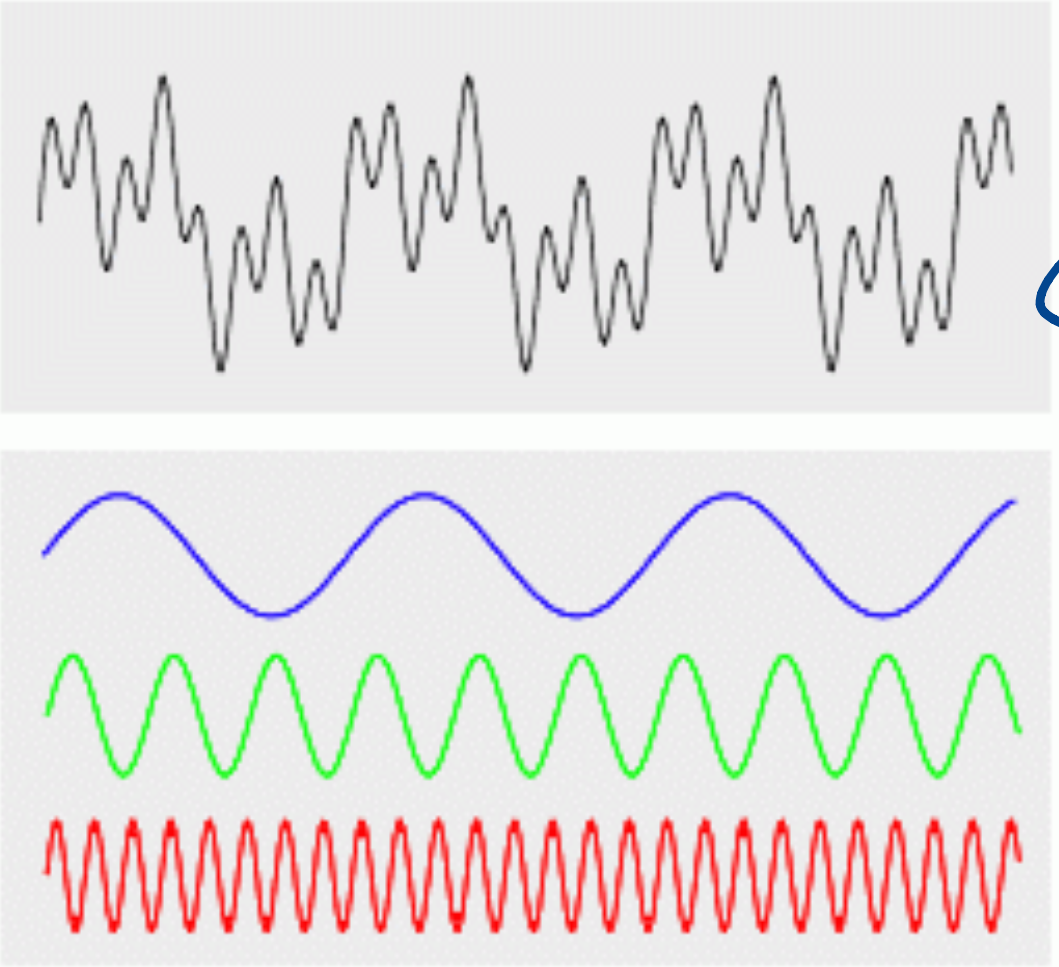
Fourier analysis

- Fourier analysis attempts to represent a set of data with a series of sines and cosines with different periods, amplitudes, and phases.
- This representation is done by a mathematical process called a transform. the data measurements in the time domain are *transformed* into the period or frequency domain.

$$F(\nu) = \int f(t) exp(i2\pi\nu t) dt$$

allow the extraction of the individual frequency components of a complex wave form.

Fourier Analysis



Fourier Analysis:
The complex wave (upper) can be decomposed into the sum of the three simple sine waves (lower).

AnaesthesiaUK

Questions..

A sound represented by a completely regular sine wave is called a pure tone.
The frequency of a wave informs us about the pitch of the sound, which can be low or high.
The amplitude of a wave informs us about how loud the sound is.
The wavelength of a light ray informs us about its colour.

- What kind of sound is represented by a completely regular sine wave ? pure tone
- What does the frequency of a wave tell us about the sound it represents ? pitch
- What does the amplitude of a wave tell us about the sound it represents ? the loudness
- What information is contained in the wavelength of a light ray ? colour

Analogue-to-digital conversion

= converting the continuous phenomena of images, sound, and motion into a discrete representation that can be handled by a computer.

Advantages:

1. Digitised pictures and sound can now be captured in fine detail.
2. Digital data communication is less vulnerable to noise than is analogue.
3. Digital data can be communicated more compactly than analogue… when compressed!

Sampling

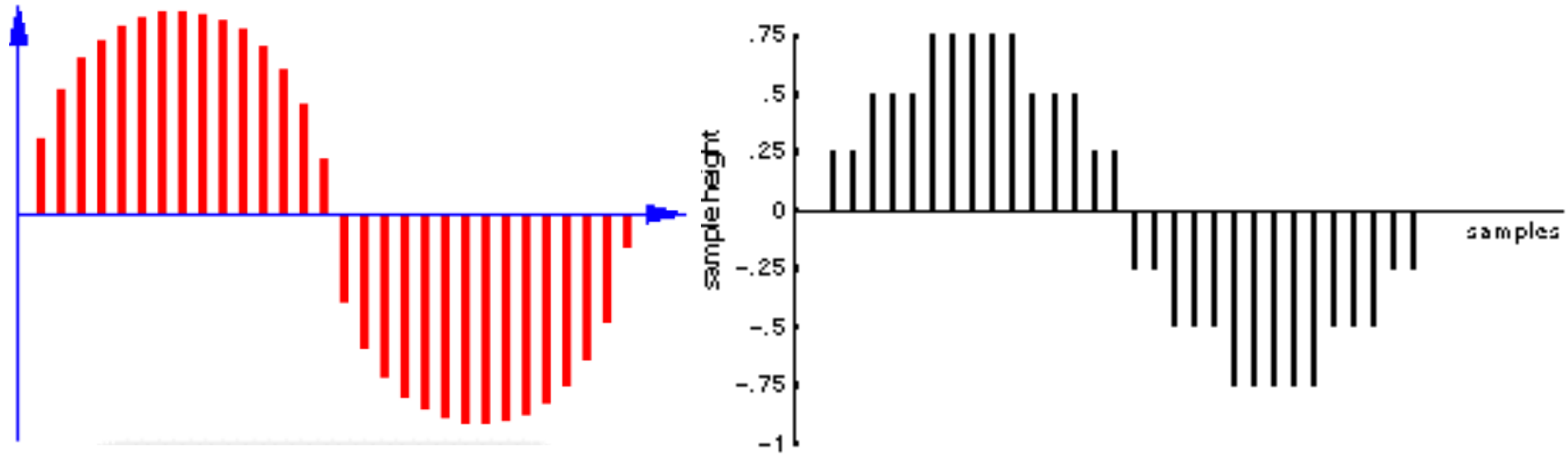
- Sampling chooses discrete points at which to measure a continuous phenomenon (a *signal*).
- In the case of images, the sample points are evenly separated in space.
- In the case of sound, the sample points are evenly separated in time.
- The number of samples taken per unit time or unit space is called the ***sampling rate.*** 取樣率

CD: 44.1 KHz
Music: 16 bits
Speech: 8 bits
Good quality color image: 24 bits

Quantisation

- Quantisation requires that each sample be represented in a fixed number of bits, called the ***sample size*** or the ***bit depth***. 比特位深
- In the case of an image, the bit depth is called the ***colour depth***.
- The bit depth limits the precision with which each sample can be represented.

Quantisation



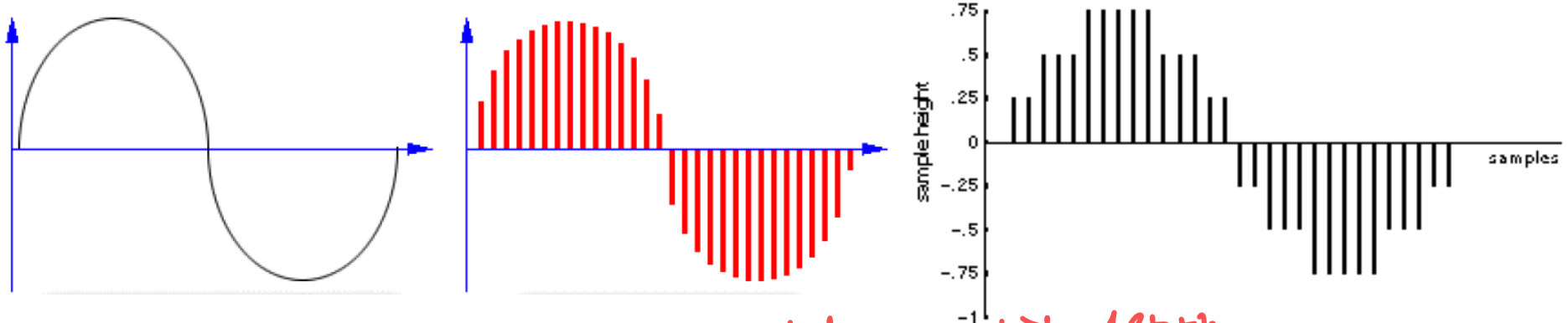
Let n be the number of bits used to quantize a digital sample. Then the maximum number of different values that can be represented, m, is

$$m = 2^n$$

n bits
可表示 2^n 个 value

Sampling + Quantisation

- The sampling rate relates directly to the frequency of a wave.
- Quantisation, on the other hand, relates more closely to the amplitude of a wave.



The higher sampling rate / larger bit depth,
The bigger size of image

What are the implications (positive and negative) of using a low sampling rate?

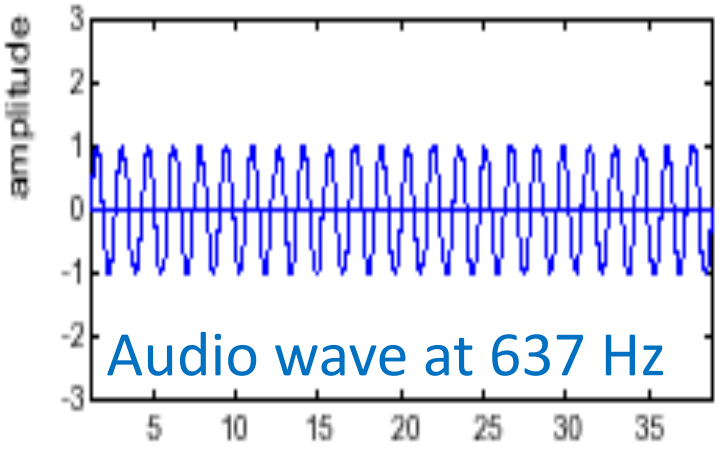
Negative: It causes aliasing. The sampled signal can not be reconstructed well. (Distortions happened because of low resolution)
Positive: Smaller files, easy to transmit and store, especially applying to small screen.

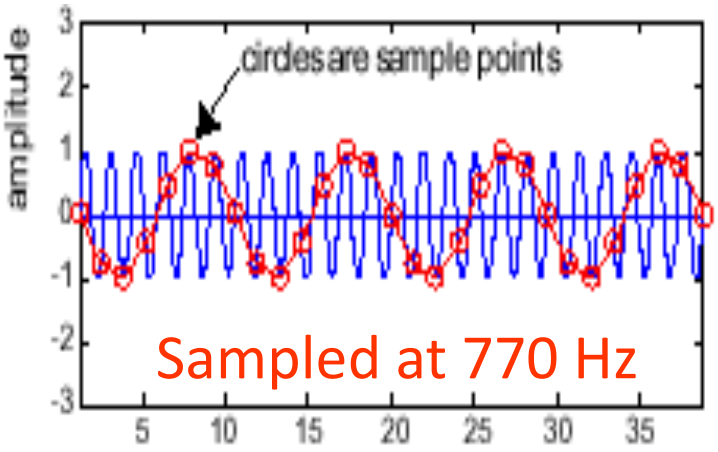
What are the implications(positive and negative of using a low bit depth?

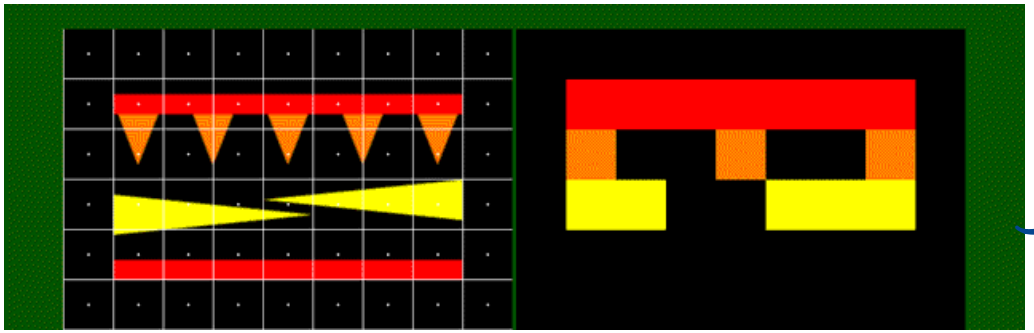
Negative: A low bit depth will cause loss of details (less precision) because the quantisation error will be big. （不要用distortion描述，可以参看电信系统中distortion定义）
Positive: fewer bits to store, smaller files, easy to transmit. 对于不在意quality时可以减少: speech 8 bits compared to music 16 bits. It dosen't matter for us to recognize the word through speech when using low bit depth.

混疊

Aliasing (sampling error)

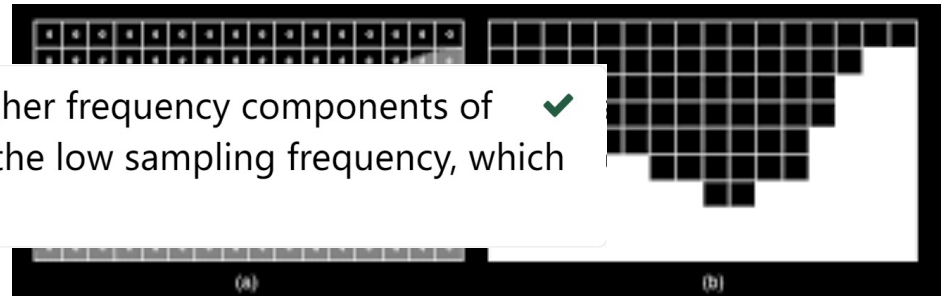






$f \gtrless 2f_{original}$

→ distortion in shapes and patterns



With aliasing, what happens is that the higher frequency components of the signal cannot be captured because of the low sampling frequency, which results in overlap in the spectrum. ✔

Quantisation error

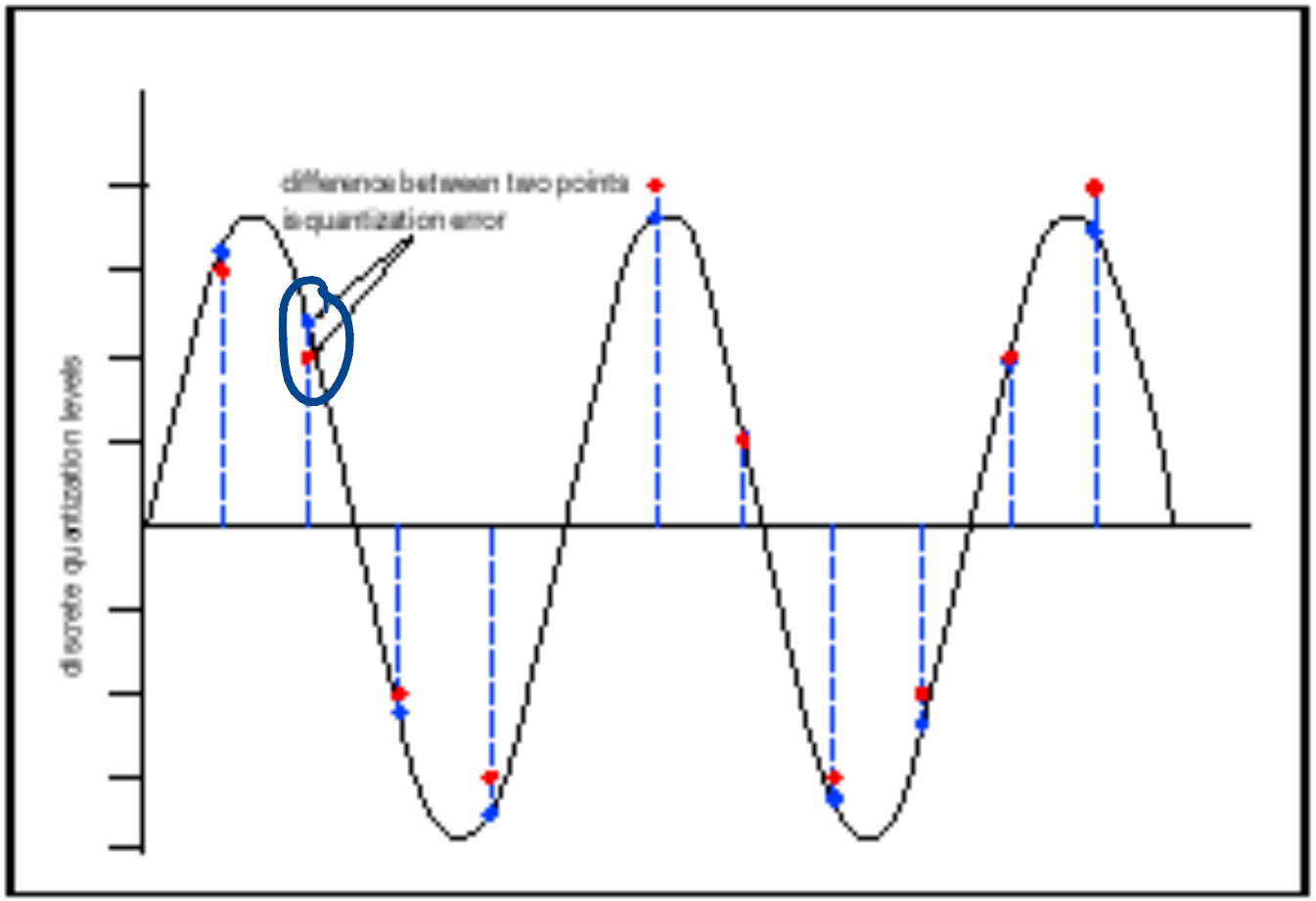
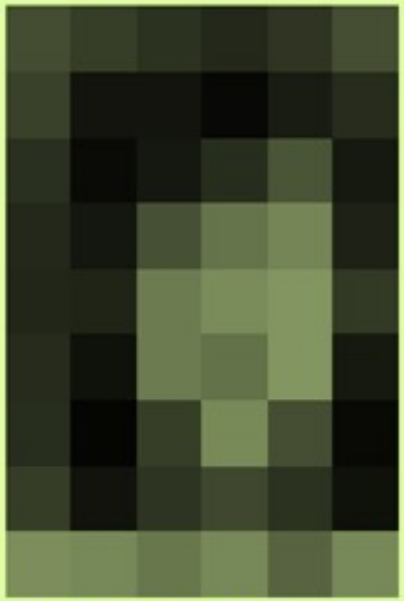


Image of girl

Reduction by 4

Reduction by 16



Reduction by 32

Image of girl: 256 levels

8 levels

4 levels

2 levels

Nyquist theorem

Sample twice as often as the highest frequency you want to capture

Let *f* be the frequency of a sine wave. Let *r* be the minimum sampling rate that can be used in the digitisation process such that the resulting digitised wave is not aliased. Then:

$$r = 2f$$

r is called the ***Nyquist rate***.

≠ Nyquist frequency

信噪比

Signal to Noise Ratio (SNR)

- SNR: the ratio of the meaningful content of a signal versus the associated noise.
- In analogue data communication, SNR is defined as the ratio of the average power in the signal versus the power in the noise level.
- For a digitised image or sound, SNR, also called ***signal-to-quantisation-noise ratio*** (***SQNR***), is defined as the ratio of the maximum sample value versus the maximum quantisation error.

$$SQNR = 20 \log_{10}\left(\frac{\max(quantization\ value)}{\max(quantization\ error)}\right)$$

SQNR

- SQNR is measured in decibels.
- SQNR is directly related to ***dynamic range:*** the ratio of the largest sound amplitude (or colour, for images) and the smallest that can be represented with a given bit depth.

Let n be the bit depth of a digitised media file (e.g. digital audio). Then the signal-to-quantisation noise ratio ***SQNR*** is:

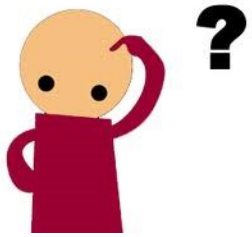
$$SQNR = 20 \log_{10}(2^n)$$

Summary so far …

- Analogue sounds and images must be digitised
- Digitisation includes two processes: sampling and quantisation
- Undersampling causes aliasing (the original data cannot be reproduced)
- Quantisation errors cause loss of details and determine the Signal to Quantisation Noise ratio
- sampling ✔ errors, also called aliasing ✔ incur distortions ✔ that can modify the shapes of objects in an image; quantisation ✔ errors incur a loss of details ✔, for example less colour ✔ available in the reconstructed image.

Can't be reconstruct entirely
be poor quality. differ from original

Question



You have a digital image in 8 bit colour which lacks the subtle gradations from one colour to the next that you would like to see. Is this a matter of aliasing or quantisation error? Explain.

Data size (example): audio file

Sampling rate: 44.1 kHh (44,100 samples/s)
Bit depth: 32 bits per sample (16 for each of two stereo channels)
Number of minutes: 1 minute
Total number of bits:

44,100 * 32 * 60 = 84,672,000 bits

84,672,000 bits = 10,584,000 bytes

File size: ~ 10.58 Mbytes for 1 minute
Data rate of the file: ~ 1.41 Mbits/s

Data size : image file

Example:

Sampling: 1024 pixels x 768 pixels (samples)

Bits per pixel: 24 (sample)

1024 x 768 = 786,432 pixels

786,432 x 24 = 18,874,368 bits

÷8

18,874,368 / 8 = 2,359,296 bytes

File size ~ 2.36 MB

Data size : video file

Sampling: 720 pixels x 480 pixels
Bits per pixel: 24
Frame rate : 30 frames/s
Length: 1 minute
Audio : 44,1 KHz, 32 bits

image + audio

pixels per frame : 720 x 480 = 345,600 pixels
bits per frame : 345,600 x 24 = 8,294,400 bits
Video images size/s : 8,294,400 x 30 = 248,832,000 bits
Total video image size : 248,832,000 x 60 =
14,929,920,000 bits

1 image

Data size : video file

Sampling: 720 pixels x 480 pixels
Bits per pixel: 24
Frame rate : 30 frames/s
Length: 1 minute
Audio : 44,1 KHz, 32 bits

Audio size/s : 44,100 x 32 = 1,411,200 bits
Total audio size : 1,411,200 x 60 = 84,672,000 bits

audio

Data size : video file

Sampling: 720 pixels x 480 pixels
Bits per pixel: 24
Frame rate : 30 frames/s
Length: 1 minute
Audio : 44,1 KHz, 32 bits

Audio size/s : 44,100 x 32 = 1,411,200 bits
Total audio size : 1,411,200 x 60 = 84,672,000 bits

Total video size : images + audio = 15,014,592,000 bits
File size ~ 1.88 GB !!!

Exercise

720×576×60×25×3

Compute the number of bytes needed for 1 minute of video that has 720 x 576 pixels per frame, 25 frames per second, 3 bytes per pixel, and CD-quality stereo audio.

+ audio: 16×44.1k×60 / 8 bytes. } ÷8

60×720×576×3×25

CD quality: 16 bits/sample, 44100 samples/s

- mono ×1
- Stereo ×2

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Multimedia Fundamentals

Colours

Learning Objectives (Colours)

- relate colour properties with light.
- compare colour models.
- use the RGB model to encode a colour.
- use the HSV model to describe a colour.

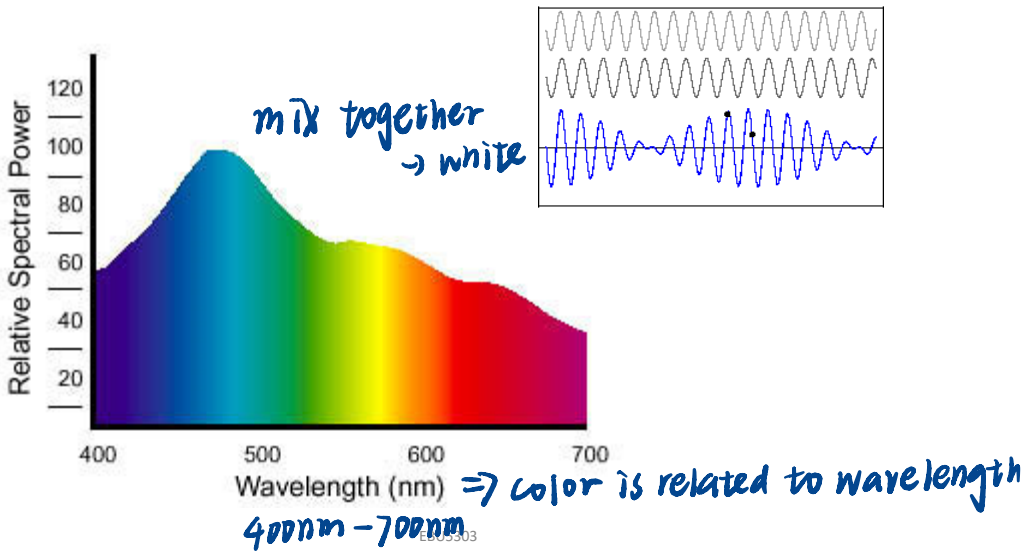
Agenda

- Colour is a property of light
- Colour vision is a complex phenomenon
- There are different colour models
- RGB is used in computer monitors
- YUV/YCbCr are used for image compression
- CMYK is used for printing
- HSV is an intuitive colour model

5

Colour is a property of [light], which is an analogue phenomenon characterised by its [wavelength] components. The [visible spectrum] is between 400 nm and 700 nm. Colours that correspond to [narrow] wavelength bands are called [pure spectral] colours.

Light



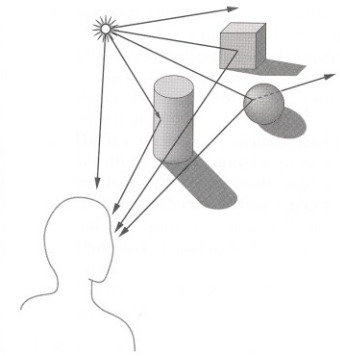
6

Properties of Colours

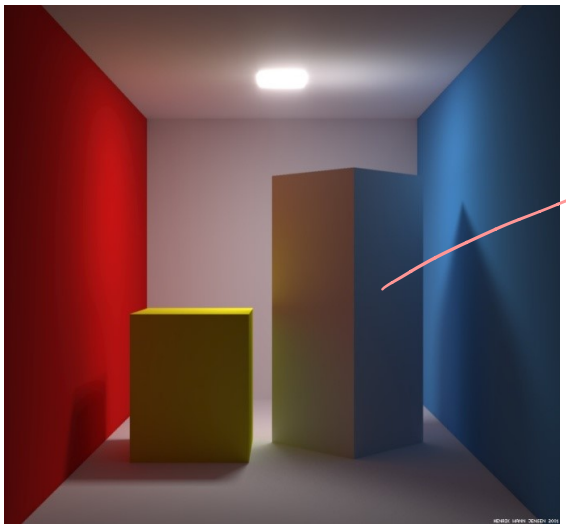
Colour is a property of light, not an object in itself.

Objects have no colour of their own, but merely the ability to reflect a certain section of the visible spectrum.

Objects reflect what we perceive, and absorb what we don't see.



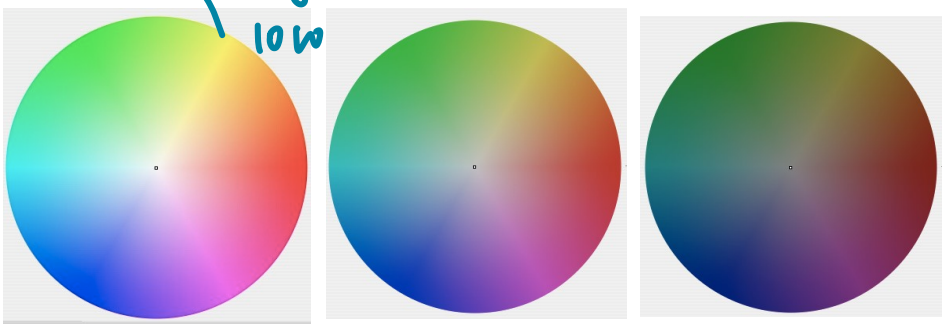
Properties of Colours



The hue of a colour is dependent on its dominant wavelength and independent of intensity. Colour saturation is the expression of the bandwidth of light from a source. It describes the purity of the colour. When fully saturated, the colour is considered in its purest version. Brightness describes how much a source appears to be radiating or reflecting light.

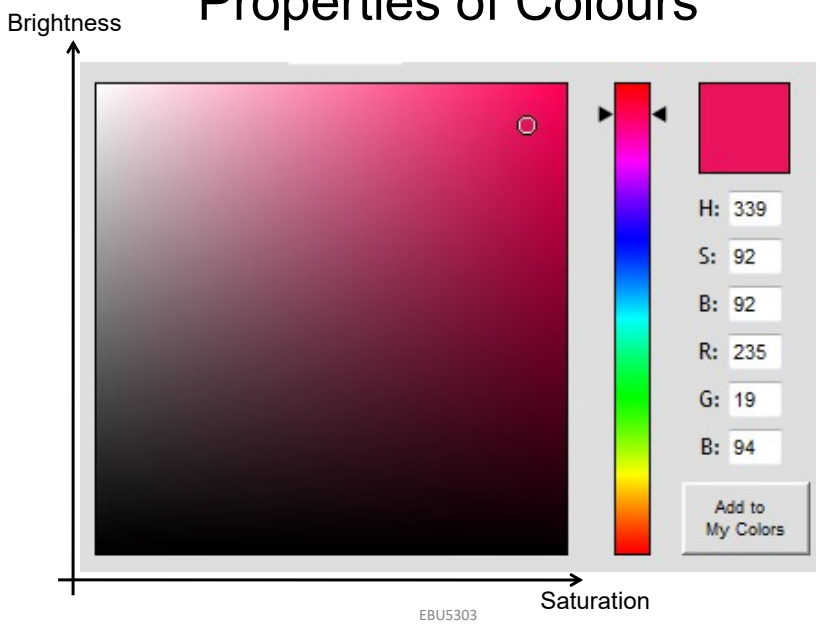
Properties of Colours

- Hue is the name of the colour (dominant wavelength).
- Saturation: a fully saturated colour is one with no mixture of white, and it contains a narrow range of wavelength.
- Brightness is the extent to which an area appears to emit light.



9

Properties of Colours



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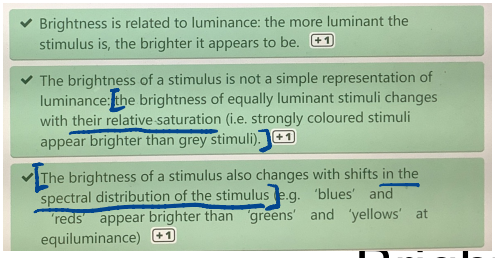
10

In the (R, G, B) model, we can increase the saturation of a colour by decreasing the value that does not represent the colour. In the (C, M, Y) model, we can decrease the saturation of a colour by adding white into the colour, which means decreasing the value that represents the colour.



Brightness and Luminance

知觉亮度有关，反映发光面发光强弱

- ***Brightness*** is a matter of subjective perception and has no precise mathematical definition.
- ***Luminance*** has a mathematical definition that relates a light's wavelength and power to how bright it is perceived to be.

受照面被照明程度的物理量

Interestingly, lights of equal power but different wavelengths do not appear equally bright. The brightest wavelengths are about 550 nm.

Summary so far …

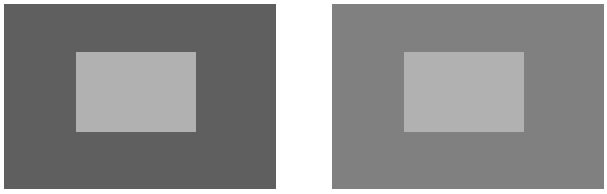
- Colour is a property of light
- Objects reflect what we perceive
- They absorb what we don't see
- The hue is the name of the colour
- The saturation is the degree of mixture with white
- The brightness relates to the quantity of emitted light

Colour perception is subjective

The brain has a major role in interpreting colour: perception of colour may be affected by previous experience and functioning of The brain.

Perception

- Colour perception is difficult because:
  - It varies from person to person
  - It is affected by adaptation
  - It is affected by surrounding colours

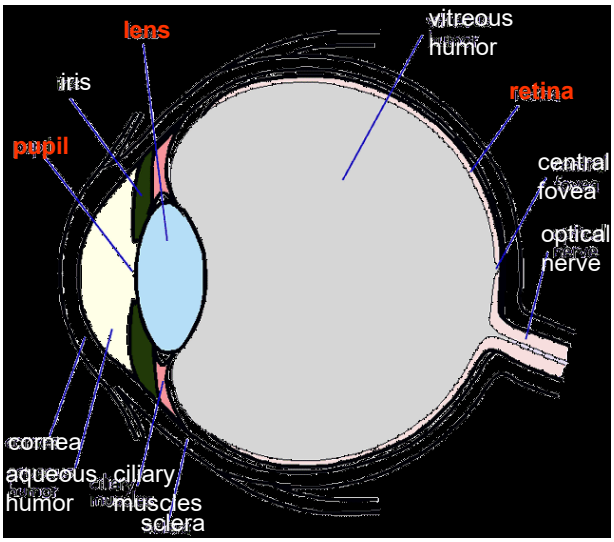


Physiology of vision

The retina
- Rods — night vision
- Cones
  - Color!

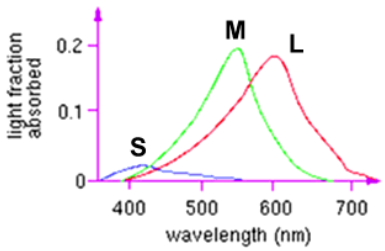
color vision → respond differently to light of different λ.

but it doesn't help to discriminate between light & dark colors



Physiology of vision: cones

- Cones → three types
  - **L** or **R**, most sensitive to red light (610 nm)
  - **M** or **G**, most sensitive to green light (560 nm)
  - **S** or **B**, most sensitive to blue light (430 nm)



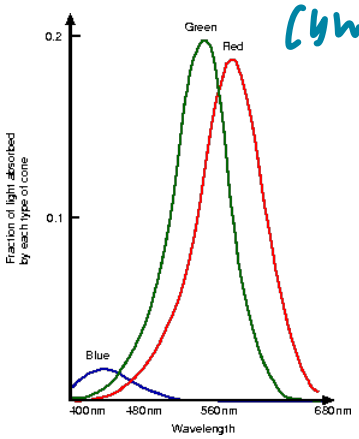
  - Colour blindness results from missing cone type(s)

The Tristimulus Theory of Colour Vision

thanks to cones, not about the brain

(subjective X)

- Human perception of colour derives from the eye's response to 3 different groups of wavelengths, i.e. those corresponding to red, green and blue (RGB).
- Therefore any sensation of colour can be produced by mixing together suitable amounts of these colours.



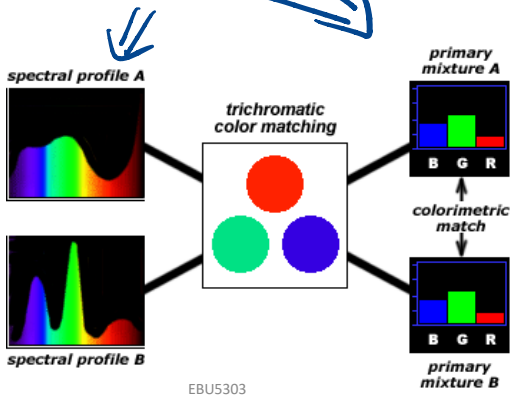
22

Metamers are colour stimuli that have different spectral power distributions but are perceived as identical for a given observer. ✔

Metamers

一种特定的颜色感知来自三种的刺激

- A given perceptual sensation of colour derives from the stimulus of all three cone types.
- Identical perceptions of colour can thus be caused by very different spectra.



23

Summary so far …

- Colour vision is a complex phenomenon
- Colour perception is the result of sensations in the eyes and brain activity
- Colour perception is subjective
- The eye is sensitive to 3 groups of wavelengths : Red, Green and Blue (Tristimulus theory)
- Any colour can be created by mixing some R, G and B

Agenda

- Colour is a property of light
- Colour vision is a complex phenomenon
- There are different colour models
- RGB is used in computer monitors
- YUV/YCbCr are used for image compression
- CMYK is used for printing
- HSV is an intuitive colour model

Colour models

- What is a colour model?
  - A colour model is an abstract mathematical model describing the way colours can be represented as sets of numbers

Colour models

- Why do we need colour models ?
  - Colour is a difficult thing to describe accurately.
  - In a model each colour has a numerical code, which can be used to reproduce exactly the colour intended.
  - With colour codes we can pass instructions to computers.

精确复现

Colour models

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(fully) saturated – RGB中至少有一个是0（反过来，只要有一个是0，此色必定饱和）
non-saturated – 增加该颜色补色的值（e.g. 某饱和蓝(0,0,200)变成不饱和蓝就是(10,10,200)；某饱和青(0,220,220)变成不饱和青就是(40,220,220)）
bright – RGB中至少有一个是255
light – 建议RGB三个值都大于127
dark – 建议RGB三个值都小于127

例：dark grey：(50,50,50)；bright orange：(255,128,0)；unsaturated dark blue：(20,20,100)

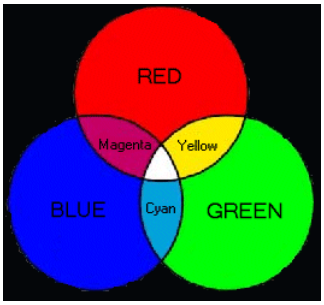
brightest unsaturated yellow：(255,255,199)

unsaturated bright green：(50,255,50)

Using the RGB Scheme in Computers

- In the RGB model, we can define any colour value in terms of the proportions of R, G and B it contains.
- Any colour can be represented in the form (r, g, b).
- The number of bits used to hold a colour value is called the colour depth or pixel depth.

E.g. if 8 bits are used, then:
  - (0, 0, 0) is black
  - (255, 0, 0) is red
  - (255, 255, 255) is white
  - (100, 100, 100) is gray



Additive Primary Colours

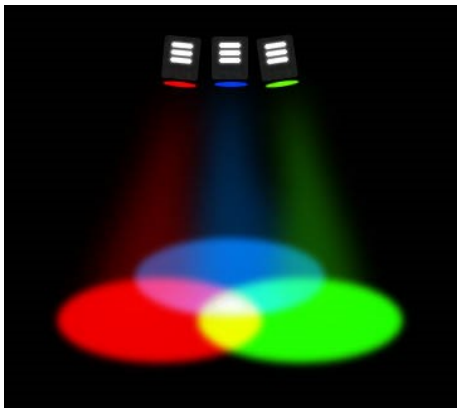
additive

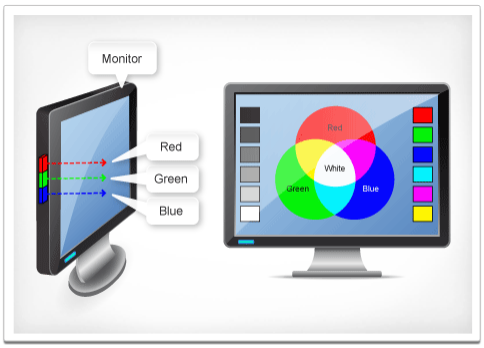
pink：non-saturated red

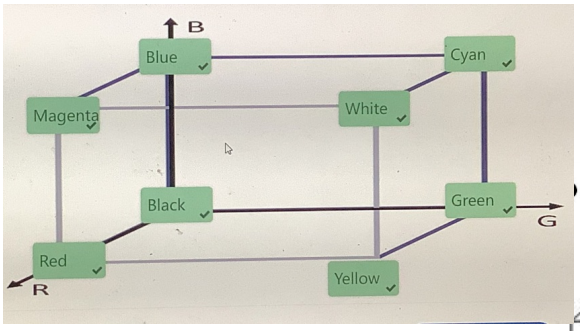
- The RGB model is used for colour monitors and most video cameras.

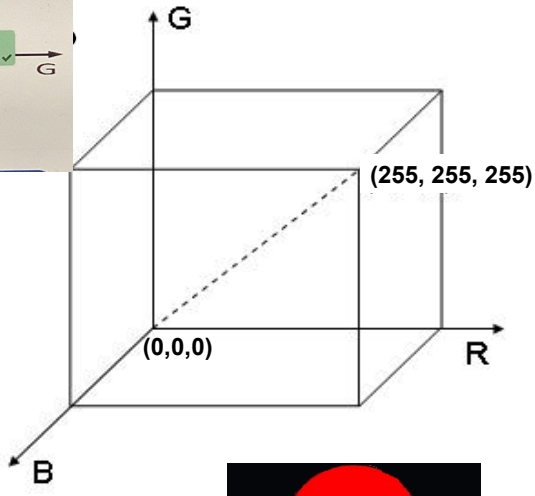
dark pink：(100,20,20)

Using the RGB Scheme in Computers



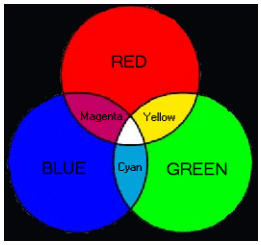






- What colours correspond to the eight corners of this cube?

- What does a straight line running from the origin (0,0,0) to (255,255,255) represent? grayscale



From R,G,B to Grayscale

Let an RGB color pixel be given by (R, G, B), where R, G, and B are the red, green, and blue colour components, respectively. Then the corresponding grayscale value is given by (L, L, L), where:

$$L = 0.30R + 0.59G + 0.11B$$

YUV

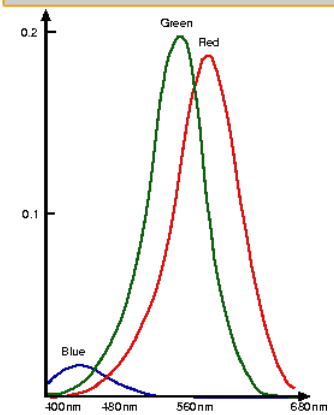
- The YUV model is a simple translation of the RGB model, separating all the luminance information (Y) from the colour (or *chrominance*) information (U, V).
- We will see later how YUV (in fact YCbCr) is used for image compression (JPEG).

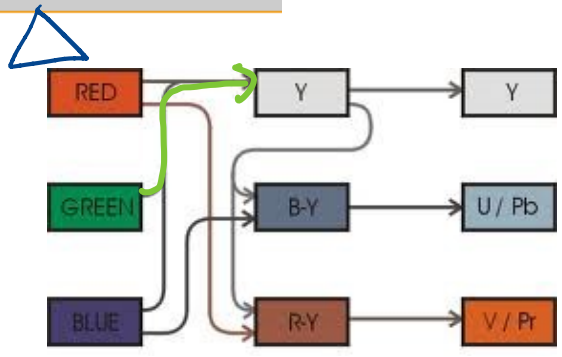
RGB to YUV

$$\begin{bmatrix} Y \\ U \\ V \end{bmatrix} = \begin{bmatrix} 0.299 & 0.587 & 0.114 \\ -0.147 & -0.289 & 0.436 \\ 0.615 & -0.515 & -0.100 \end{bmatrix} \cdot \begin{bmatrix} R \\ G \\ B \end{bmatrix}$$

150
20
100

97.89
1.07
133.45





B→U

R→V

CMY(K) for printing

e.g.	C	M	Y	K
(pure) Blue	255	255	0	0
(dark) Cyan	255	0	0	220
White	0	0	0	0
Black	0	0	0	255
Grey	0	0	0	127

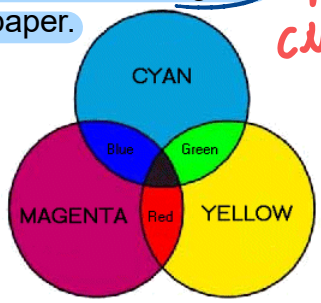
每个值range: [0,255]

Subtractive Primaries CMYK

- The CMYK model is a subtractive model that is used in printing: C = W-R ; M = W-G ; Y = W-B
- Whereas the RGB model depends on a light source to create colour, the CMYK model is based on the light-absorbing quality of ink printed on paper.

  It uses the subtractive primaries Cyan, Magenta and Yellow.

  In addition, because it is impossible to produce a pure black from these primaries, a Black (K) primary is added thus giving the CMYK model.



Subtractive Primary Colours

3 inks must be put on paper in large quantity which is a waste of ink and does not allow a perfect black in practice.

A dark unsaturated green color : (C,M,Y,K) = (200,0,200,180)

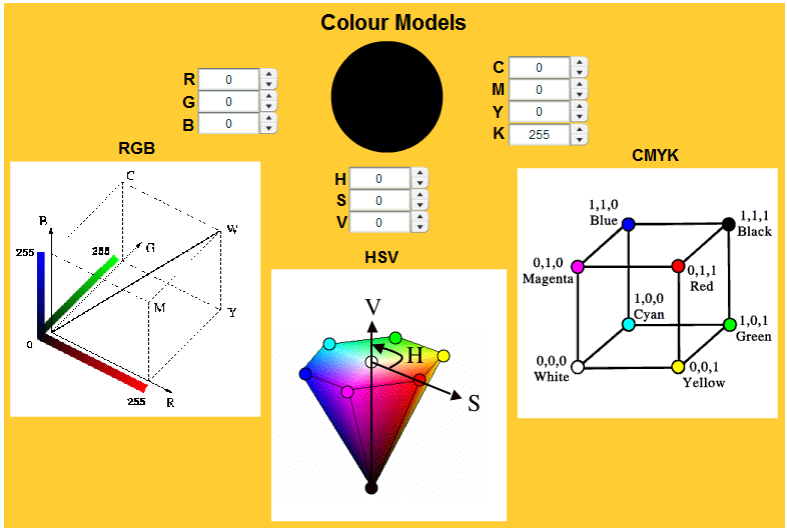
A fully saturated bright green color : (C,M,Y) = (255,0,255)

A fully saturated dark red color : (C,M,Y,K) = (0,255,255,50)

bright orange color : (C,M,Y,K) = (0,127,255,0)

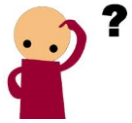
"Saturated Red is a mixture of equal amounts of M and Y to make it dark, some amount of K should be added."

Colour Converter



做法：所有光分解成红/绿/蓝光
照的光若能被涂在纸上的颜色反射，即能呈现.

Question



Colour inkjet printers use the CMY model. When the cyan ink colour is sprayed onto a sheet of white paper, what colour would it appear under a green light? Why?

Blue + green

green

反射 Blue, green, red 光

Yellow ink is spread onto a white sheet of paper. What colour will you see if the paper is illuminated with a blue light?

Yellow absorbs blue, so the paper will appear black.

Blue

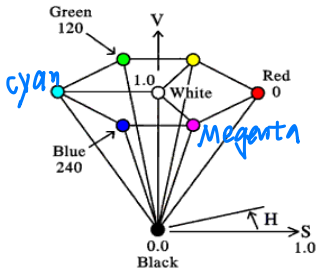
反射 yellow 光

In HSV model, grayscale is represented along the vertical V axis.
Black is at the bottom, white at the top.
H=0, S=0

Select the propositions where both the RGB values and the HSV values represent light grey.

- [x] RGB = (200, 200, 200) ; HSV = (0, 0, 78)
- [x] RGB = (230, 230, 230) ; HSV = (300, 0, 90)
- [ ] RGB = (200, 250, 200) ; HSV = (120, 0, 98)
- [ ] RGB = (230, 230, 230) ; HSV = (300, 50, 90)

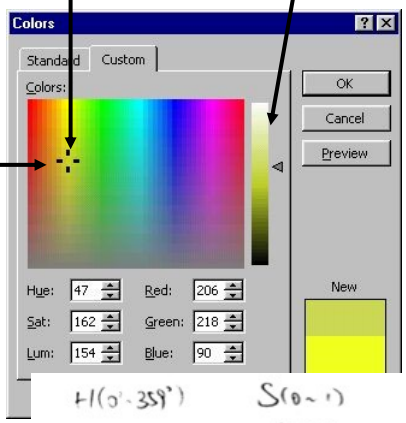
HSV

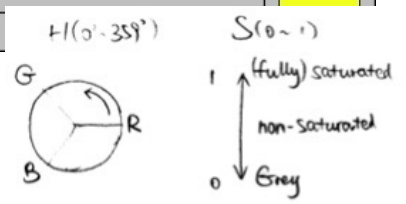


Hue

Saturation

Value





'Light': V=1, 0<S<1

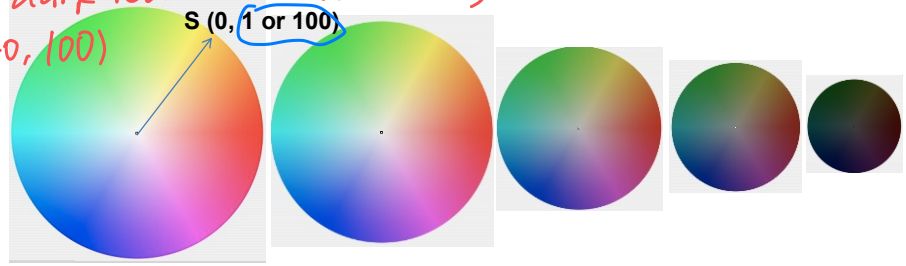
dark: 5-45

Select the proposition where both the RGB values and the HSV values represent bright non saturated red.

- [ ] RGB = (200, 0, 0) ; HSV = (0, 100, 78)
- [x] RGB = (255, 55, 55) ; HSV = (0, 78, 100)
- [ ] RGB = (120, 55, 55) ; HSV = (0, 54, 47)
- [ ] RGB = (255, 55, 20) ; HSV = (9, 92, 100)

bright orange: (30, 100, 100) (70)
bright grey: (whatever H, 0, 0.7)
unsaturated dark red color: (0, 50, 50)
pink: (0, 40, 100)

HSV



V (0, 1 or 100)

Consider the following three colours expressed using 24 bits, in the RGB colour model: Colour A = (220, 30, 20); Colour B= (100, 100, 100); Colour C = (50, 0, 50). Estimate the H, S, and V values for each of these colours in the HSV colour model.

- [x] A = (3, 91, 86) ; B = (0, 0, 39) ; C = (300, 100, 20)
- [ ] = (3, 91, 86) ; B = (0, 20, 39) ; C = (300, 100, 20)
- [ ] = (0, 91, 86) ; B = (0, 0, 39) ; C = (300, 100, 20)
- [ ] A = (3, 91, 86) ; B = (0, 0, 39) ; C = (320, 100, 20)

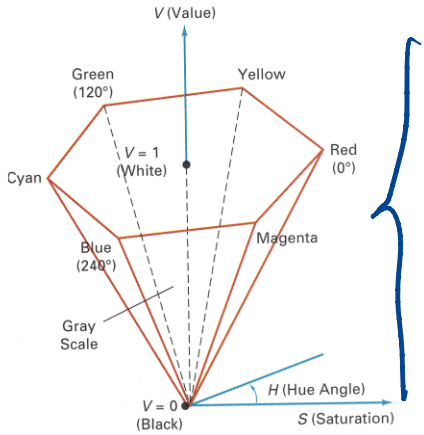
EBU53C

Select the propositions where both the RGB values and the HSV values represent bright magenta.

red = blue

- [ ] RGB = (170, 0, 255) ; HSV = (280, 100, 100)
- [x] RGB = (255, 0, 255) ; HSV = (300, 100, 100)
- [x] RGB = (225, 50, 225) ; HSV = (300, 78, 98)
- [ ] RGB = (50, 0, 50) ; HSV = (300, 100, 20)

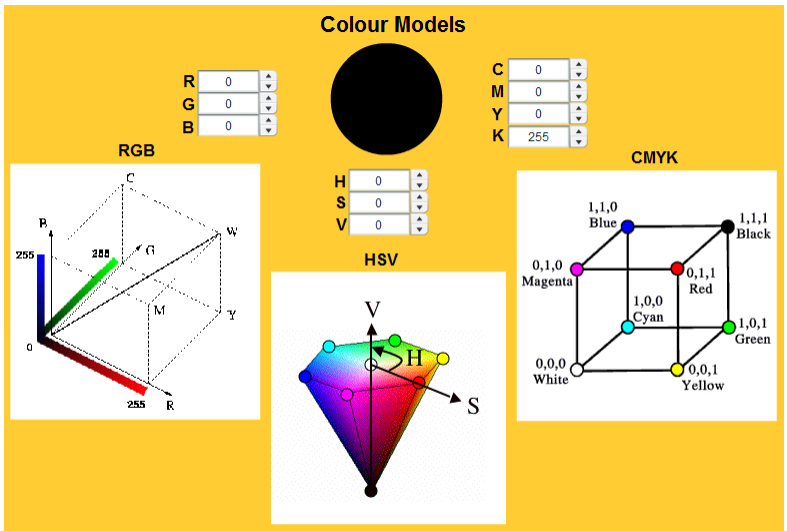
HSV



H	S	V	Color
0	1.0	1.0	Red
120	1.0	1.0	Green
240	1.0	1.0	Blue
*	0.0	1.0	White
*	0.0	0.5	Gray
*	*	0.0	Black
60	1.0	1.0	
270	0.5	1.0	
270	0.0	0.7	

50

Colour Converter



51

EBU5303

Multimedia Fundamentals

Digital Images

Agenda

- Images can be stored as bitmaps or be vector-based
- The points at which an image is sampled are known as picture elements (pixels)
- Colour bitmap images can be true-colour or index-based

attention: vector-based 图像已经够外，无需压缩!!!

Bitmapped Vs. Vector-based

- All images are displayed on a computer screen as a grid of “pixels” of various colours. The image files that contain these images store that image data in one of two fundamentally different ways:
- Bitmapped image files store image data as a map of individual pixels (e.g. GIF and JPEG).
- Vector-based image files store image data as a set of mathematical formulas that instruct the computer how to draw the image.
- Some file formats, such as PNG, contain both bitmapped and vector-based image data.

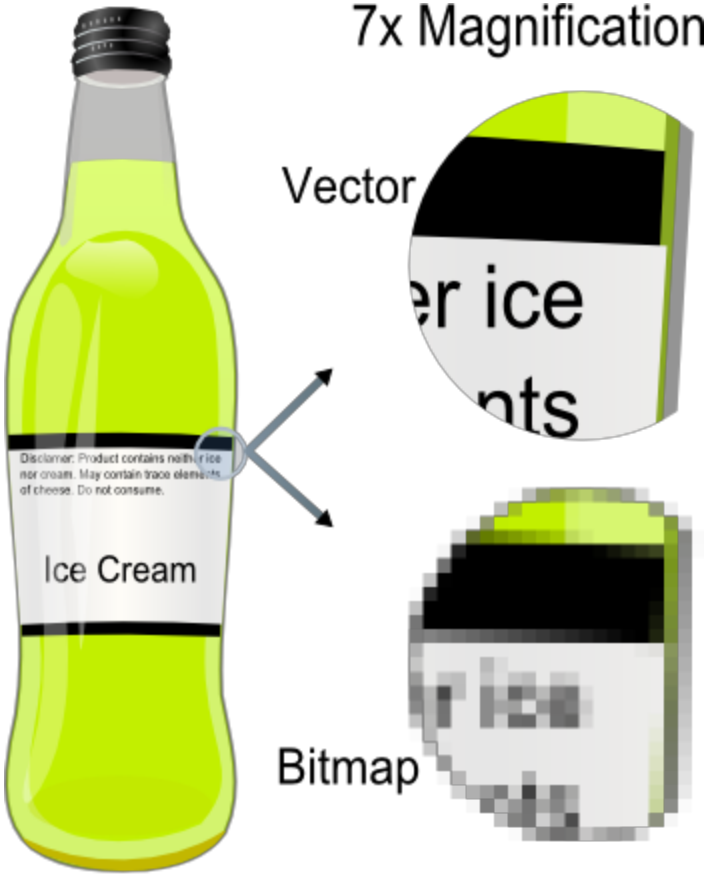
Bitmapped Vs. Vector-based

- ***Bitmap images*** (also called ***pixmaps*** or ***raster graphics***) are created with a pixel-by-pixel specification of points of colour. ***Bitmaps*** are commonly created by digital cameras, scanners, paint programs like Corel Paint Shop Pro, and image processing programs like Adobe Photoshop.

- ***Vector graphic images***—created in programs such as Adobe Illustrator and Corel Draw—use object specifications and mathematical equations to describe shapes to which colours are applied.

is made of points, lines, and curves that are based upon mathematical equations.

Bitmapped Vs. Vector-based



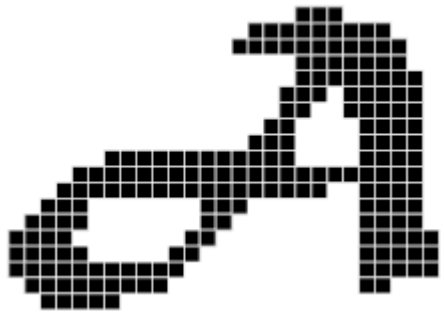


Image Sampling and Quantisation (grayscale images)

- A digital image is represented by a matrix of numeric values, each representing a quantised intensity value. e.g. I (r, c)

- The intensity at each pixel is represented by an integer and is determined from the continuous image by averaging over a small neighbourhood around the pixel location.

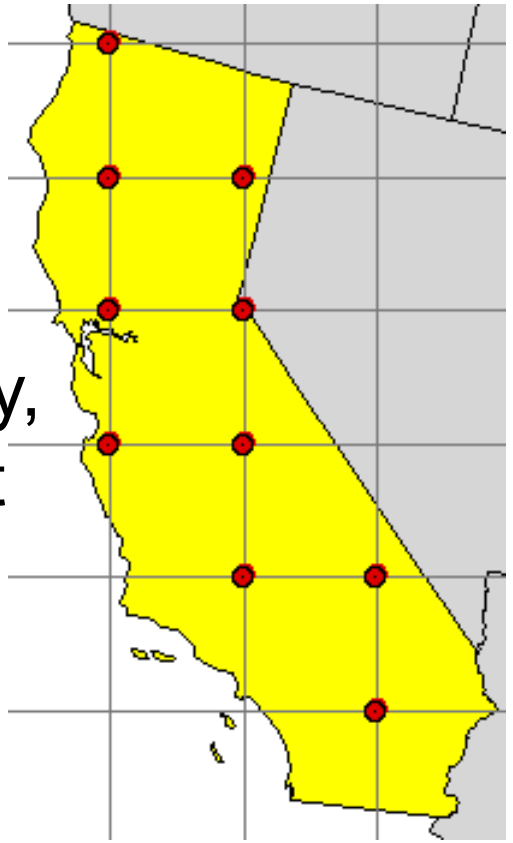
- E.g. when 8-bit integers are used to store each pixel value, the grayscale levels range from 0 (black) to 255 (white).

Image sampling

don't require prior knowledge of the content of the image

It is common to use a square sampling grid with pixels equally spaced along the two sides of the grid.

Both sampling and quantisation can introduce error in the sense that the image captured does not represent, with perfect fidelity, the original scene or objects that were photographed.



The effects of sampling and quantisation

quantization rate: image depth

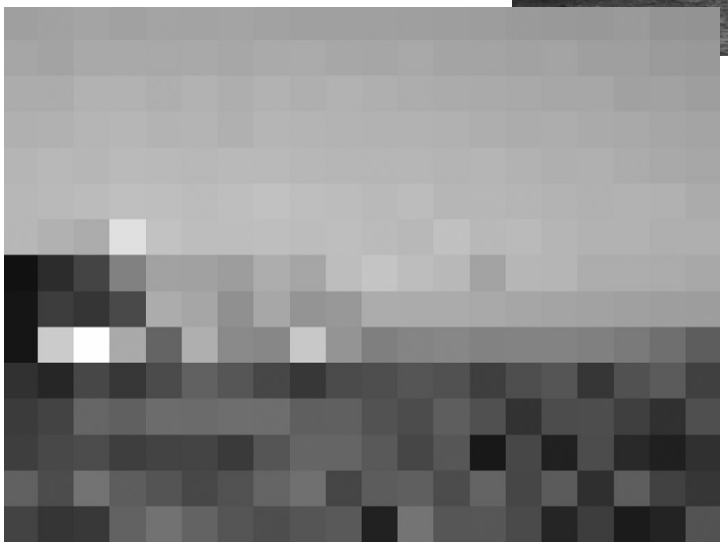


Image undersampled

distortion / blurred

Reduced bit depth

lose details / precision

Digital cameras

- Digital cameras use the same digitisation process discussed already: sampling and quantisation.
- Sampling rate is a matter of how many points of colour are sampled and recorded in each dimension of the image.
- A digital camera might allow you to choose from:

  1600x1200, 1280x960, 1024x768, 640x480

Digital cameras

- Quantisation is a matter of the colour model used and the corresponding bit depth.
  - Digital cameras generally use RGB, which saves each pixel in three bytes, one for each of the colour channels: red, green, and blue. Since three bytes is 24 bits, this makes it possible for 2^{24} = 16,777,216 colours to be represented.

8 bits for red
8 bits for green
8 bits for blue

Pixel dimensions

- For an image file, ***pixel dimensions*** is defined as the number of pixels (the "logical pixels") horizontally and vertically, e.g. 1600x1200.
- Similarly, your computer screen has a fixed maximum pixel dimensions (the "physical pixels"), e.g. 1024x768 or 1400x1050.
- When you display a bitmap image on your computer, the logical pixel is mapped to a physical pixel on the computer screen.

Image resolution 分辨率

- The number of pixels per inch is the image resolution.
- It is assumed that the same number of pixels are used in the horizontal and vertical directions.
- Typically, monitors have a screen resolution of 72 ppi (pixels per inch).

256 * 256 = 65536

Image resolution & image size

When you zoom in on a bitmap image, you can see the individual pixels that make up that image

resolution = pixel dimension / size (inch)

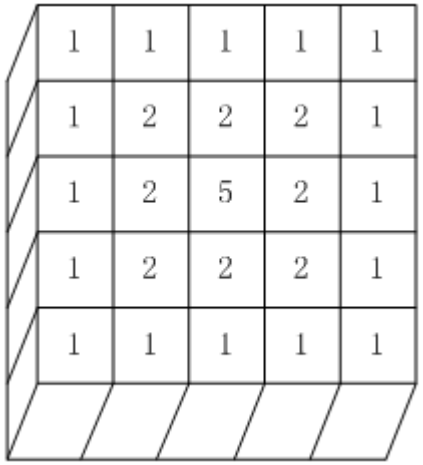
size ↑
resolution ↓

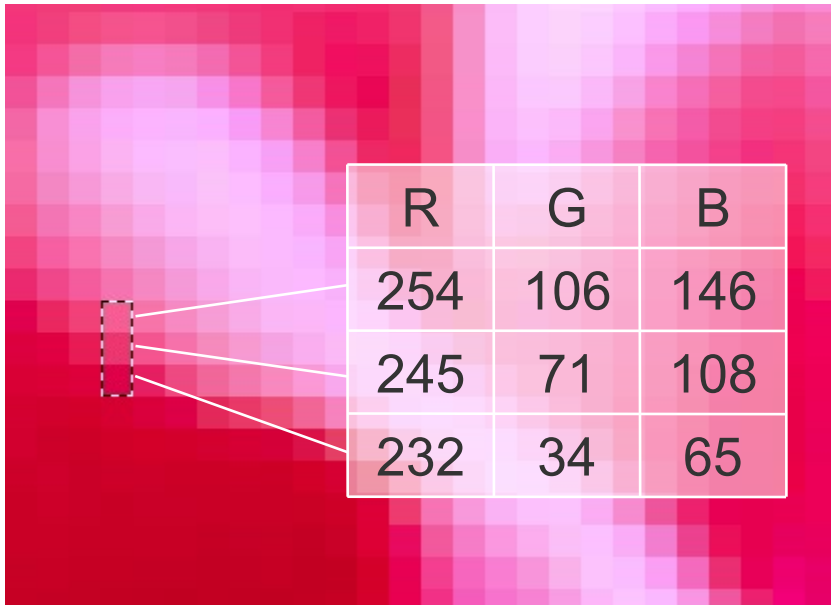
The image resolution varies with its size

less scalable than vector image

Image quantisation

The quantisation level is called image depth.



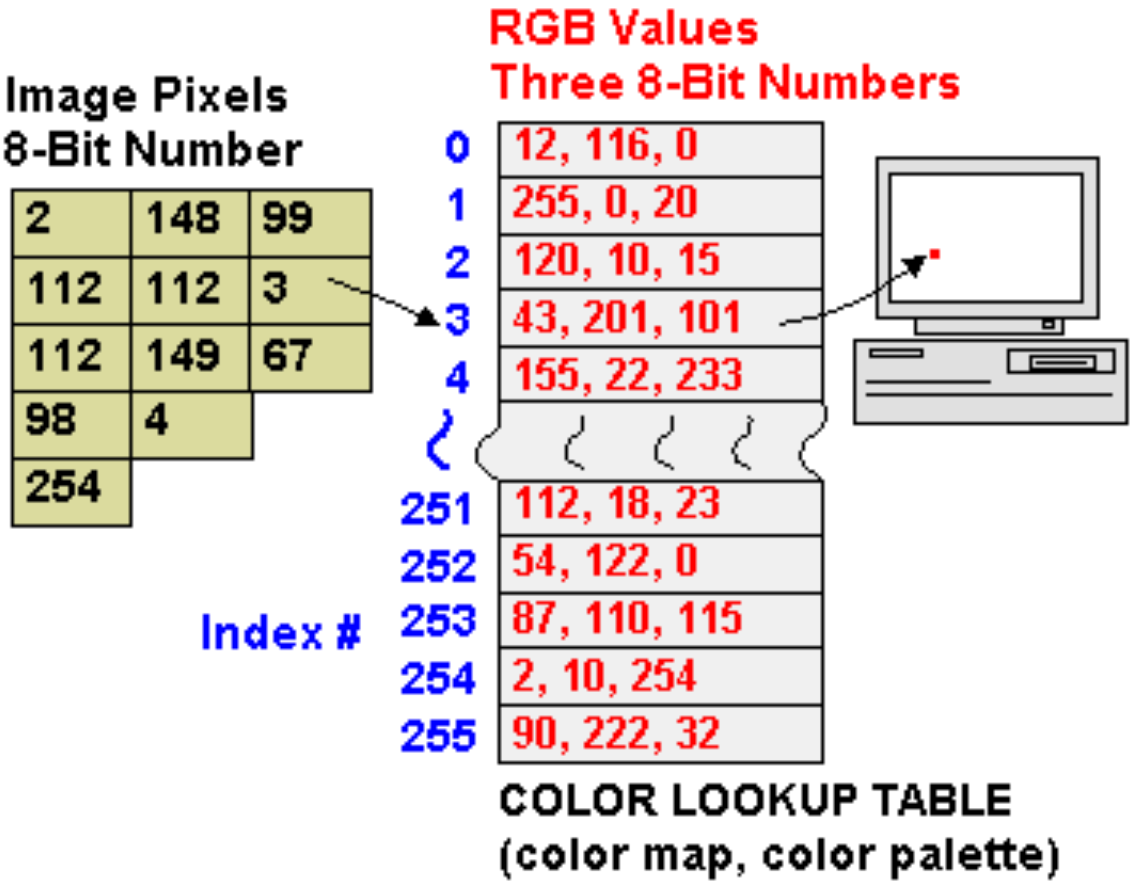


Stored Image Formats

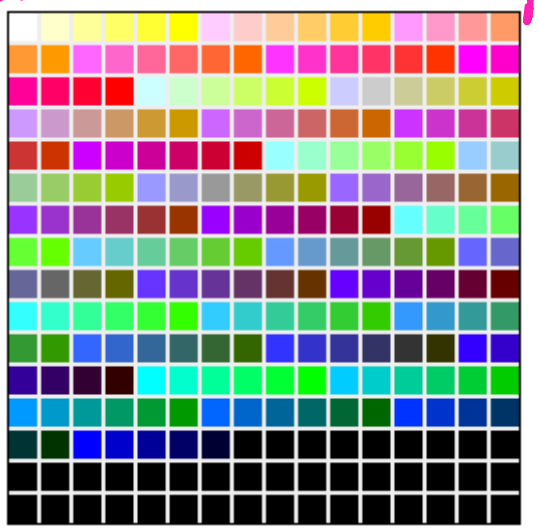
- RGB formats, also known as true-colour, use 8 bits of data for each Red, Green, and Blue value. Together, this forms a 24-bit pixel palette which has 16.7 million colours.
- Indexed formats are mapped to a smaller colour palette (CLUT) : 256-colours or less (normally).
  - The indexed image's palette contains all of the colours that are available for the image.

Each pixel is represented by an index, which refers to a color in the CLUT. The number of bits needed for each pixels depends on the size of CLUT.

Indexed format



可能会改变颜色.
因为不一定能找到原始图的颜色



CLUT (Colour Look Up Table)

depends on the images
many colors may be missing

Exercise

A 200 by 300 pixel image is stored in different formats. Calculate its size in Kbytes in the following cases:

- True colour image

  200×300×24 = 1,440,000 bits (180 KB)

  RGBA: 24→32

- Grayscale (8 bits image) ∵ RGB shows same value

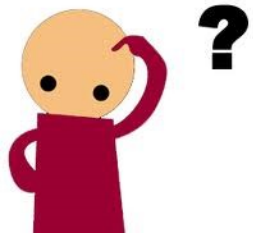
  256 different shades of grey.

  200×300×8 = 480,000 bits (60 KB)

- 4 bits indexed format

  200×300×4 = 240,000 bits (30 KB)

Question



Give two possible strategies for reducing the colour depth of a true colour image to just a third of its original one.

Convert the images to grayscale or use a 256 entries (8 bits) CLUT.

Grayscale images are entirely sufficient for many tasks and there is often no need to use more complex color images.

Exercise

A Colour Look-Up Table with 256 entries is used.

- Calculate the reduction in the size of the bitmap when a CLUT is used instead of true colour.
- Comment on how use of the CLUT will affect the appearance of the image on the screen.
- If the first entry in the CLUT has value 0,0,0, what colour would be stored there?

How would you avoid too much quality loss when converting from true colour to index-based format?
先要提一嘴true color是啥、答一下index-based只存了图片里用的
unavoidable maybe, depending on differenct images.
just to minimize.
有很多方法，比如
挑选最重要、出现最频繁的colours，用来决定图片中某些重要的细节
挑选最重要的区域(attract viewers' attention)，优先保存那部分的所有颜色，其他区域颜色适当丢弃

Exercise

960000
12 x 8

A 300 by 400 pixel image has a size of 120 Kbytes. How is it encoded?

300 x 400 x y = 120 x 8 = 960K

8 bits CLUT or grayscale

Devise a process to transform a true colour image into a binary image (open ended question).
Convert the true color into grayscale/index based
Set up a threshold value(maybe the middle value for grayscale like 127) the higher one will be classified as 1, the lower one will be considered as 0.

Devise a process to transform a true colour image into an index-based image (open ended question).
Analyse what colors you have in the image, pick repesentative colors,
give the color depth you're going to use,
(Identify all the colours used in this image and use these colours as a colour table
Give each colour a index and replace the original colour with the corresponding index)

Digital Image File Types

- If you take a picture with a digital camera or scan a photograph with a digital scanner, you'll have a choice of file types in which to save the image.
- Not all colour models can be accommodated by all file types, and some file types require that the image be compressed while some do not.
- The four most important things to know about a bitmap filetype are: its colour model; its bit depth; its compression type, if any; and the operating systems, browsers, and application software that support it.
- The possible bit depths for bitmap images include 1, 4, 8, 16, 24, 32, 48, and 64 bits.

Summary

- Bitmap images can be true-colour or index-based
- A true colour image pixel is represented with 24 bits (8 bits for each colour channel)
- 16.7 million colours can be represented with 24 bits
- A grayscale image has an image depth of 8
- An indexed image uses a Colour Look Up Table (CLUT)
- In an indexed image, each pixel is represented by an index, which refers to a colour in the CLUT
- The number of bits needed for each pixel depends on the size of the CLUT

EBU5303

Multimedia Fundamentals

Digital Video and Audio

Learning Objectives (Video and Audio)

- apply the Nyquist theorem to avoid digital audio aliasing.
- relate quantisation level and dynamic range of an audio file.
- calculate decibels from air pressure.
- calculate the signal-to-quantisation noise ratio.
- interpret the spectral analysis of an audio wave.
- describe the MIDI format.

Agenda

- A video is a sequence of images
- A sound is characterised by its frequency (pitch) and amplitude (loudness)
- CD standard quality is 44,100 Hz (sampling) and 16 bits (quantisation)
- Speech signals contain 3 types of sound, some of them are used for speech recognition
- MIDI format for music stores information such as instrument specification, beginning and end of a note, basic frequency, etc.

Video - definitions

- **Video** is the technology of electronically capturing, recording, processing, storing, transmitting, and reconstructing a sequence of still images representing scenes in motion.
- **Frame rate**: the number of still pictures per unit of time of video. 每單位时间的靜止圖片數
- **Analog video**: video recording method that stores continuous waves of red, green and blue intensities.
- **Digital video**: video recording system that works by using a digital rather than an analog video signal.

Refresh rate and frame rate

- The **refresh rate** is the number of times in a second that the display hardware draws the data (i.e. repeated drawing of identical frames).
- The **frame rate** measures how often a video source can feed an entire frame of new data to a display.
- Typical rates: 24, 25 or 30 frames per second (frame rates) ; 60, 75 or 120 Hz (refresh rates).

刷新率

幀率

刷新率 < 幀率 ⇒ 次圖沒法顯示所有幀

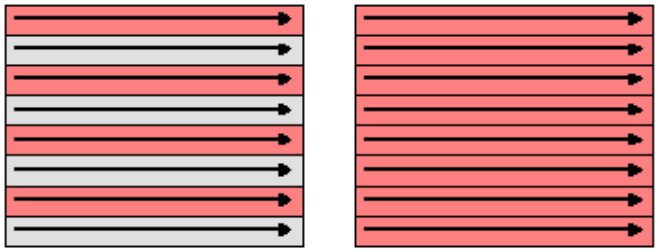
Frame Rates

Video Type	Frames Per Second (fps)
NTSC	29.97
PAL	25
SECAM	25
Motion Picture Film	24

NTSC was 30 fps for black-and-white TV, Frame rate was lowered to 29.97 fps to accommodate for color encoding.

Interlaced vs Progressive

- **Interlaced** scanning displays alternating sets of lines. Because each field happens so quickly we are given the illusion of a whole image.
- **Progressive** video displays the entire image.



9

Exercise

A 30fps digital video uses 352 by 255 pixels video frames with a pixel depth of 8.

i) Calculate the size of 1 second of data.

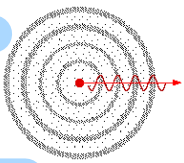
ii) What compression ratio would be needed to transmit 1 second of data in real-time over a 64 Kbps communication channel?

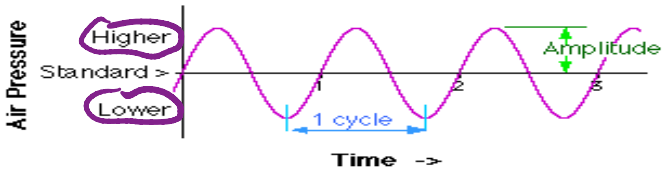
Agenda

- A video is a sequence of images
- A sound is characterised by its frequency (pitch) and amplitude (loudness)
- CD standard quality is 44,100 Hz (sampling) and 16 bits (quantisation)
- Speech signals contain 3 types of sound, some of them are used for speech recognition
- MIDI format for music stores information such as instrument specification, beginning and end of a note, basic frequency, etc.

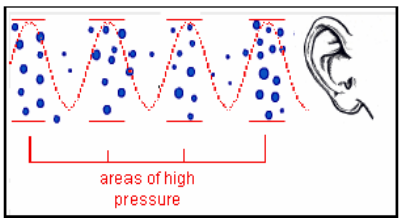
Sound

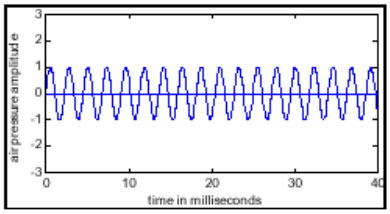
- Sound is a physical phenomenon produced by the vibration of matter, such as a violin string, or a block of wood.
- As the matter vibrates, pressure variations are created in the air surrounding it.
- This alteration of high and low pressure is propagated through the air in a wave-like motion.

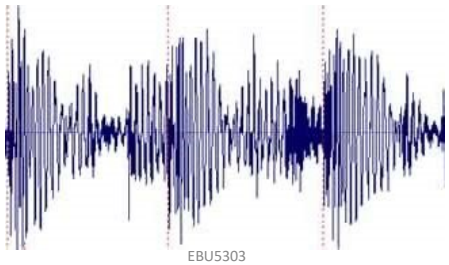




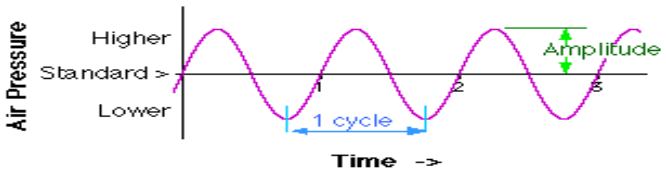
Sound in the analogue domain







Characteristics of Sound Waveforms



- Frequency determines the pitch
  (higher frequency = higher pitch)
  - Infra-sound: from 0 to 20 Hz
  - Human hearing frequency range: 20 Hz – 20 kHz
  - Ultrasound: from 20 kHz to 1 GHz
- Amplitude of the wave determines the volume or intensity (a property subjectively heard as loudness).

Agenda

- A video is a sequence of images
- A sound is characterised by its frequency (pitch) and amplitude (loudness)
- CD standard quality is 44,100 Hz (sampling) and 16 bits (quantisation)
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- MIDI format for music stores information such as instrument specification, beginning and end of a note, basic frequency, etc.

15

Computer Representation of Sound - Sampling -

- A computer measures the amplitude of the waveform at regular time intervals to produce a series of number (sampling). This is done by an ADC (*Analog-to-Digital Converter*)
- Sampling rate: the rate at which a waveform is sampled. e.g. *the CD standard sampling rate of 44100 Hz means that the waveform is sampled 44100 times / second.*

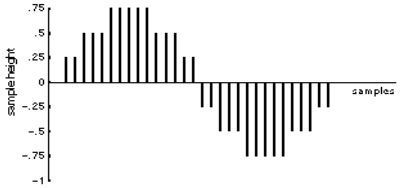
Sampled waveform

16

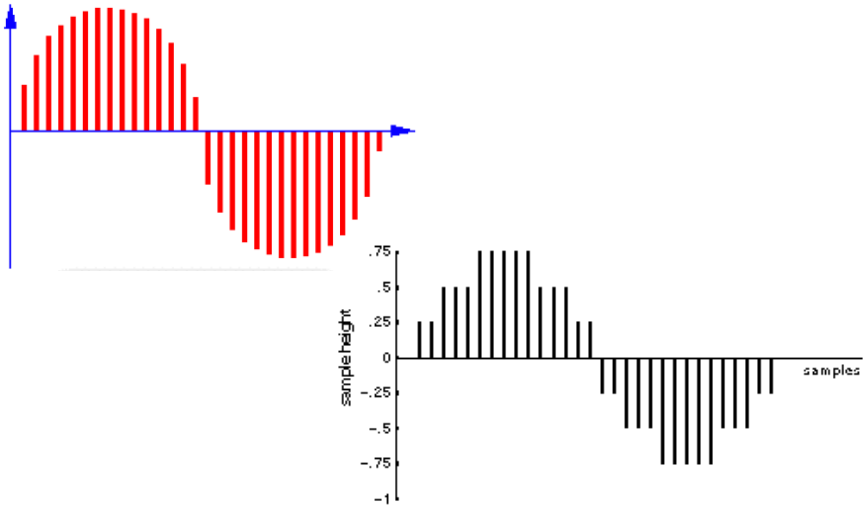
Computer Representation of Sound
- Quantisation -

- Quantisation: the resolution or quantisation of a sample value depends on the number of bits used in measuring the height of the waveform (*usually 8-bit or 16-bit*)

3-bit quantisation



Digitisation of Sound



Nyquist Rate and Nyquist Frequency

- Given an actual frequency to be sampled, the ***Nyquist rate*** is the lowest sampling rate that will permit accurate reconstruction of an analog digital signal.
- Given a sampling rate, the ***Nyquist frequency*** is the highest actual frequency component that can be sampled at the given rate without aliasing.
- Based on the Nyquist theorem, the Nyquist frequency is half the given sampling rate.

Nyquist Rate and Nyquist Frequency

KEY EQUATION

Given f_{max}, the frequency of the highest-frequency component in an audio signal to be sampled, then the ***Nyquist rate***, f_{nr}, is defined as

$$f_{nr} = 2f_{max}$$

KEY EQUATION

Given a sampling frequency f_{samp} to be used to sample an audio signal, then the ***Nyquist frequency***, f_{nf}, is defined as

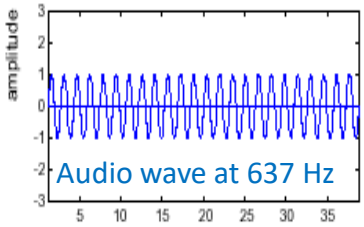
$$f_{nf} = \frac{1}{2} f_{samp}$$

Exercise

The bandwidth of a music signal is between 15 Hz and 20 KHz, assuming the Nyquist sampling rate is used, with 16 bits per sample:

- Derive the bit rate that is generated by the digitisation procedure
- What is the memory in Mbytes required to store a 10 minute passage of stereophonic music?

Aliasing (sampling error)



- The reason a too-low sampling rate results in aliasing is that there aren't enough sample points from which to accurately interpolate the sinusoidal form of the original wave.
- If we take ***more*** than two samples per cycle on an analog wave, the wave can be precisely reconstructed from the samples.

Measuring Sound Amplitude in Decibels

- A decibel is not an absolute unit of measurement.
- A decibel is always based upon some agreed-upon reference point, and the reference point varies according to the phenomenon being measured.
- For sound, the reference point is the *air pressure amplitude for the threshold of hearing*.
- A decibel in the context of sound pressure level is called ***decibels-sound-pressure-level*** (***dB_SPL***).

Measuring Sound Amplitude in Decibels

KEY EQUATION

Let E be the pressure amplitude of the sound being measured and E_0 be the sound pressure level of the threshold of hearing. Then ***decibels-sound-pressure-level***, (***dB_SPL***) is defined as

$$dB_SPL = 20 \log_{10}\left(\frac{E}{E_0}\right)$$

E_0 = 0.00002 Pa

Exercise

- What would be the amplitude (in decibels) of the audio threshold of pain, given as 30 Pa?
- What would be the pressure amplitude of normal conversation, given as 60 dB?

$dB\text{-}SPL = 20\log_{10}\left(\frac{30}{0.00002}\right) = 20 \times 6.17 \approx 123$

0.02Pa

Measuring Sound Amplitude in Decibels

- dB_SPL is an appropriate unit for measuring sound because the values increase logarithmically rather than linearly.
- This is a better match for the way humans perceive sound.
- Experimentally, it has been determined that if you increase the amplitude of an audio recording by 10 dB, it will sound about twice as loud.
- For most humans, a 3 dB change in amplitude is the smallest perceptible change.

对数式增长

可注意到的改变

Dynamic Range → SQNR

KEY EQUATION

Let n be the bit depth of a digital audio file. Then the ***dynamic range of the audio file***, d, in decibels, is defined as

$$d = 20n \log_{10}(2) \approx 6n$$

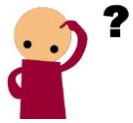
- You can estimate that an *n*-bit digital audio file has a dynamic range (or, equivalently, a signal-to-noise-ratio) of $6n$ dB.
- Dynamic range is a relative measurement—the relative difference between the loudest and softest parts representable in a digital audio file, as a function of the bit depth.

Exercise

$20\log_{10}2^n \approx 6n$

- What is the dynamic range (SQNR) of a 16 bit digital audio file? 16×6
- How about a 8 bit digital audio file? 8×6

Question



A sound file encoded with a 8 bits quantisation rate is likely to be:

- A piece of music
- Natural sound (e.g. rain)
- Speech
- A song

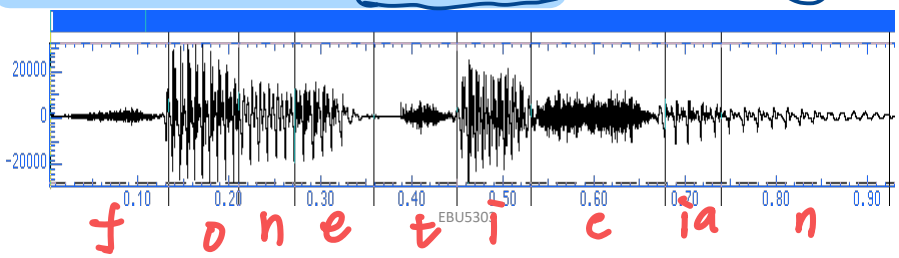
good enough to let listeners understand the speech.

Quantisation Error

- While an insufficient sampling rate can lead to aliasing, an insufficient bit depth can create quantisation error.
- ***Audio dithering*** is a way to compensate for quantisation error. The way to do this is to add small random values to samples in order to mask quantisation error.
- ***Noise shaping*** is another way to compensate for the quantisation error: it redistributes the quantisation error so that the noise is concentrated in the higher frequencies, where human hearing is less sensitive

Types of Speech Sounds

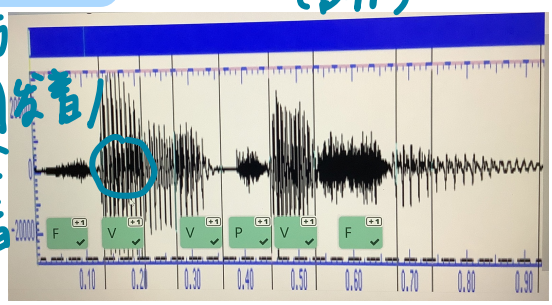
- Voiced sounds : the vocal chords are vibrated, which can be felt in the throat. All vowels are voiced.
- Fricatives (unvoiced sounds) : a consonant, such as *f* or *s* in English, produced by the forcing of air through a constricted passage.
- Plosives (also unvoiced sounds) : a speech sound produced by complete closure of the oral passage and subsequent release accompanied by a burst of air, as in the sound (d) in *dog*.



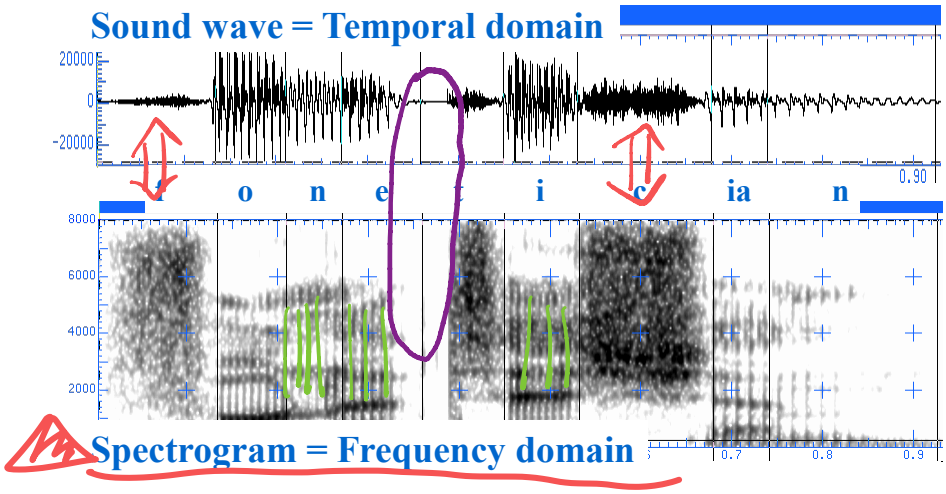
Voiced Speech Sounds

Voiced speech sounds have two properties which can be used in speech processing:

- Speech signals show during certain time intervals almost periodic behaviours. These signals are *quasi-stationary signals* for around 30 ms.
- The spectrum of speech signals (voiced sounds) shows characteristic maxima. These maxima, called formants, occur because of resonances of the vocal tract.



Temporal and Frequency Domains



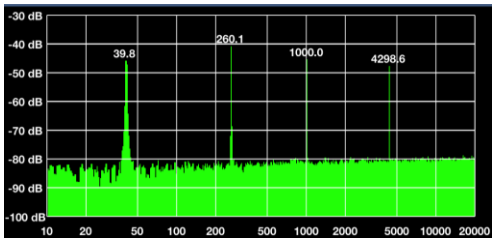
audio histograms audio spectrograms

Temporal and Frequency Domains

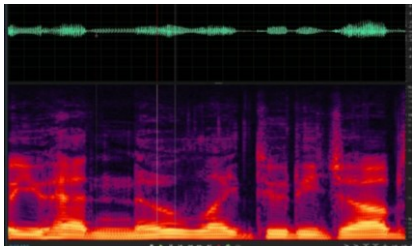
- Sound can be represented either over the time domain or the frequency domain.
- The transform that is used most frequently in digital audio processing is the Fourier transform.
- A complex waveform is equal to an infinite sum of simple sinusoidal waves, beginning with a ***fundamental frequency*** and going through frequencies that are integer multiples of the fundamental frequency.
- These integer multiples are called ***harmonic frequencies***.
- In the ***frequency domain***, data is stored as the amplitudes of frequency components

Frequency Domain

A Power Spectrum is a 2-dimensional representation (frequency(x) / amplitude(y); a Spectrogram is a 3-dimensional representation (time(x) / frequency(y) / amplitude(colour).



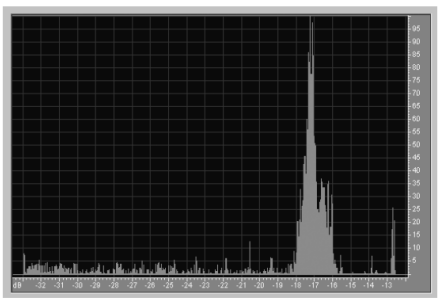
Power Spectrum



Spectrogram

Audio Histogram

- Audio processing programs sometimes offer a statistical analysis of your audio files, which analyse sample values in the **time domain**.
- An ***audio histogram*** shows how many samples there are at each amplitude level in the audio selection.



X axis: amplitude in decibels
Y axis: number of samples

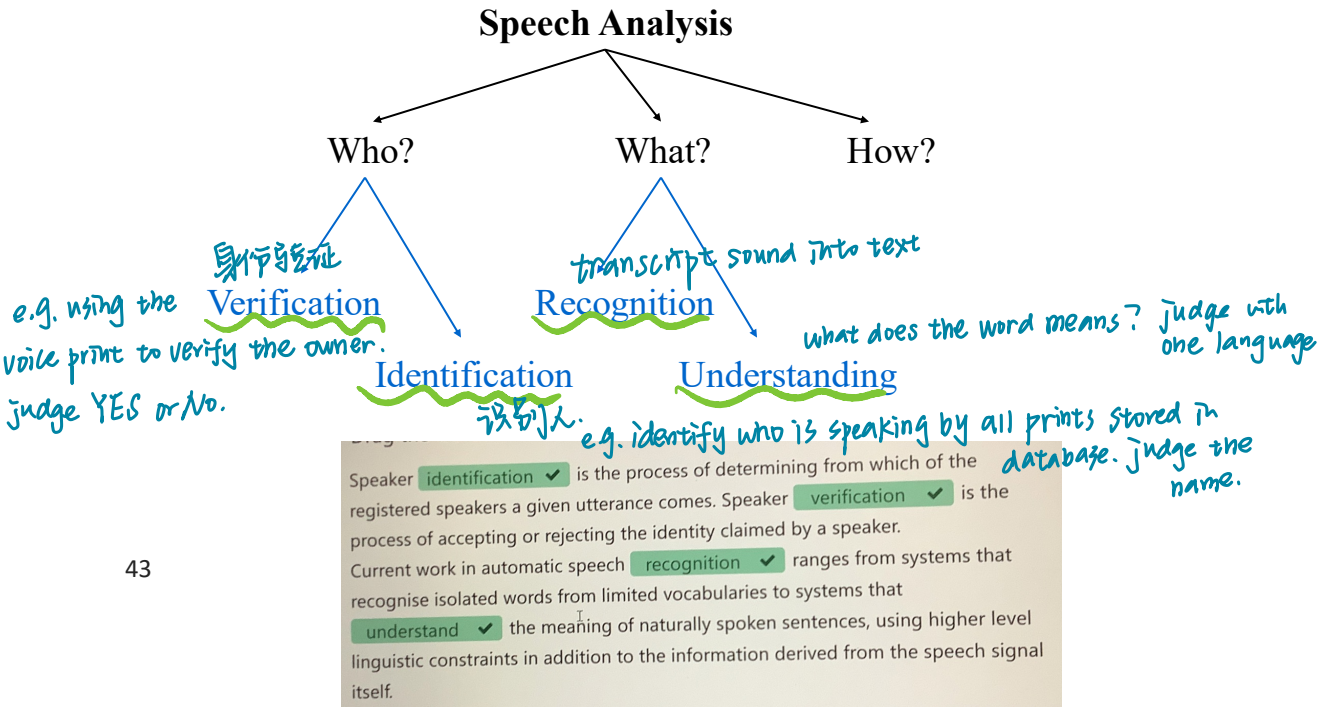
An audio histogram is in the time domain. The X axis represents sample values such as amplitude levels, and the Y axis represents their number of occurrences or probability.

4/4

Continue

The difference between speech recognition and speech understanding
speech recognition basically take the sound and try to map the sound with language
And then transcribe the sound into text.
speech understanding will check the text that has been produced is correct.So speech understanding will have a lot more knowledge.

Speech Processing Applications



43

Agenda

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- MIDI format for music stores information such as instrument specification, beginning and end of a note, basic frequency, etc.

Music

• Thus far, we've been considering digital audio that is created from sampling analog sound waves and quantising the sample values.

•There's another way to store sound in digital form: ***MIDI (Musical Instrument Digital Interface)***.

• MIDI stores "sound events" or "human performances of sound" rather than sound itself. ∴ It's smaller than other sound files

• The difference between sampled digital audio and MIDI is analogous to the difference between bitmapped graphics and vector graphics

MIDI Data Format

• A MIDI file contains messages indicating when a note begins (Note On), when a note ends (Note Off), what the note is, how hard it is pressed (Velocity), how hard it is held down (Aftertouch), what instrument is played, and so forth.

• Each MIDI message communicate one musical event.

e.g. when a musician presses a piano key, the MIDI interface creates a MIDI message where the beginning of the note with its stroke intensity is encoded.

• The MIDI standard identifies 128 instruments (including noise effects) with unique numbers (e.g. 41 for the violin).

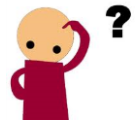
MIDI Hardware and Software

• Hardware devices that generate MIDI messages are called ***MIDI controllers***.

• Devices that read MIDI messages and turn them into audio signals are called ***MIDI synthesizers***. Two methods for synthesizing sound are ***frequency modulation synthesis*** and ***wavetable synthesis***.

• A ***MIDI sequencer*** is a hardware device or software application program that allows you to receive, store, and edit MIDI data.

Questions



- Why are MIDI encoded music signals very small?
- What other advantage to MIDI audio is there compared to sampled digital audio?
- Is there any disadvantage to MIDI audio compared to sampled digital audio?

→ 저용량 sound samples, just like vector-based image.

① small file size
② ease of modification & manipulation
③ a wide choice of electronic instrument and synthesisers.
④ no background noise

Musical Acoustics and Notation

• In Western music notation, musical sounds, called ***tones,*** are characterized by their pitch, timbre, and loudness.

• With the addition of onset and duration, a musical sound is called a ***note***.

• The ***pitch*** of a note is how high or low it sounds to the human ear.

• The ***timbre*** of a musical sound is its "tone color".

• The lowest frequency of a given sound produced by a particular instrument is its **fundamental frequency**. Then there are other frequencies combined in the sound, which are integer multiples of the fundamental frequency, referred to as ***harmonics.***

Exercise

- A musical note played on an instrument consists of a fundamental frequency and, depending on the instrument, different numbers of harmonics. Each harmonic is an integer multiple of the fundamental frequency.
- Given a note from a musical instrument, which contains only the following frequency components: 100Hz, 200Hz, 300Hz, and 400Hz, **at what rate would you need to sample this sound to ensure that the sampled audio was of the same fidelity as the original note?**
- Assuming that the amplitude of each harmonic is half the amplitude of the previous harmonic, **sketch the signal in the frequency domain for the above note.**

Musical Acoustics and Notation

If the frequency of one note is 2^n times of the frequency of another, where n is an integer, the two notes sound "the same" to the human ear, except that the first is higher-pitched than the second.

声音相同，第一个音符音调高于第二个音符

KEY EQUATION

Let g be the frequency of a musical note. Let h be the frequency of a musical tone n ***octaves*** higher than g. Then

$$h = 2^n g$$

Exercise

If the frequency of a note A is about 440 Hz, what is the frequency of an A two octaves below the 440 Hz A?

EBU5303

Introduction to Data Compression

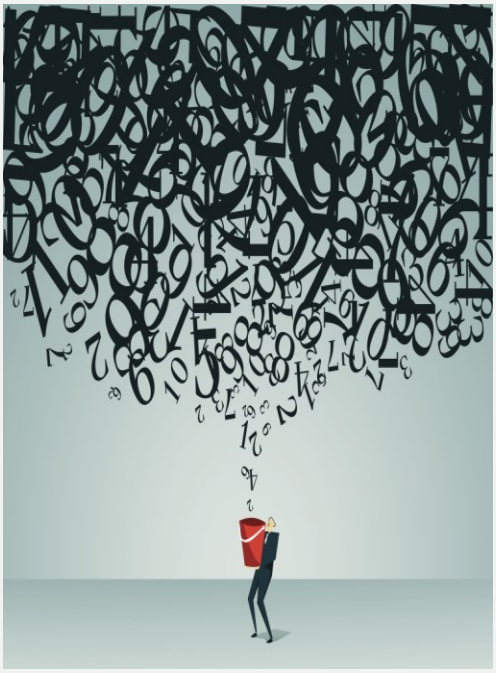


Image Credits: erhui1979 / Getty Images

1

Learning Objectives

- Explain what data compression is
- Discuss why it is important
- Appraise how it can be done
- Describe examples of compression techniques for image, music and video data

2

2

How does the brain deal with the enormous amount of information constantly coming through the senses?

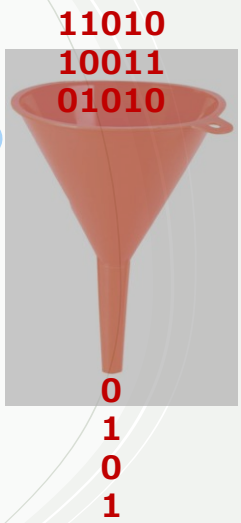
A. It throws away most of it
B. It groups information into meaningful chunks
C. It simplifies the representations by removing large amount of data from the stimuli
D. It stores large complex information using short codes

9

Digital Data Compression

- Compression: the process of encoding information using less data than the original representation.
- Encoding = representing information in binary format (bits).

➢ Digital data compression: the process of removing some of the bits …



10

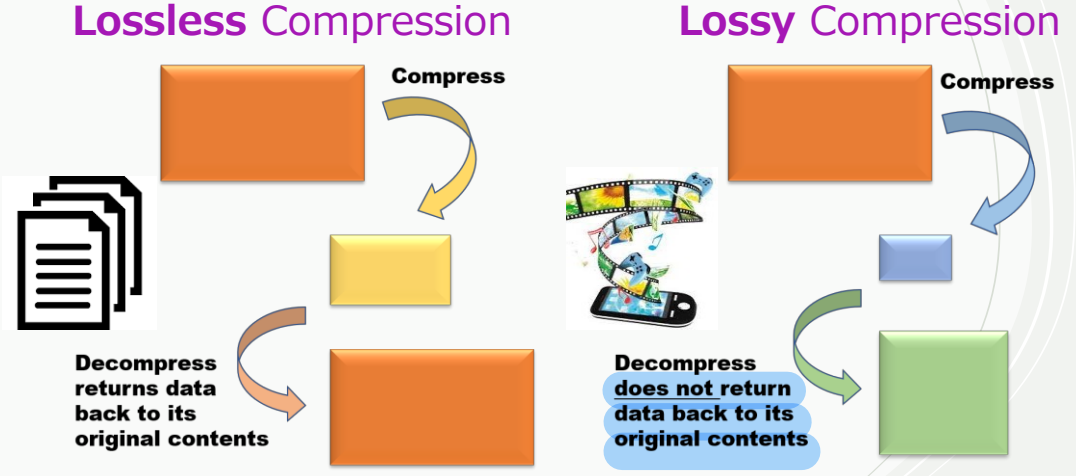
In your opinion, which of these applications are data intensive? Rank them from the most data intensive (1st) to the least (6th).

Video conferencing; Video on demand; Music streaming; Storing personal images; Reading eBooks; Sharing images with friends

19

19

How does compression work?



20

20

is the content / is the carrier

Information and Data

- Information carries meaning, it is useful, it contributes to knowledge.
- Data are representations of information, e.g. symbols, characters, bits …

➢ Compression strives to reduce data while preserving information

23

Redundancy

- Information that is expressed (represented) more than once
- Data that do not carry new information / Repetition of the same information
  - *Spatial redundancy*
  - *Statistical redundancy*
  - *Perceptual redundancy*
  - *Temporal redundancy*

24

run length encoding

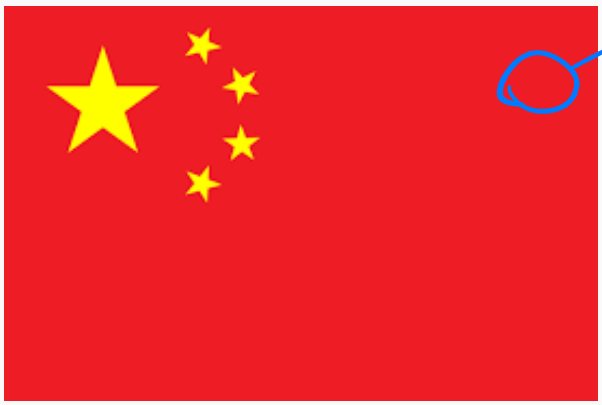
	Lossless	Lossy
Image	RLE (spatial) Huffman	Jpeg
Sound		Mp3
Video		Mpeg

25

25

Spatial redundancy

the color is organised



same color / pixels cause redundancy

26

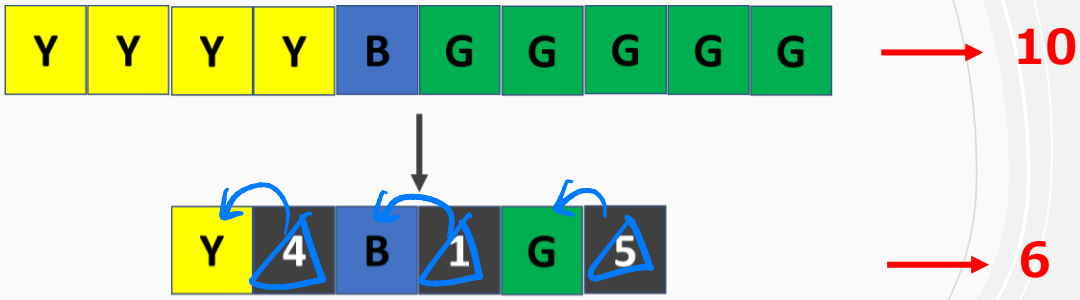
26

Ex 1: compressing images with RLE (lossless)

- RLE = Run Length Encoding
- Exploits spatial redundancy in images
- Reduces any type of repeating sequence (run), once the sequence reaches a predefined number of occurrences (length)
- Instead of storing each pixel as an individual value, stores number pairs *(c, n)*, where *c* indicates the pixel colour and *n* is how many consecutive pixels have that colour.

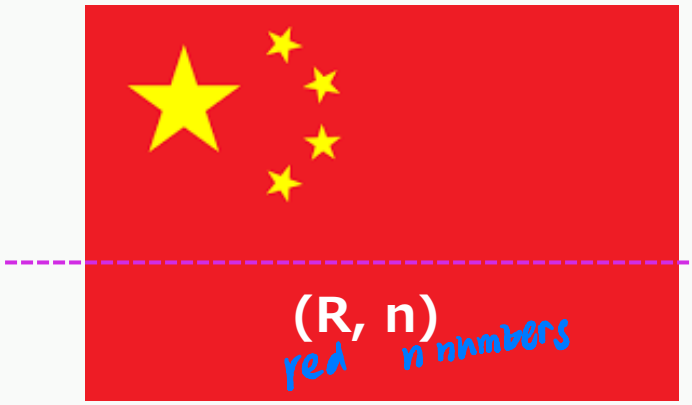
27

Ex 1: compressing images with RLE (lossless)



28

Ex 1: compressing images with RLE (lossless)



29

29

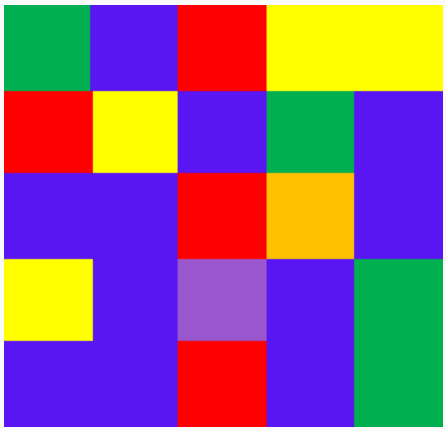
	Lossless	Lossy
Image	RLE (spatial) **Huffman (statistical)**	Jpeg
Sound		Mp3
Video		Mpeg

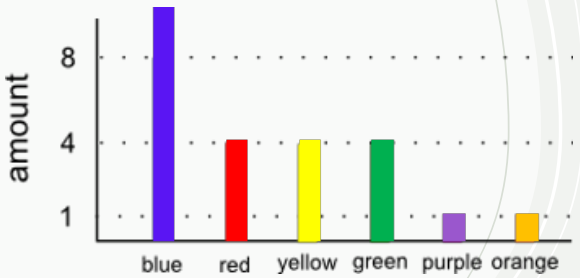
30

30

tomorrow → tmrw

Statistical redundancy





31

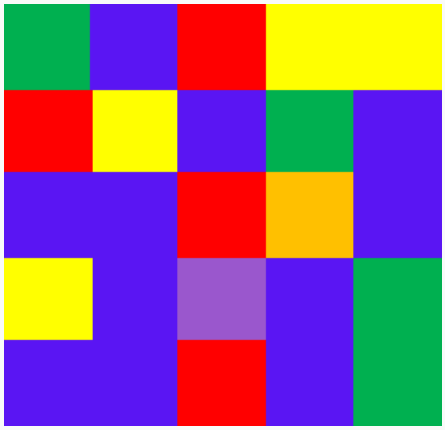
randomly appearing

Ex 2: compressing images with Huffman (lossless)

- Exploits statistical redundancy in images
- Is an example of entropy compression
- Encodes frequent colours with short codes and infrequent colours with longer codes
- Fixed-length inputs become variable-length outputs
- Effective when probabilities vary widely

32

Ex 2: compressing images with Huffman (lossless)



Colour	freq.	bit code
	11	1
	4	011
	4	010
	4	001
	1	0001
	1	0000

1 bits

3 bits

4 bits

25 pixels x 3 bits = **75 bits**

Huffman (variable length) encoding:

(11x1)+(4x3)+(4X3)+(4x3)+4+4 = 55 bits

33

33

Which of the following statements about spatial redundancy is true?

An image has spatial redundancy when …

A. it contains a small number of colours

B. the colours are almost identical from one pixel to the next

C. it contains large areas of uniform colour

34

34

In which case will Huffman be the most efficient for image compression?

A. When the colours' probabilities are all the same

B. When the colours' probabilities vary widely

C. When there is many different colours

35

35

	Lossless	Lossy
Image	RLE (spatial) Huffman (statistical)	**Jpeg (perceptual)**
Sound		Mp3
Video		Mpeg

36

36

Perceptual redundancy

original

Compression: 10:1

37

Ex 3: compressing images with jpeg (lossy)

- Jpeg exploits perceptual redundancy
- Is an example of transform encoding
- Pixel data is converted to frequency coefficients
- Compression is done by removing the high frequency values

do in the frequency domain

38

Transform encoding

去除人类感知不到的细节

- The idea is that changing the representation (domain) of data can sometimes make it possible to extract details that won't be missed because they are beyond the acuity of human perception.
- High frequency components in images correspond to quick fluctuations of colour in a short space—changes that aren't easy for the human eye to see.
- Once the high frequency components of an image have been separated out, they can be removed.
- Jpeg uses the ***discrete cosine transform*** (***DCT***).

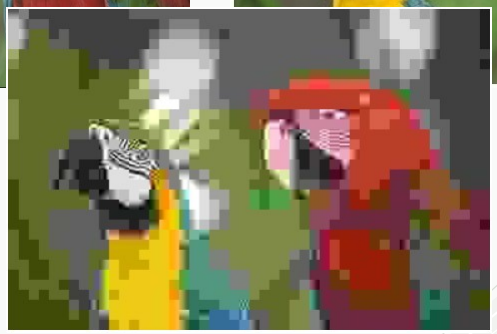
39

39

Ex 3: compressing images with jpeg (lossy)

original

Compression: 10:1



Compression: 100:1

40

40

	Lossless	Lossy
Image	RLE (spatial) Huffman (statistical)	Jpeg (perceptual)
Sound		**Mp3 (perceptual)**
Video		Mpeg

41

41

Ex 4: compressing music with mp3 (lossy)

- Mp3 also exploits perceptual redundancy
- Uses knowledge from psychoacoustics to find inaudible data
- Psychoacoustics is the study of how the human ears and brain perceive sound
- Compression is done by removing the sounds that are either below the "*threshold of hearing*" or that are "*masked*" by other sounds.

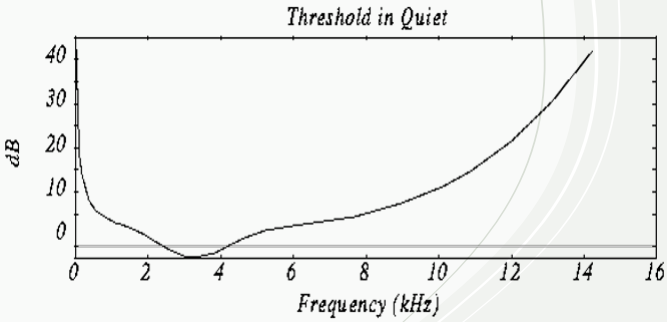
42

42

Threshold of hearing

- Human's sensitivity to sound varies with frequencies: humans hear best in the range of about 1 to 5 kHz, which is close to the range of the human voice.
- Threshold of hearing = minimal sound level at which sound can be heard

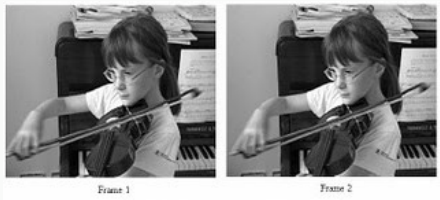
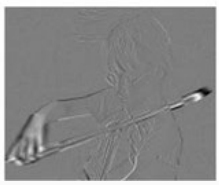
definition



43

	Lossless	Lossy
Image	RLE (spatial) Huffman (statistical)	Jpeg (perceptual)
Sound		Mp3 (perceptual)
Video		**Mpeg (temporal)**

44

Temporal redundancy



difference frame

45

Ex 5: compressing video with mpeg (lossy)

- Mpeg exploits temporal redundancy
- It looks for similarities between consecutive frames 连续的
- Only the changes from one frame to the next are encoded

46

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Multimedia Fundamentals

Lossless Compression

Agenda

- Compression can be lossless or lossy
- Lossless techniques include:
  - RLE
  - LZW
  - Huffman

Basics of Media Compression

- **Lossless encoding**
  - redundant data is not encoded
  - no information is lost in compression and decompression cycle
  - possible because some data appear more often that other and some data normally appear together
- **Lossy encoding**
  - details not perceived by users are discarded; e.g. small change in lightness (luminance) noticed more than change of colour details
  - approximation of the real data

Lossless versus Lossy

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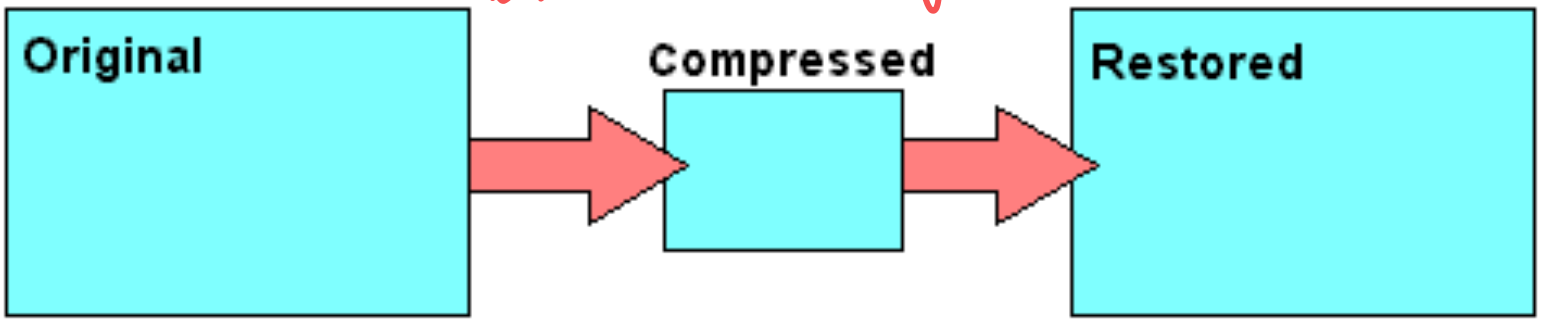
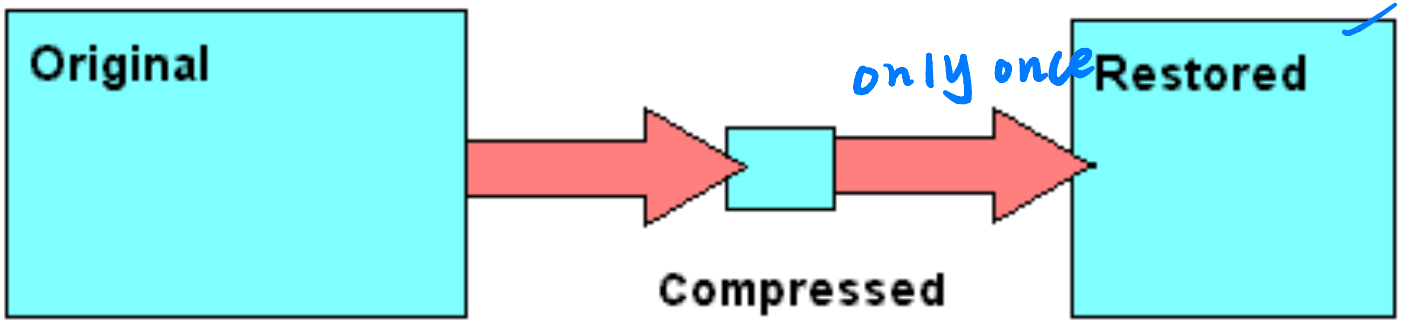
LOSSLESS can be ecoded/decoded many times, as the decoded image is the same as the original one.



LOSSY



only once

reconstruction

can't be repeated.

Compression rate

- The ***compression rate*** is the ratio of the original file size *a* to the size of the compressed file *b*, expressed as *a*:*b*.
- Alternatively, you can speak of the ratio of *b* to *a* as a percentage.
- For example, for a file that is reduced by compression to half its original size, you could say that 50% compression is achieved, or alternatively, that the compression rate is 2:1.

Types of Compression Algorithms

- Types of compression algorithm include ***dictionary-based, entropy,*** *arithmetic, adaptive, perceptual* and *differential compression methods*.
- *Dictionary-based methods* (e.g. **LZW compression**) use a look-up table of fixed-length codes, where one code word may correspond to a string of symbols rather than to a single symbol.
- *Entropy compression* (e.g. **Huffman encoding**) uses a statistical analysis of the frequency of symbols and achieves compression by encoding more frequently-occurring symbols with shorter code words.

comparison .

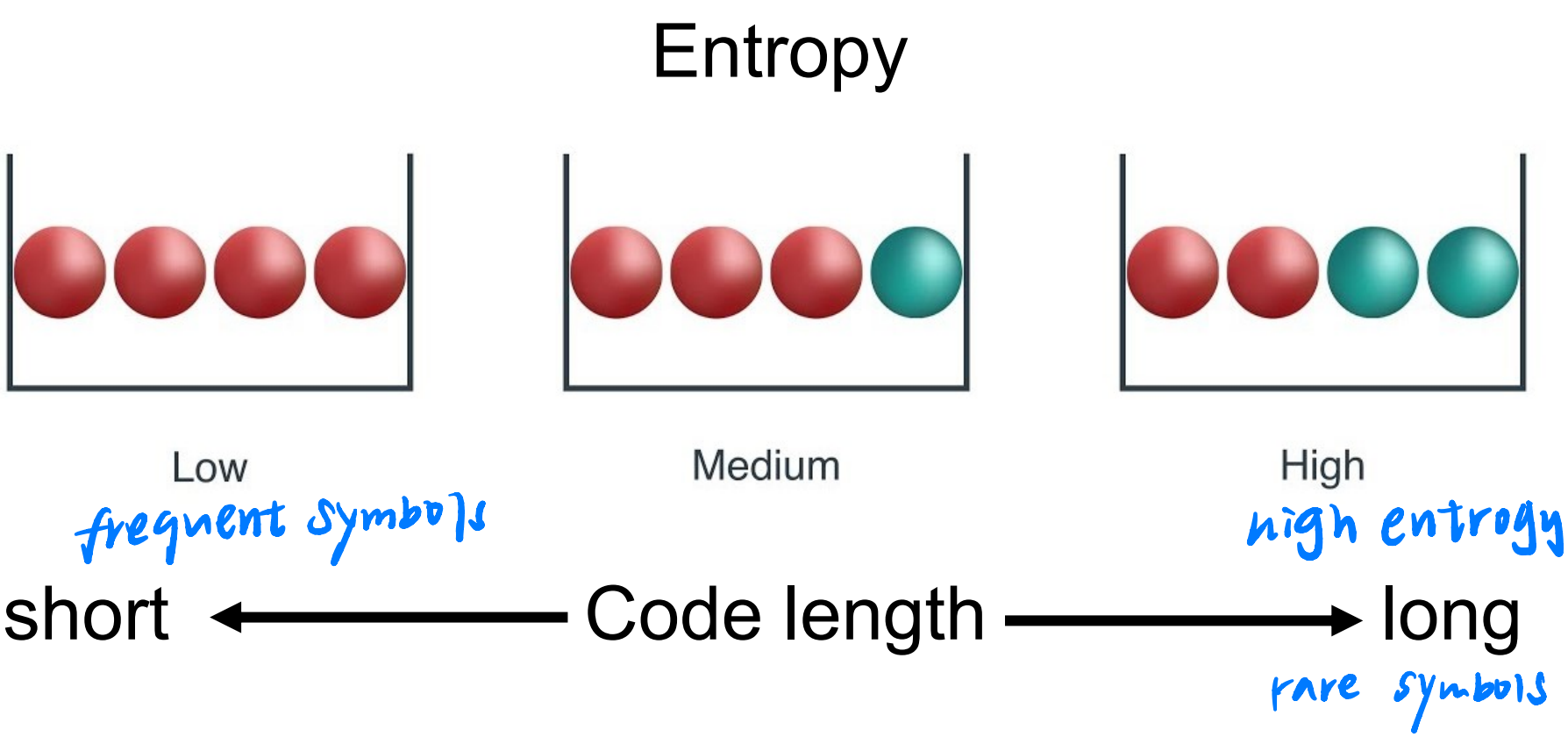
Dictionary-based algorithms shrink files by finding common elements that match a "catalog" of data (i.e. dictionary). Entropy encoding compresses images by looking for patterns in the data, and replacing those that occur frequently with codes that are smaller in size.

- ◉ True ✔
- ○ False

You got 1 of 1 points

▶ Continue

Entropy Compression



✔ Entropy encoding is a term used to denote some compression algorithms, which work based on the probability distribution of source symbols. +1

With entropy encoding, the most common symbols use the longest codes.

✔ Entropy coding is based on the frequency of occurance of a data item (pixel in images). +1

✔ Entropy encoders compress data by replacing each fixed-length input symbol with a variable-length output codeword. +1

3/3

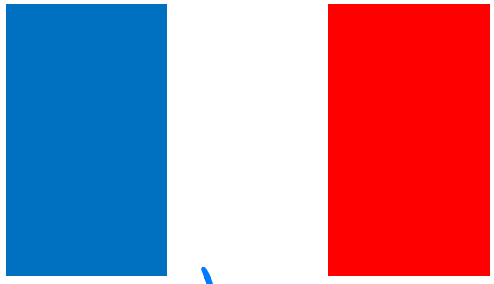
Continue

例: !4A "!" 占一个Byte; 1,2,3 无需RLE, 4~255 1Byte 255~$2^{16}-1$ 2Byte.

Run-Length Encoding (RLE)

字母代表color. true color: 3 byte. grayscale: 1 Byte

- Physically reduce any type of repeating sequence (**run**), once the sequence reach a predefined number of occurrence (**length**)
  - E.g. area of solid colour



always have uniform colors

- Lossless
- Exploits **spatial redundancy**
- The simple idea is that instead of storing each pixel as an individual value, it *can be* more concise to store number pairs (c, n), where *c* indicates the pixel (color) value and *n* is how many consecutive pixels have that value. length

RLE Example

- For example, say that in a 10,000-pixel grayscale image, the first 20 pixels are:

  255 255 255 255 255 255 242 242 242 242 238 238 238 238 238 255 255 255 255

- The run-length encoding of this sequence would be:

  (255, 6), (242, 4), (238, 6), (255, 4)

- Without RLE, 20 pixels require:

  20 pixels * 1 byte/pixel = 20 bytes

- With RLE: 8 bytes
- Compression rate: 20:8 = **5:2**

RLE: Calculating the length

- The run-length encoding algorithm scans the image file in a preprocessing step to determine the size of the largest run of colors, r.
- The formula for figuring out how many bytes you need to represent a number that can be anywhere between 0 and r is:

$$b = \left\lceil \frac{\log_2(r + 1)}{8} \right\rceil$$

向上取整

RLE Example revisited

- Let's consider this sequence of pixels again:

  255 255 255 255 255 255 242 242 242 242 238
  238 238 238 238 238 255 255 255 255

- The run-length encoding of the sequence is:

  (255, 6), (242, 4), (238, 6), (255, 4)

- If after pre-processing the image, the longer run is found to be r= 300, 2 bytes will be needed to store *n*.
- With RLE, the 20 pixels now require 12 bytes.
- Compression rate: 20:12

4+8=12

Calculating the length (another way)

- Rather than determining the largest *n* in advance and then setting the bit depth accordingly, we could choose a bit depth *d* in advance (e.g. 8 bits).
- Then if more than 2^d consecutive pixels of the same colour are encountered, the runs are divided into blocks.
- For example, if 1 byte is used to represent each *n*, then the largest value for *n* is 255. If 1000 consecutive whites exist in the file they would be represented as 4 different runs:

(255, 255), (255, 255), (255, 255), (255, 235)

Exercise

- How would you encode the following sequence of symbols using RLE, assuming each symbol represents a grayscale pixel value (i.e. 1 byte) ?

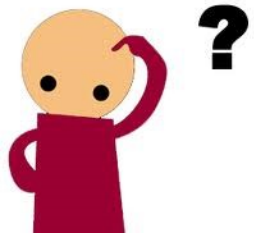
A(2)B(1)C(1)A(6)B(5)C(11)D(1)

AABCAAAAAABBBBBCCCCCCCCCCCCD

27 : 14

- What is the compression rate?

Question



Images of country flags can usually be compressed very efficiently using RLE. Why? Choose one answer.

- Because they contain a small number of colours.
- Because the colours are almost identical from one pixel to the next.
- Because of their statistical properties.
- Because they contain large areas of uniform colour.

Exercise

Consider the 8 bits gray scale image shown in the Figure below. It contains 1050 x 780 pixels. Calculate the maximum compression rate you can achieve on that image by using Run Length Encoding. Explain your calculations.

uncompressed size: $1050 \times 780 \times 1 = 819000$ bytes

compressed size: $3 \times 3 \times 780 = 7020$ bytes

$819000 / 7020 = 350 : 3$

3 bytes 3 bytes 3 bytes

$\frac{\log_2(350+1)}{8} \approx 2$ bytes

1 byte for the pixel value

2 bytes for the size

LZW

- LZW stands for Lempel-Ziv-Welch, the creators of the algorithm.
- Lossless and exploits *spatial redundancy* (like RLE)
- The method is commonly applied to GIF and TIFF image files.
- The LZW algorithm is based on the observation that sequences of color in an image file (or sequences of characters in a text file, i.e. words) are often repeated.
- LZW is an example of *dictionary-based encoding*.

Dictionary-based encoding

- Dictionary-based algorithms encode variable-length strings of symbols as single tokens (indexes).
- The tokens form an index into a phrase dictionary.
- If the tokens are smaller than the phrases they replace, compression occurs.
- Dictionary-based compression uses a strategy that programmers are familiar with, i.e. using indexes into databases to retrieve information from large amounts of storage.

Select all the correct statements.

Dictionary based compression techniques replace individual symbols with a variable length codewords.

✔ With dictionary-based algorithms compression is achieved by replacing long strings with shorter codewords. +1

✔ Dictionary-based compression algorithms encode variable-length strings of symbols as single tokens. +1

✔ Dictionary based compression can be described as the compression of a file into a smaller file using a table-based lookup algorithm. +1

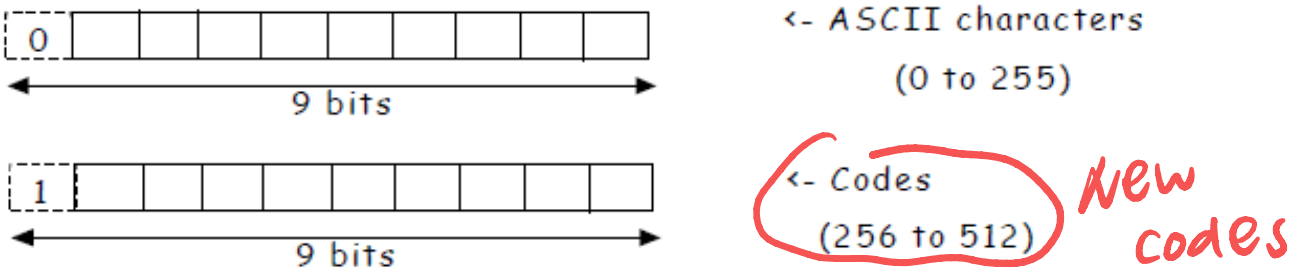
3/3

Continue

LZW for text

8 bits

- For text, the dictionary is initialised with 256 entries (indexed with ASCII codes 0 through 255) representing the ASCII table.
- Instead of using 2 or more bytes per code, we could use 9 bits:



- With 9 bits we can only have a maximum of 256 codes for strings of length 2 or above (with the first 256 entries for ASCII characters).
- Original LZW uses dictionary with 4K entries, with the length of each symbol/code being 12 bits.

Exercise

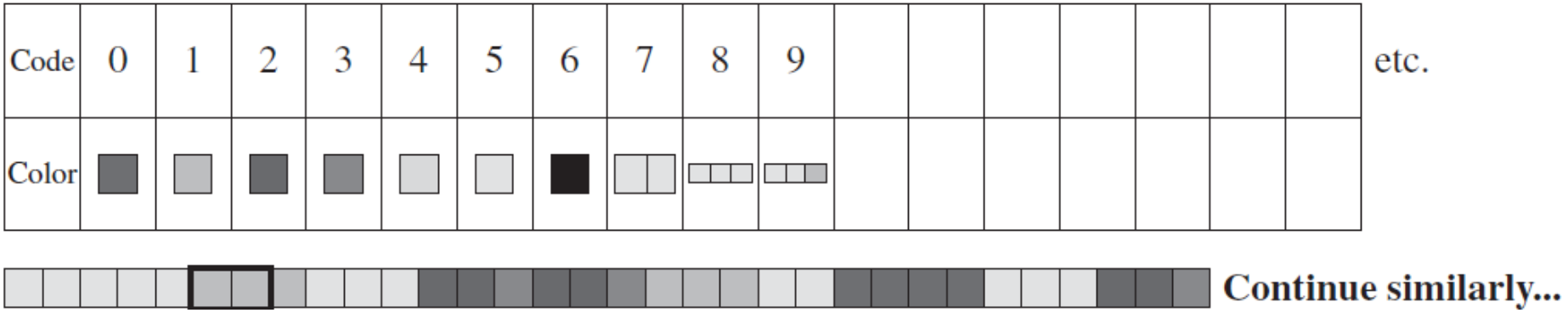
- What is the encoded string? (66)(65)(256)(257)(65)(260)(65)
- What is the original string? B|A|BA|AB|A|AA
- What is the compression rate? 10×8 : 7×9 LZW

bits bits

Current Char	Next Char	CurrentChar + NextChar is in the dictionary ?	Output Index	[New Index] New String
"B"	"A"	No	66	[256]"BA"
"A"	"B"	No	65	[257]"AB"
"B"	"A"	Yes	-	-
"BA"	"A"	No	256	[258]"BAA"
"A"	"B"	Yes	-	-
"AB"	"A"	No	257	[259]"ABA"
"A"	"A"	No	65	[260]"AA"
"A"	"A"	Yes	-	-
"AA"	"A"	No	260	[261]"AAA"
"A"	EOF	-	65	-

LZW for images

- With a first pass over the image file, the code table is initialised to contain all the individual colours that exist in the image file.
- These colours are encoded in consecutive integers.



Huffman encoding

- Encode frequent bit patterns with short code, infrequent bit patterns with longer codes
- Example of *statistical coding* or *entropy compression*
- Effective when probabilities vary widely
- Lossless
- Fixed-length inputs become variable-length outputs

statistical redundancy

utilize statistical property, i.e. the fact that some colors appear more often than others.

entropy↑ uncertainty↑ information↑

Entropy Compression

Shannon's source coding: the optimal code length for a symbol

- Entropy encoding works by means of variable-length codes, using fewer bits to encode symbols that occur more frequently, while using more bits for symbols that occur infrequently.
- Shannon's entropy equation:

$\log_2 P$

the probability of the input symbol

KEY EQUATION

Let S be a string of symbols and p_i be the frequency of the i^{th} symbol in the string. (p_i can equivalently be defined as the probability that the i^{th} symbol will appear at any given position in the string.) Then

$$H(S) = \eta = \sum_i p_i \log_2\left(\frac{1}{p_i}\right)$$

Entropy and bit depth

For an image file that has exactly 256 pixels in it, each pixel being of a different colour, then the frequency of each colour is 1/256. The average number of bits needed to encode each colour is 8.

$$\sum_{0}^{255} \frac{1}{256} \left(\log_2 \left(\frac{1}{\frac{1}{256}} \right) \right) = \sum_{0}^{255} \frac{1}{256} (\log_2(256)) = \sum_{0}^{255} \frac{1}{256} (8) = 8$$

Entropy and bit depth

If you now have an image file of 256 pixels and only 8 colours in the image with the following frequencies:

Color	Frequency
black	100
white	100
yellow	20
orange	5
red	5
purple	3
blue	20
green	3

Shannon's equation tells us that the *minimum value for the average number of bits* required to represent each colour in this file is 2.006

$$\frac{100}{256}\log_2\left(\frac{256}{100}\right) + \frac{100}{256}\log_2\left(\frac{256}{100}\right) + \frac{20}{256}\log_2\left(\frac{256}{20}\right) + \frac{5}{256}\log_2\left(\frac{256}{5}\right)$$

$$+ \frac{5}{256}\log_2\left(\frac{256}{5}\right) + \frac{3}{256}\log_2\left(\frac{256}{3}\right) + \frac{20}{256}\log_2\left(\frac{256}{20}\right) + \frac{3}{256}\log_2\left(\frac{256}{3}\right)$$

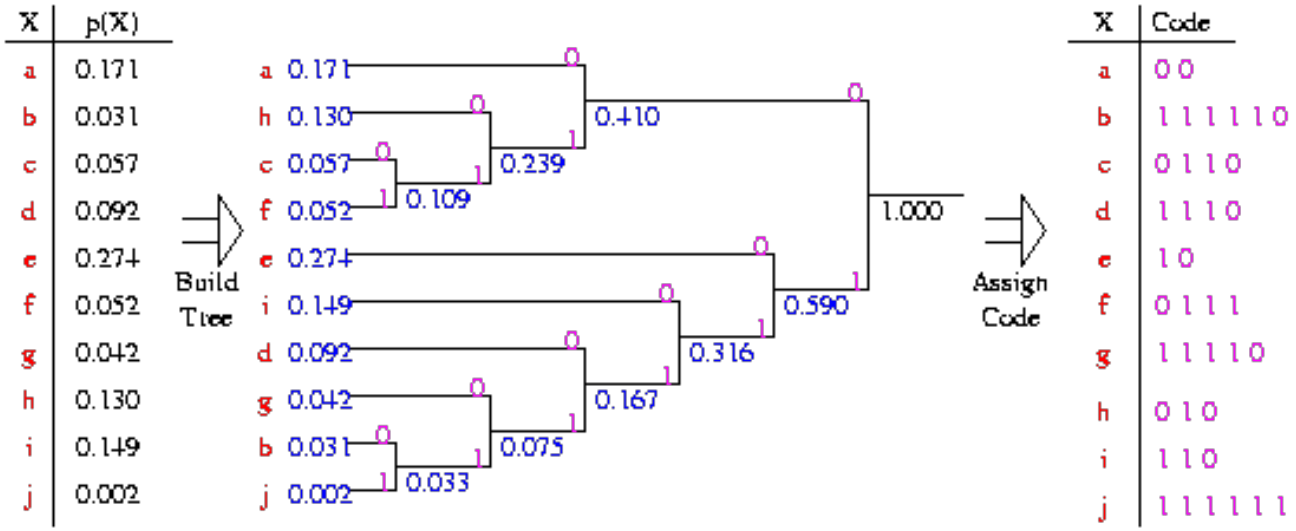
$$\approx 0.530 + 0.530 + 0.287 + 0.111 + 0.111 + 0.075 + 0.287 + 0.075 \approx 2.006$$

Entropy and bit depth

- The implication in Shannon's equation is that we don't necessarily need to use the same number of bits to represent each symbol.
- A better compression ratio is achieved if we use fewer bits to represent symbols that appear more frequently in the file.

Huffman encoding

- The Huffman encoding algorithm requires two passes: (1) determining the codes for the colours and (2) compressing the image file by replacing each colour with its code.
- To determine the codes, a tree data structure is built from the colour frequency table.

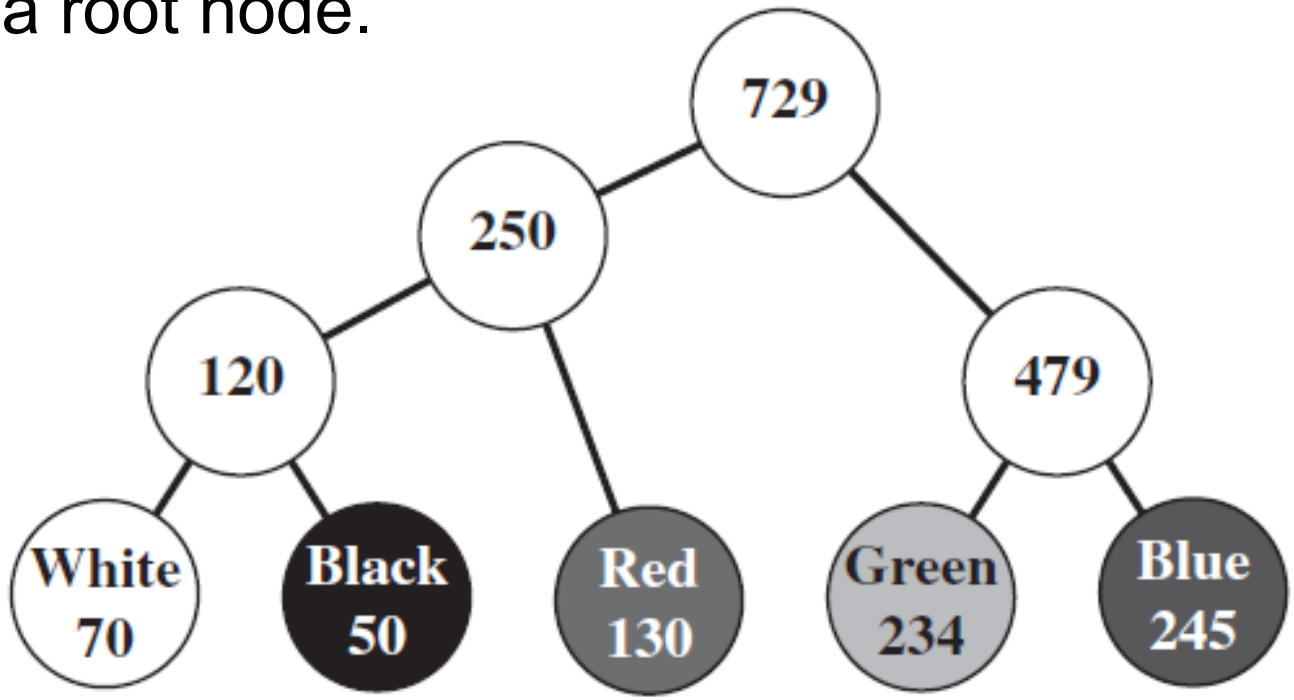


Huffman encoding

A node is created for each of the colours in the image, with the frequency of that colour's appearance stored in the node. These nodes will be the leaves of the code tree.

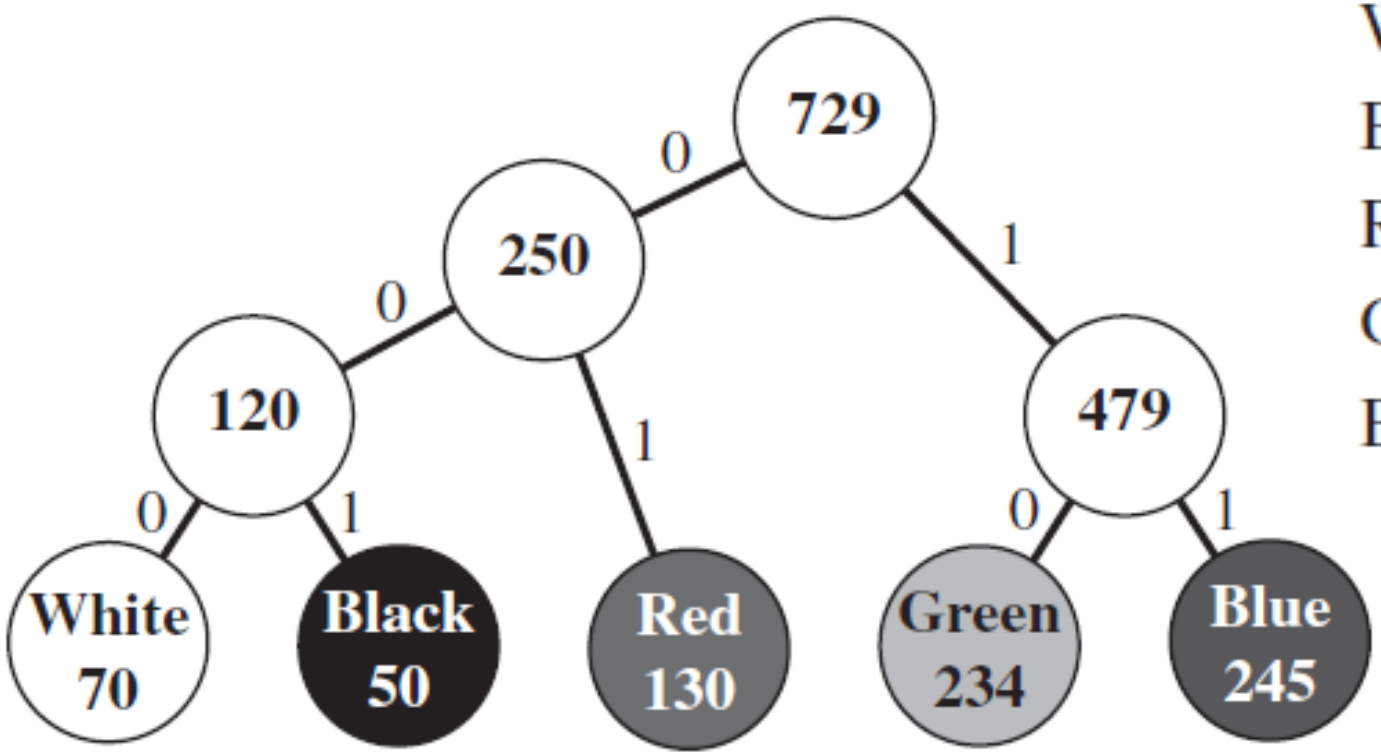
Huffman encoding

The two nodes with the smallest frequency values are joined such that they are the children of a common parent node, and the parent node's frequency value is set to the sum of the frequency values in the children nodes. This node-combining process repeats until you arrive at the creation of a root node.



Huffman encoding

Once the tree has been created, the branches are labeled with 0s on the left and 1s on the right. Then for each leaf node, you traverse the tree from the root to the leaf, gathering the code on the way down the tree.



White	000
Black	001
Red	01
Green	10
Blue	11

Exercise

- Find a Huffman code using the frequency table below (there may be more than one solution).
- Then encode the following sequence:

  ABBAAADCBAA 0 100 100 0 0 0 110 101 100 0 0 21 bits

  3×11 = 33 bits
- Finally, calculate the compression rate.

  33 : 21 = 11 : 7

Symbol/Colour	probability	code
A	0.40	? 0
B	0.20	? 100
C	0.20	? 101
D	0.10	? 110
E	0.10	? 111

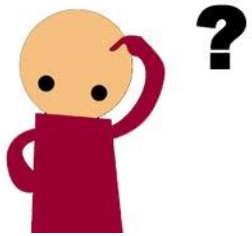
5 4 → 3 bits

Exercise

Consider the following sequence of letters (each letter represents one byte):
AABBBBCDDDAAAAA

- Encode the sequence using RLE (Run Length Encoding) and calculate the compression rate.
- Considering the following Huffman codes: 00, 01, 10, 110; how would you encode the sequence of bytes using Huffman encoding? What is the compression rate you achieve?
- Which of the two techniques works best on this data?

Question



Which of the following statements about lossless image encoding is not true?

- T With lossless encoding, redundant data is not encoded.
- T No information is lost in a compression and decompression cycle.
- F For encoding to remain lossless, an image should be compressed only once.
- T Lossless encoding is possible when some data appear more often than other.

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Multimedia Fundamentals

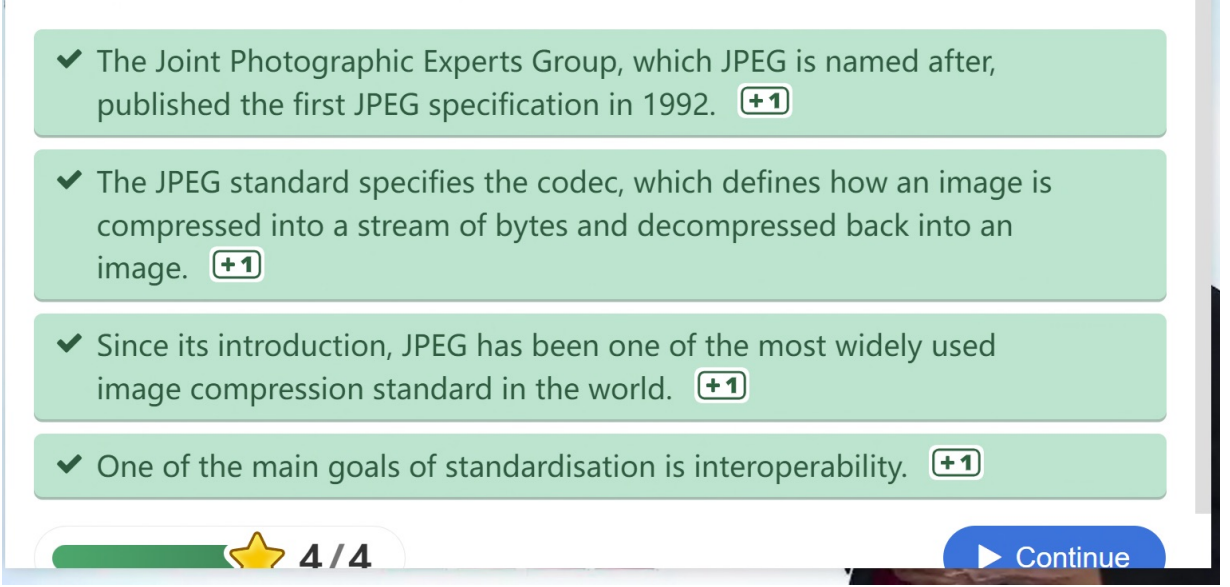
JPEG

Agenda

- JPEG is lossy
- JPEG is an example of transform encoding
- Images are pre-processed: colour conversion, chroma-subsampling, 8x8 blocks
- Pixel data is converted to frequency coefficients
- Compression is done by quantisation of the frequency values
- Lossless encoding is also used to compress further the frequency values

JPEG

- Standard for compressing and decompressing still images
- Based on DCT and applies to colour and grayscale still images
  - Independent of image sizes
  - Applicable to any aspect ratio (W/H)
  - Colour space and number of colours independent
  - Image content of any complexity, with any spatial and statistical properties
  - Run on many standard processors
  - Compression ratio dictated by user or application



Drag the words into the correct boxes

Well-designed lossy compression ✔ technology often reduces file sizes significantly before degradation ✔ is noticed by the end-user.
For example, JPEG can cause an issue called artifacting ✔, in which parts of the image become noticeably blocky ✔. This typically happens when a high ✔ compression setting is used to save the image.
Lossy methods are most often used for compressing sound, images or videos because these types of data are intended for human interpretation ✔ where the mind can easily "fill in the blanks" or see past very minor errors ✔.

Transform Encoding

- The idea is that changing the representation (domain) of data can sometimes make it possible to **extract details** that won't be missed because they are **beyond the acuity of human perception**.
- Two of the most commonly used transforms in digital media are the ***discrete cosine transform*** (***DCT***) and the ***discrete Fourier transform*** (***DFT***).
- High frequency components correspond to quick fluctuations of colour in a short space—changes that aren't easy for the human eye to see.
- Once you have separated out the high frequency components of an image, you can remove them.

To summarise: transform encoding is a type of data compression for data like audio signals or photographic images. The transformation is lossless (perfectly reversible) on its own but is used to enable better (i.e. more targeted) quantisation, which then results in a lower quality copy of the original input (lossy compression).

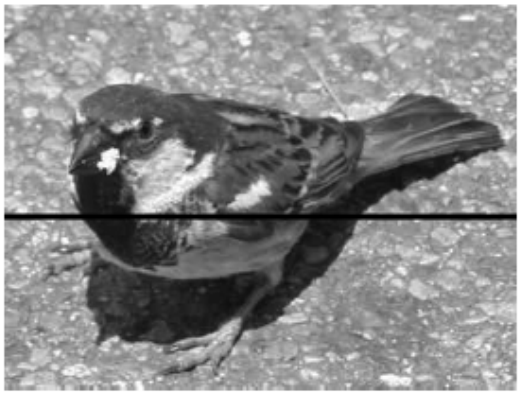
◉ True ✔ ○ False

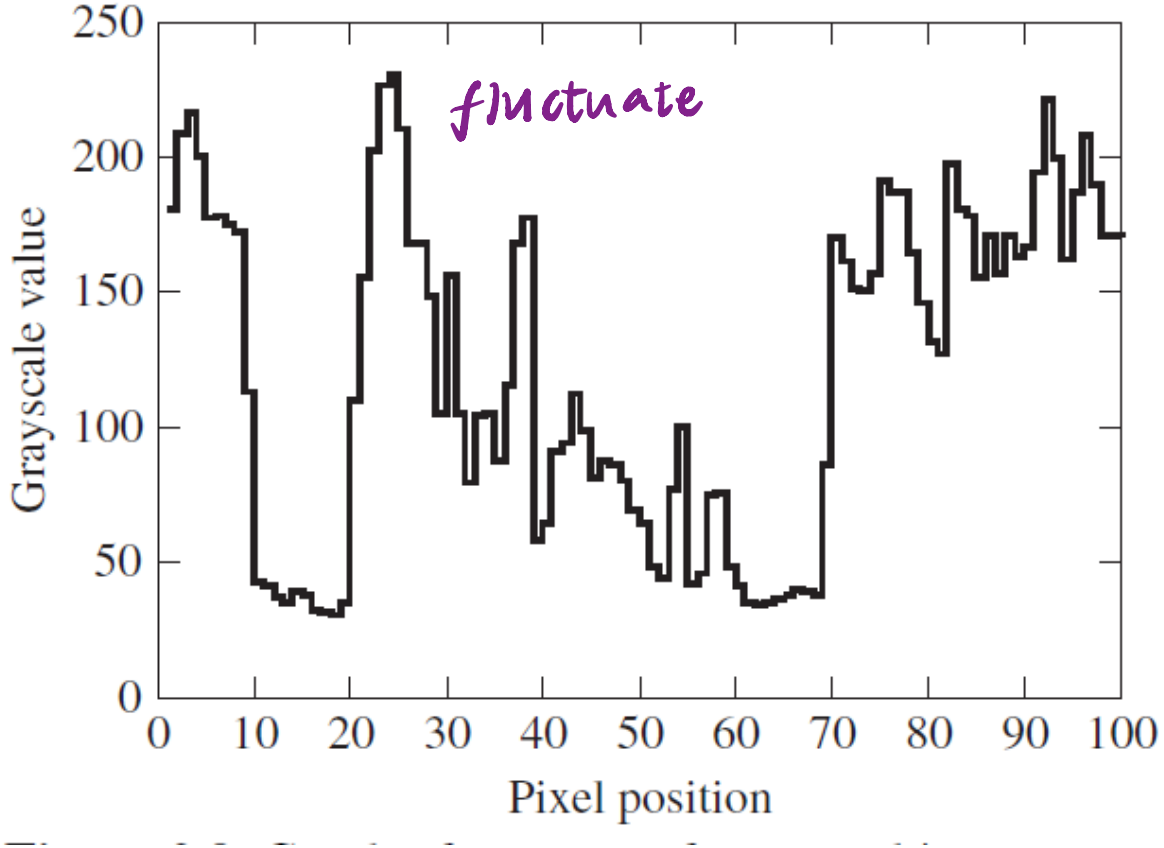
Fill in the missing words

For an image, transform encoding is used to transform the pixel ✔ values to spatial frequency ✔ values. During lossy ✔ compression, the less important high frequencies are removed ✔.

Frequency in Digital Images

- For digital imaging, *frequency refers to the rate at which color values change*.
- All bitmap images can be viewed as waveforms.





The Discrete Cosine Transform

- DCT tells how to transform an image from the spatial domain (i.e. pixel values), to the frequency domain (coefficients by which the frequency components should be multiplied).
- Rather than being applied to a full MxN image, the DCT is generally applied to 8x8 pixel blocks.

DCT transforms the image from the spatial to the frequency domain to obtain 8x8 blocks of DCT coefficients. The coefficients are arranged from lowest frequency (average value for the block) to highest frequency, i.e. fine details.

筛出低频去高频

KEY EQUATION

Let $f(r, s)$ be the pixel value at row r and column s of a bitmap. $F(u, v)$ is the coefficient of the frequency component at (u, v), where $0 \leq r, u \leq M - 1$ and $0 \leq s, v \leq N - 1$.

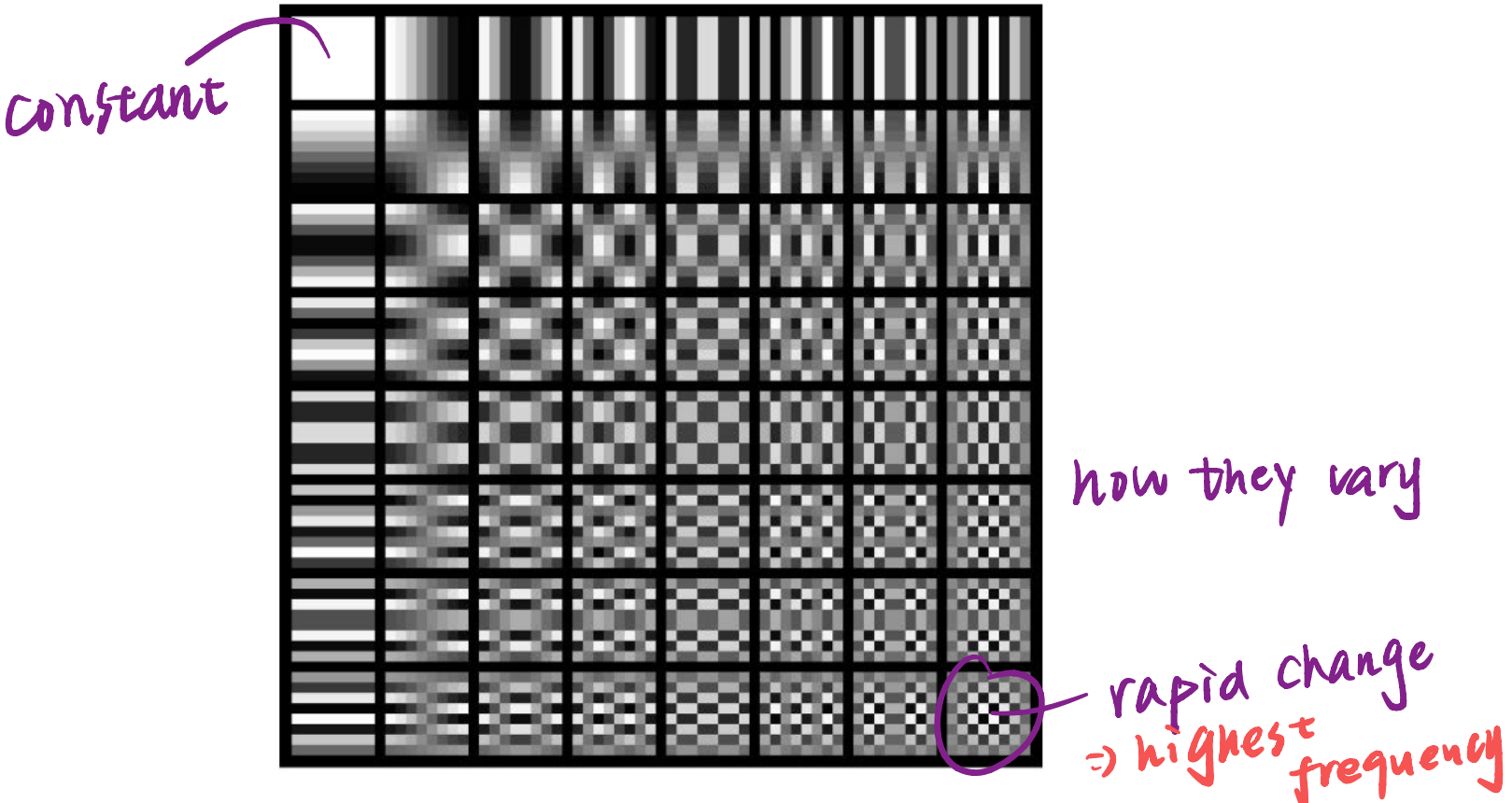
$$F(u, v) = \sum_{r=0}^{M-1} \sum_{s=0}^{N-1} \frac{2C(u)C(v)}{\sqrt{MN}} f(r, s) \cos\left(\frac{(2r + 1)u\pi}{2M}\right) \cos\left(\frac{(2s + 1)v\pi}{2N}\right)$$

$$\text{where } C(\delta) = \frac{\sqrt{2}}{2} \text{ if } \delta = 0 \text{ otherwise } C(\delta) = 1$$

Base Frequencies

Each graph at position (u,v) represents the function:

$$\cos\left(\frac{(2r+1)u\pi}{16}\right)\cos\left(\frac{(2s+1)v\pi}{16}\right)$$



The Discrete Cosine Transform

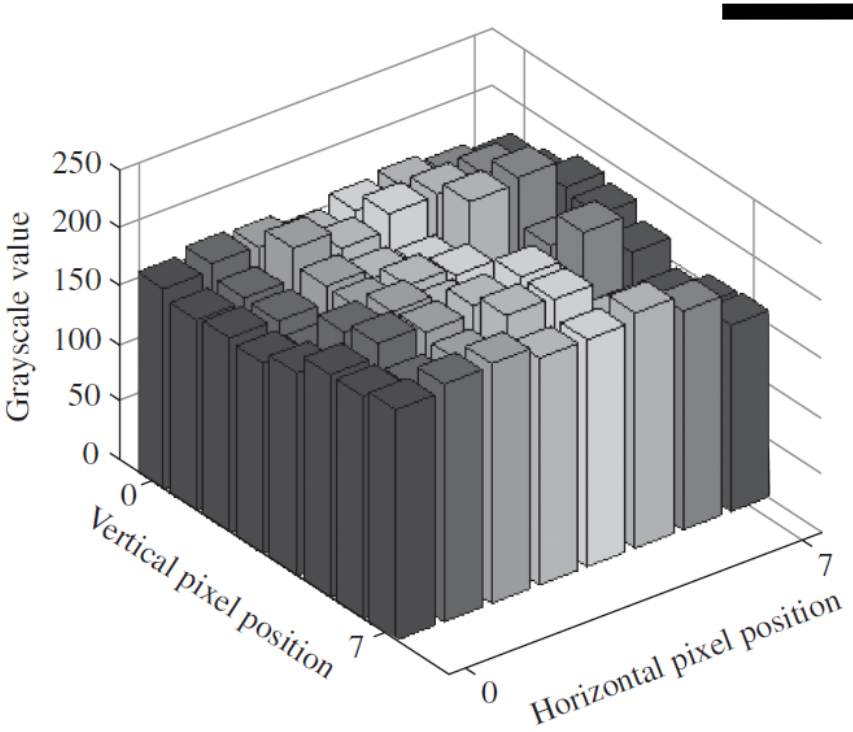
- DCT takes a bitmap image in the form of a matrix of color values – $\boldsymbol{f}(r, s)$ – and returns the frequency components of the bitmap in the form of a matrix of coefficients – $\boldsymbol{F}(u,v)$.
- The coefficients give the amplitudes of the frequency components.
- The first element $\boldsymbol{F}(0,0)$—is called the ***DC component***.
- All the other components are called ***AC components***
- A negative coefficient amounts to adding the inverted waveform.
- The discrete cosine transform is invertible.

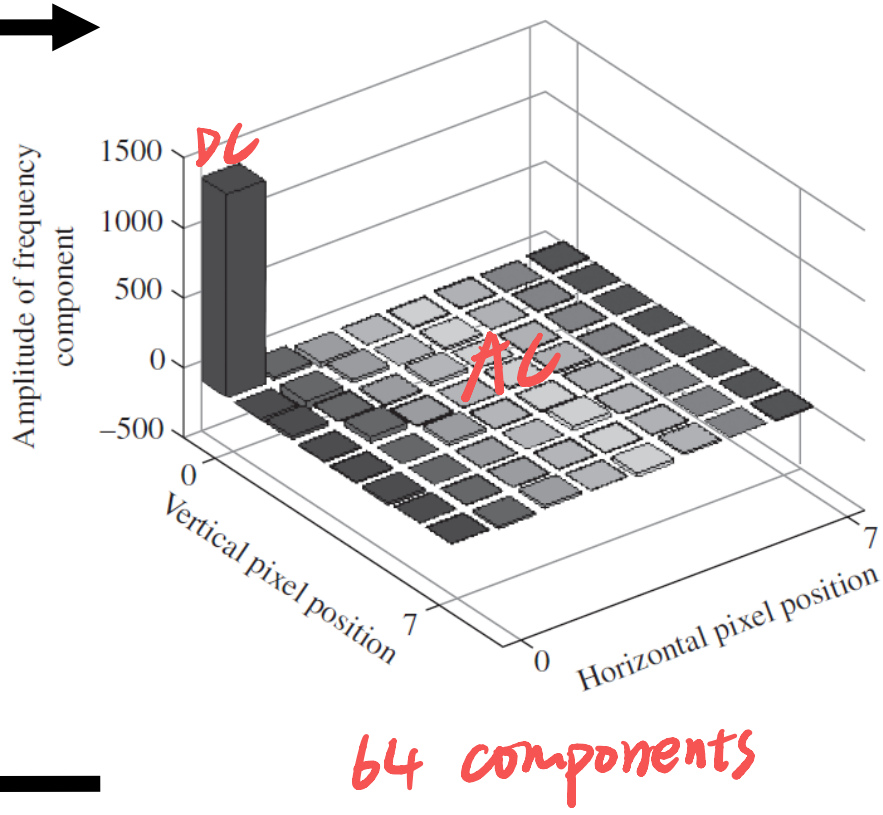
The Discrete Cosine Transform

Spatial Domain

DCT

Frequency Domain





64 components

Inverse DCT

Spatial Domain

255	255	255	255	255	255	159	159
255	0	0	0	255	255	159	159
255	0	0	0	255	255	255	255
255	0	0	0	255	255	255	255
255	255	255	255	255	255	100	255
255	255	255	255	255	255	100	255
255	255	255	255	255	255	100	255
255	255	255	255	255	255	100	255

DCT ↓ ↑ Inverse DCT

1628	−61	39	234	173	−128	171	22
−205	−163	74	222	1	74	−30	111
81	150	−95	−82	−42	−11	−6	−53
188	231	−135	−188	−53	−36	−2	−103
96	71	−42	−78	−32	2	−17	−32
25	−42	25	3	−14	27	−26	19
70	15	−6	−51	−17	7	−16	−13
94	72	−38	−87	−18	−14	−4	−40

Frequency Domain

rapid change

Steps of JPEG compression

algorithm jpeg
/*Input: A bitmap image in RGB mode.
Output: The same image, compressed.*/

{
(Convert image to Y'CbCr and do chroma sub-sampling)
Divide image into 8 x 8 pixel blocks

Use Discrete Cosine Transform (DCT) to transform the pixel data from the spatial domain to the frequency domain

Quantise frequency values

DPCM and zigzag order

Do run-length encoding

Do entropy encoding (e.g., Huffman)
}

Colour Conversion: RGB to Y'CbCr

$$
\begin{aligned}
Y' &= 0 + (0.299 \cdot R'_D) + (0.587 \cdot G'_D) + (0.114 \cdot B'_D) \\
C_B &= 128 - (0.168736 \cdot R'_D) - (0.331264 \cdot G'_D) + (0.5 \cdot B'_D) \\
C_R &= 128 + (0.5 \cdot R'_D) - (0.418688 \cdot G'_D) - (0.081312 \cdot B'_D)
\end{aligned}
$$

And back:

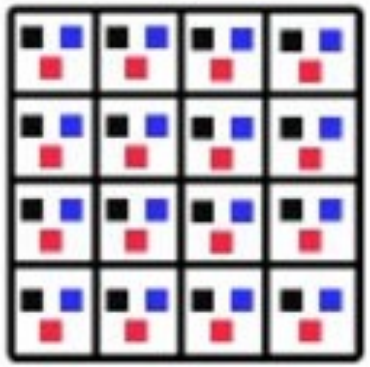
$$
\begin{aligned}
R'_D &= Y' + 1.402 \cdot (C_R - 128) \\
G'_D &= Y' - 0.344136 \cdot (C_B - 128) - 0.714136 \cdot (C_R - 128) \\
B'_D &= Y' + 1.772 \cdot (C_B - 128)
\end{aligned}
$$

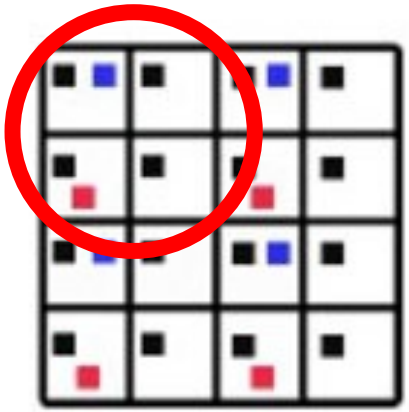
The reason to change RGB into YCbCr:
During the process we can separate luminance from chrominance because then we are going to be able to apply different processes to luminance and chrominance.
Compared to chrominance, luminance is going to be preserved completely and chrominance will be subsampled.
So as to realize compression

+ 亮度比颜色敏感

Chroma Sub-Sampling

- The human eye is more sensitive to changes in light (*i.e.*, luminance) than in colour (*i.e.*, chrominance).
- ***Chroma sub-sampling*** (also called ***chrominance downsampling***) is a process of throwing away some of the bits used to represent colour information.
- For example, we might choose to save only one Cb value and one Cr value but four Y values for every four pixel values.

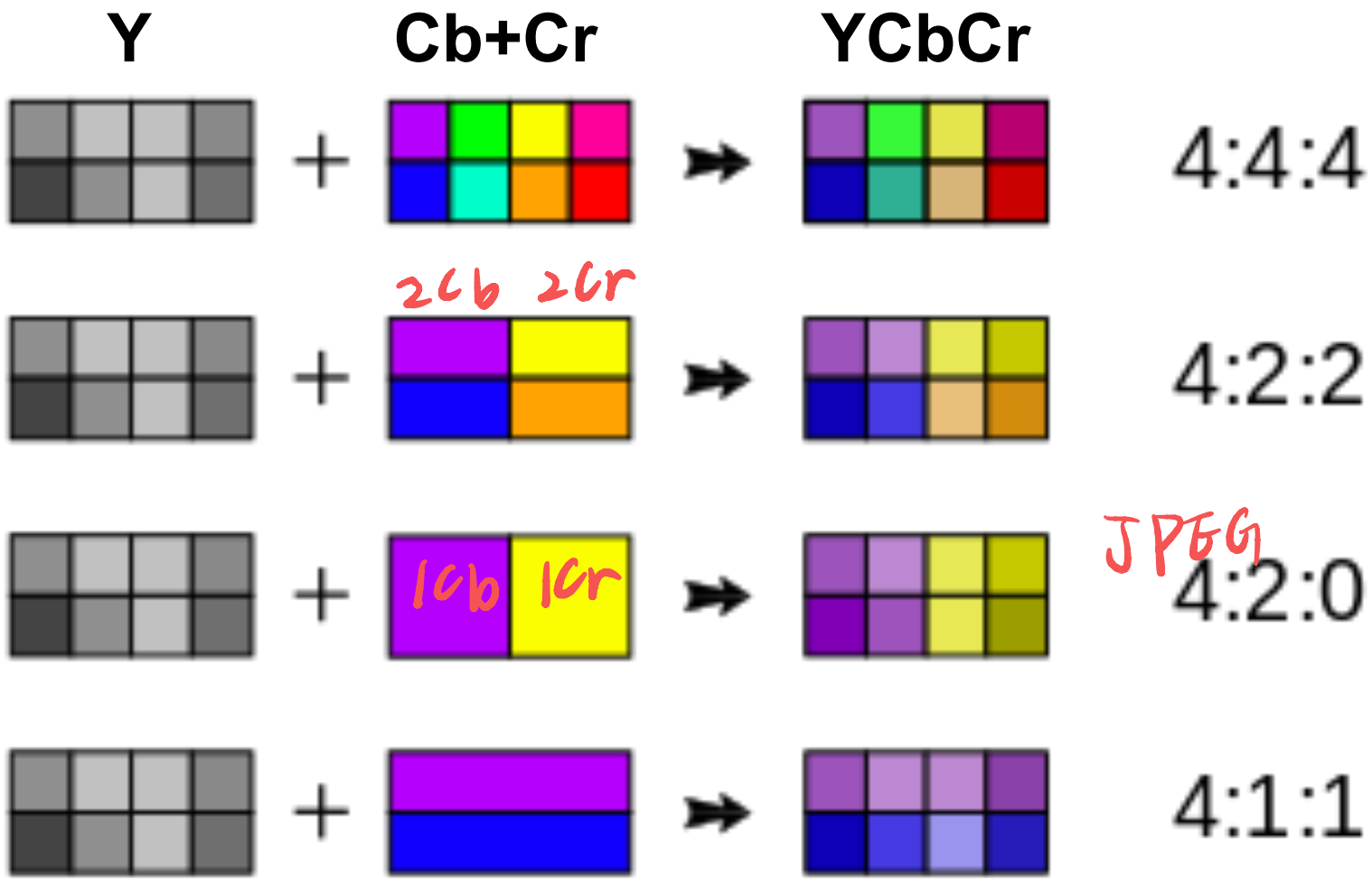




pixel數不變

don't remove pixel

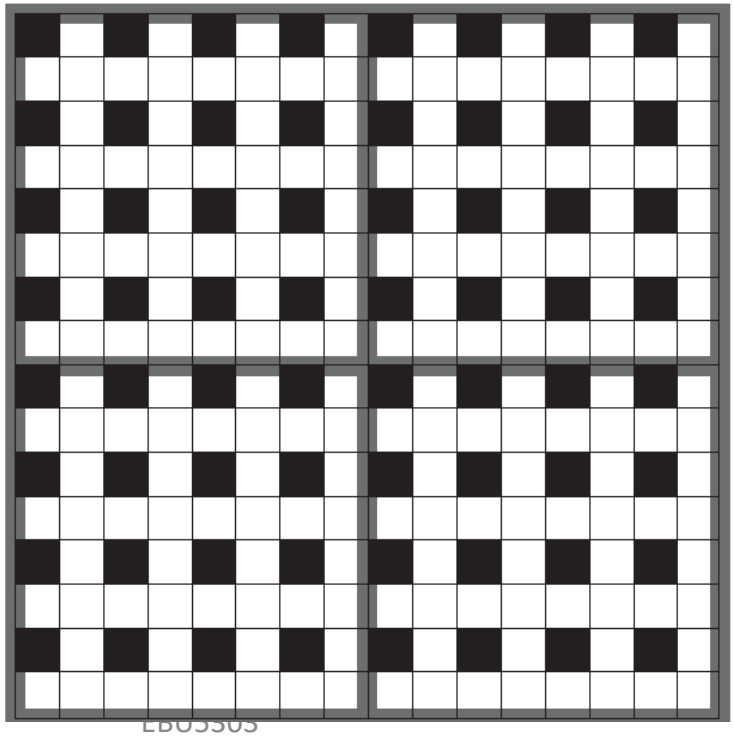
Chroma Sub-Sampling



Blocks

- With YCbCr colour mode, we begin by dividing the image into 16x16 pixel macroblocks.
- Then, with 4 : 2 : 0 chroma downsampling, we get four blocks of 8x8 Y data for every one block of 8x8 Cb and one block of 8x8 Cr data.

Black cell: YCbCr
White cell: Y only



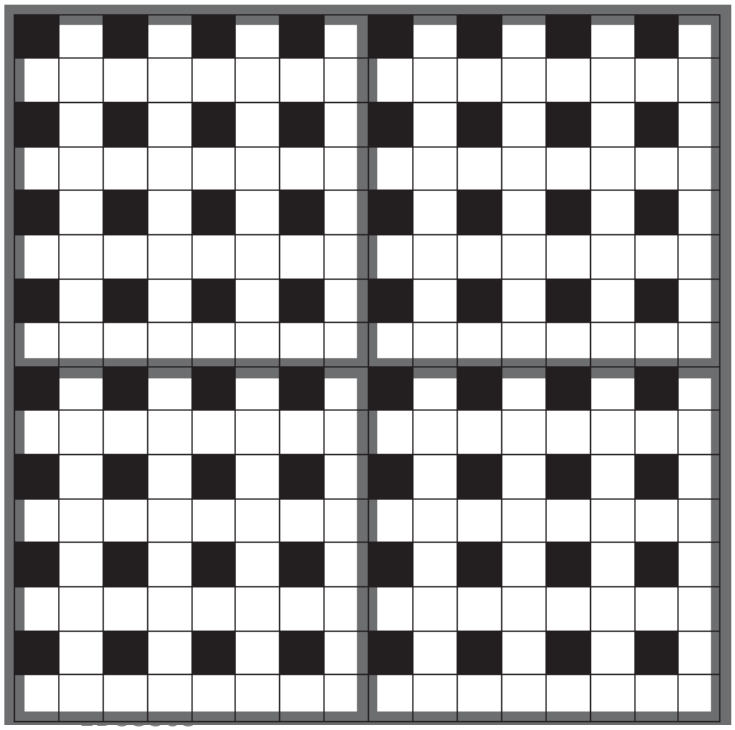
Exercise

With 4:2:0 YCbCr chrominance subsampling, a 16 x 16 macroblock yields: four 8 x 8 blocks of Y values; one 8 x 8 block of Cb values; and one 8 x 8 block of Cr values.

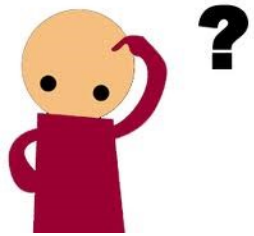
What is the compression rate achieved?

2:1

16x16x3 : 8x8x6



Question



Which of the following statements describes best what is chroma sub-sampling?

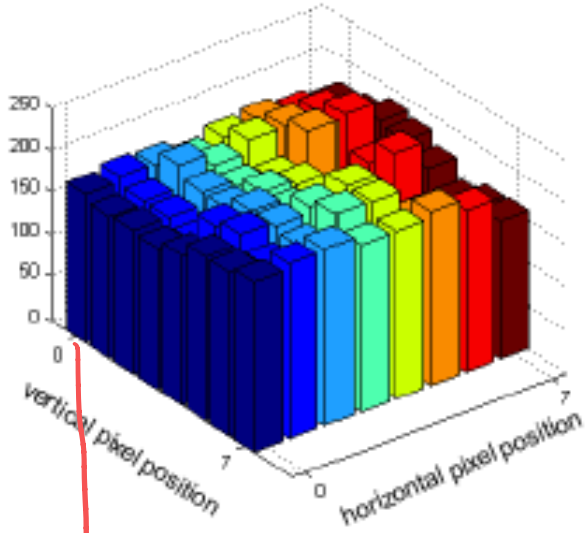
- It is a process where colours are converted to a different model.
- It is a process where some colour information is thrown away.
- It is a process to separate colour from brightness information.
- It is a process to eliminate some of the pixels.

Shifting Pixel Values

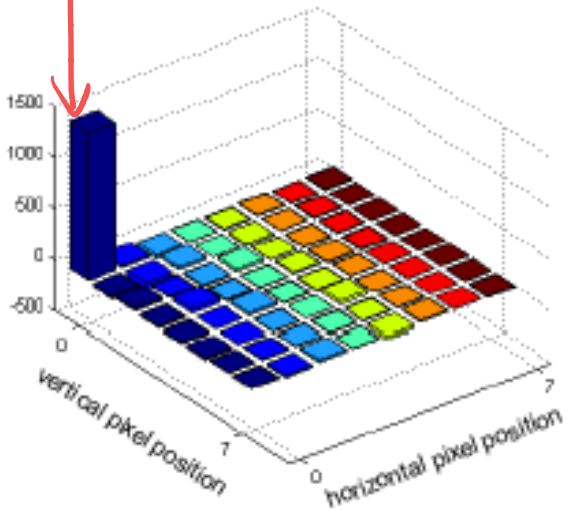
- On an intuitive level, shifting the values by –128 is like looking at the image function as a waveform that cycles through positive and negative values.
- This step is a preparation for representing the function in terms of its frequency components.

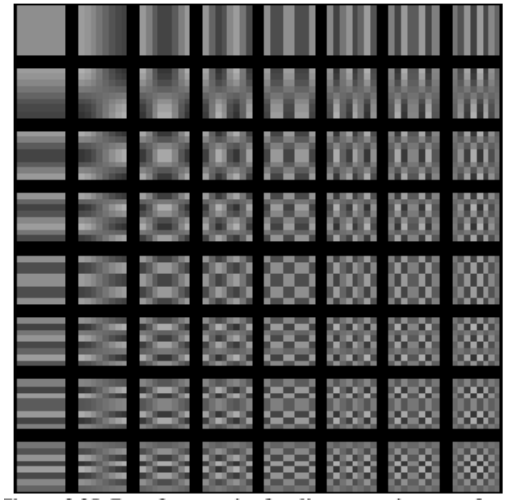
Discrete Cosine Transform (DCT)

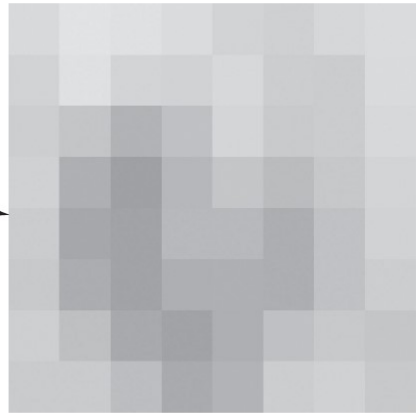
Pixel matrix (Y, Cb or Cr)



DCT coefficient matrix







Pixel values shifted by -128

94	103	101	96	88	85	92	96
88	101	89	87	93	82	81	95
83	74	55	70	90	79	81	93
86	52	36	60	75	65	77	89
81	43	38	62	62	50	71	87
78	49	38	51	52	50	71	82
84	69	45	38	51	70	78	75
80	80	67	46	56	82	86	78

DCT

585.7500	−24.5397	59.5959	21.0853	25.7500	−2.2393	−8.9907	1.8239
78.1982	12.4534	−32.6034	−19.4953	10.7193	−10.5910	−5.1086	−0.5523
57.1373	24.829	−7.5355	−13.3367	−45.0612	−10.0027	4.9142	−2.4993
−11.8655	6.9798	3.8993	−14.4061	8.5967	12.9151	−0.3122	−0.1844
5.2500	−1.7212	−1.0824	−3.2106	1.2500	9.3595	2.6131	1.1199
−5.9658	−4.0865	7.6451	13.0616	−1.1927	1.1782	−1.0733	−0.5631
−1.2074	−5.7729	−2.0858	−1.9347	1.6173	2.6671	−0.4645	0.6144
0.6362	−1.4059	−0.7191	1.6339	−0.1438	0.2755	−0.0268	−0.2255

DCT coefficients matrix

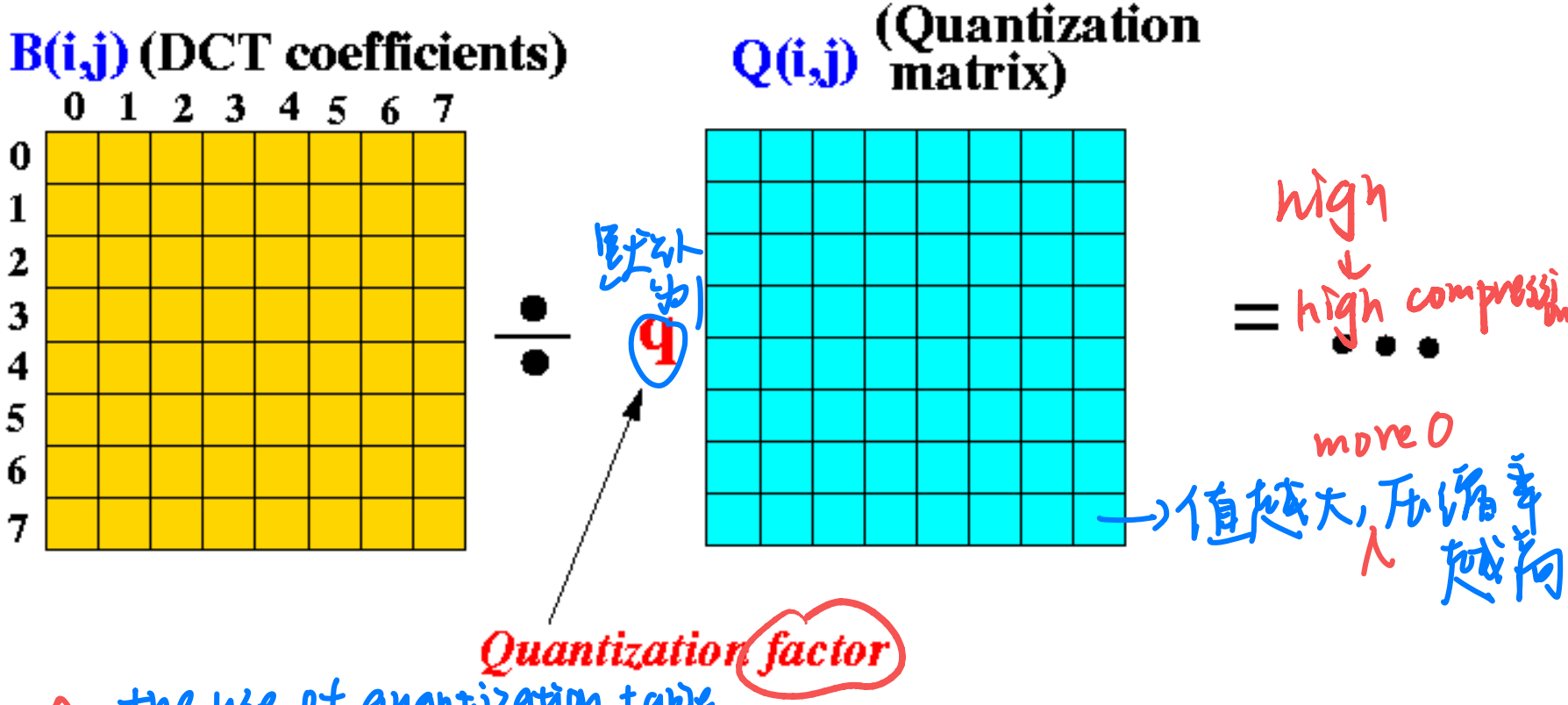
- Describes each 8x8 block in terms of how much the colour detail changes.
- Values are arranged from lowest frequencies to highest frequencies
  - Lowest frequencies represent average value for the block
  - Highest frequency represent fine detail (they can be dropped)
    - Eye unable to perceive brightness levels above or below thresholds
    - Gentle gradation of brightness of colour are more important to the eye than abrupt changes

Quantisation = Division

- Quantisation involves dividing each frequency coefficient by an integer and rounding off.
- Rounding during quantisation makes JPEG lossy.
- The coefficients for high-frequency components are typically small, so they often round down to 0—which means, in effect, that they are thrown away.
- Not every value in the matrix needs to be divided by the same integer.
- A quantisation table is be stored with the compressed image

Quantisation = Division

向下取整



the use of quantization table

The 8x8 blocks of DCT coefficients are divided by these quantisation tables. Different quantization tables are used according to the amount of compression that is needed. High quantisation values achieve greater compression but may incur a loss of quality.

the table store quantisation values. we assign different values depending on which coefficient we want to remove or preserve.

Quantisation table examples

a. Low compression

1	1	1	1	1	2	2	4
1	1	1	1	1	2	2	4
1	1	1	1	2	2	2	4
1	1	1	1	2	2	4	8
1	1	2	2	2	2	4	8
2	2	2	2	2	4	8	8
2	2	2	4	4	8	8	16
4	4	4	4	8	8	16	16

low q

b. High compression

1	2	4	8	16	32	64	128
2	4	4	8	16	32	64	128
4	4	8	16	32	64	128	128
8	8	16	32	64	128	128	256
16	16	32	64	128	128	256	256
32	32	64	128	128	256	256	256
64	64	128	128	256	256	256	256
128	128	128	256	256	256	256	256

high q

585.7500	−24.5397	59.5959	21.0853	25.7500	93	−8.9907	1.8239
78.1982	12.4534	−32.6034	−19.4953	10.	0	−5.1086	−0.5523
57.1373	24.829	−7.5355	−13.3367		10.0027	4.9142	−2.4993
−11.8655	6.9798	3.8993			12.9151	−0.3122	−0.1844
5.2500	−1.7212	−1.08		1.2500	9.3595	2.6131	1.1199
−5.9658	−4.0865		3016	−1.1927	1.1782	−1.0733	−0.5631
−1.2074	−5.7729		−1.9347	1.6173	2.6671	−0.4645	0.6144
0.6362	−1.4059	0.7191	1.6339	−0.1438	0.2755	−0.0268	−0.2255

Matrix of coefficients

Divided by Quantisation Table

8	6	6	7	6	5	8	7
7	7	9	9	8	10	12	20
13	12	11	11	12	25	18	19
15	20	29	26	31	30	29	26
28	28	32	36	46	39	32	34
44	35	28	28	40	55	41	44
48	49	52	52	52	31	39	57
61	56	50	60	46	51	52	50

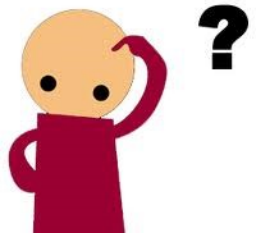
Equal

Compressed Matrix of Integer Coefficients

73	−4	10	3	4	0	−1	0
11	2	−4	−2	1	−1	0	0
4	2	−1	−1	−4	0	0	0
−1	0	0	−1	0	0	0	0
0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0

More 0 in the lower right area of the matrix. because the quantisation table contains big values in the same area in order to eliminate the coefficients that correspond to high frequencies.

Questions



- What known limitations of the human visual system are used in JPEG to achieve compression?

Fine color details are not perceived / visible

come from high f components

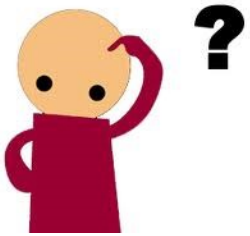
- Is JPEG always lossy?

Yes

- In general, does JPEG achieve greater compression rates on colour images or on grayscale images?

Because chroma subsampling, which only applies to color images, can achieve extra compression. It works by removing some of the color info in the image data while preserving all the brightness information. It does this after converting the image from RGB to YUV.

Question



In JPEG, what is the effect of using quantisation tables that contain small values? Choose one answer.

B

- It achieves great compression and loses much quality.
- ✓ It achieves little compression and preserves much quality.
- It achieves great compression and preserves much quality.
- It achieves little compression and loses much quality.

better quality: use smaller values to obtain less zeros during the quantisation

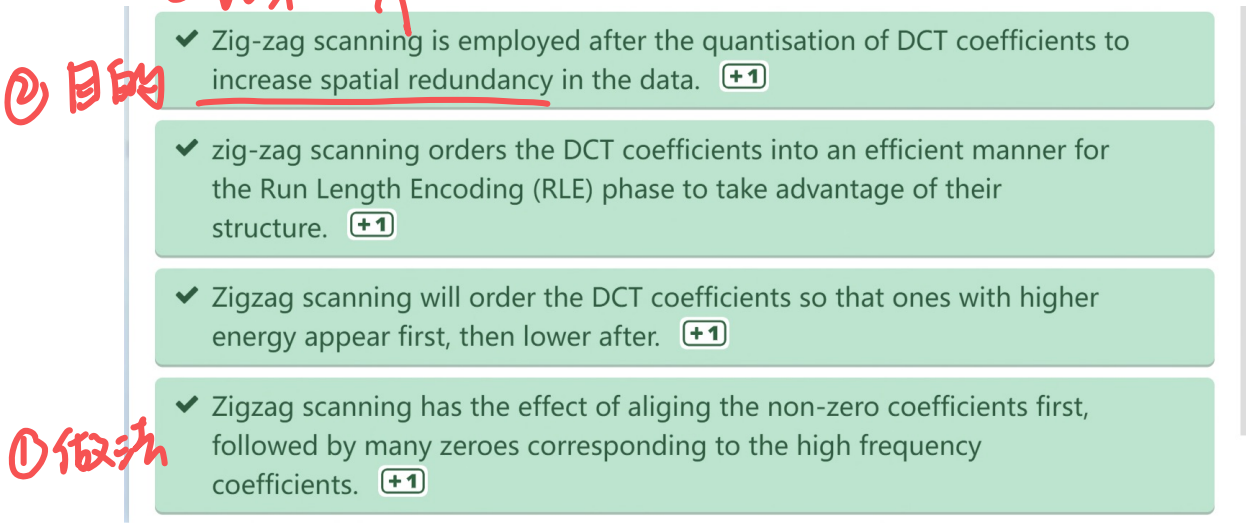
DPCM

- ***DPCM = differential pulse code modulation***
- DPCM is a compression technique that works by recording the difference between consecutive values rather than the actual values.
- It is effective in cases where consecutive values don't change very much because fewer bits are needed to record the change than the value itself.
- In JPEG, the upper leftmost value in an block (the DC component) is stored as the difference from the DC component in the previous block.

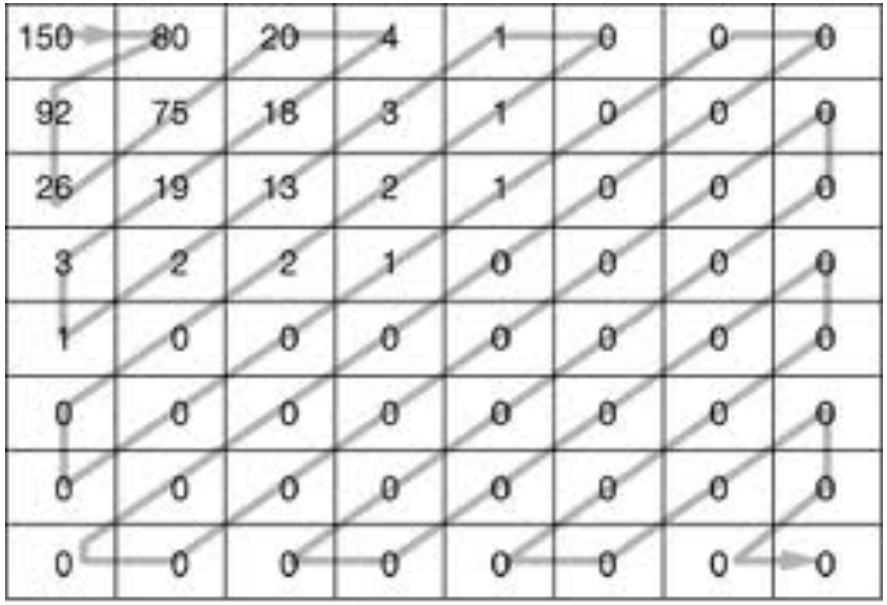
employs a simple predictive coding model, where the value of a DC coefficients is estimated from the previous one.

Zigzag Ordering

- The zigzag reordering sorts the values from low-frequency to high frequency components.
- The high-frequency coefficients are grouped together at the end.
- If many of them round to zero after quantisation, run-length encoding is even more effective.



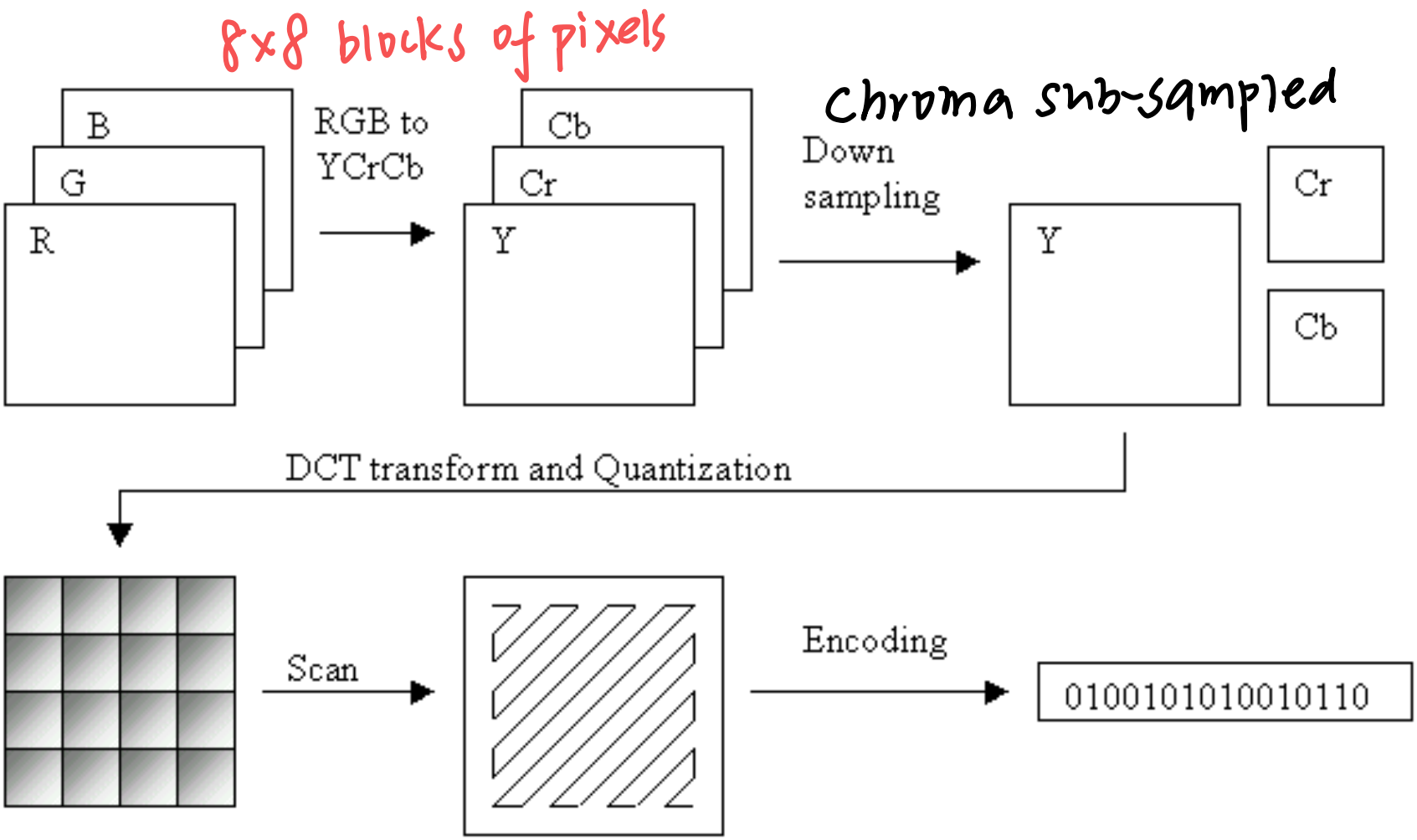
Zigzag ordering



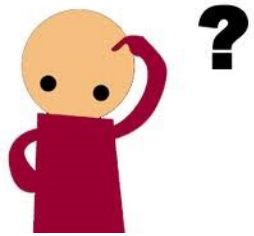
150	80	92	26	75	20	4	18	19	3	1	2	13	3	1	0	1	2
2	0	0	0	0	0	1	1	0	0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0	0								

0太多则直接用PLC

JPEG overview



Question



In JPEG, which of the following steps contribute the most to compression? Choose one answer.

- Dividing image into blocks.
- Colour conversion.
- Discrete Cosine Transform (DCT).
- DCT coefficients quantisation.

6 How is the quantisation factor used to influence the compression rate?

Answer

1.The quantization factor is used to multiply on the quantization table, that is, to scale your quantization values simultaneously.

2.If you want to increase the compression rate, you should increase your quantization factor.

3.If you want to decrease the compression rate, you should decrease your quantization factor.

2 To help image decompression, what information you think should be stored in the JPEG image file header?

Answer

1.We need to store the quatisation tables

2.We also need to store the quantisation factor.

3.We need to store the chroma-sub-sampling scheme.

4.And if we use the Huffman, we still need to store the Huffman table.

3 How does DPCM work and why is it applied to the DC coefficients?

Answer

1.DPCM will encode the differences between two values rather than each values individually.

2.DC coefficients are quite similar from one block to the next, which means the difference will be very small.

3.It will be quite economic to encode like that.

4 During the JPEG pre-processing step, images are split into small 8x8 blocks.Why?

Answer

This is the preparation process for DCT.

To make DCT more manageable in terms of the computational cost.

JPEG encoding

- After these steps have been performed, the compressed file is put into a standardised format that can be recognised by the decompressor.
- A header contains global information such as: the type of file, the width and height, one or more quantizsation tables, Huffman code tables, and an indication of any pixel-padding.
- An alternative JPEG compression method is called ***JPEG2000***, noted for its high compression rate and good quality, without some of the blocky artifacts of standard JPEG. This method represents digital image data as wavelets as an alternative to the DCT.

Summary

- JPEG is lossy
- JPEG is an example of transform encoding
- Images are pre-processed: colour conversion, chroma-subsampling, 8x8 blocks
- Pixel data is converted to frequency coefficients
- Compression is done by quantisation of the frequency values
- Lossless encoding is also used to compress further the frequency values

EBU5303

Multimedia Fundamentals

MPEG

The Moving Picture Experts Group (MPEG), a working group of ISO/IEC (officially ISO/IEC JTC 1/SC 29/WG 11) has the mission to develop standards for coded representation of digital audio, video, 3D Graphics and other data. Since its establishment in 1988, the group has developed an impressive portfolio of standards and technologies that help industry offer end users a more enjoyable digital media experience.

Agenda

- MPEG: Standardised set of algorithms
- Video Compression exploits temporal redundancy
- Two main types of compression: spatial and temporal
- Three types of frame:
  - Intra Coded Frames (I)
  - Predictive Coded Frames (P)
  - Bi-directional Predictive Coded Frames (B)
- MPEG 4 encodes media objects
- Other Codecs ..

MPEG-1

- The first MPEG compression standard for audio and video.
- Designed to allow moving pictures and sound to be encoded into the bitrate of a Compact Disc (CD).
- Commonly limited to about 1.5 Mbps (Megabits per second).
- To meet the low bit requirement, MPEG-1 downsamples the images, and uses frame rates of only 24–30 Hz.
- Only supports progressive pictures
- Includes the popular MPEG-1 Audio Layer III (MP3) audio compression format.

MPEG-2

- Transport, video and audio standards for broadcast-quality television.
- Supports interlacing and high definition.
- Chosen as the compression scheme for over-the-air digital television (e.g. DVB), digital satellite TV services (e.g. Dish Network), digital cable television signals, SVCD and DVD Video.

Video compression

- Unlike image encoding, video encoding is rarely done in lossless form.
- No storage medium has enough capacity to store a practical sized lossless video file
  - Lossless DVD video - 221 Mbps
  - Compressed DVD video - 4 Mbps
  - 50:1 compression ratio!

MPEG exploit satial, perceptual and temporal redundancy.

JPEG

P, B frames. when looking for similarities between two frames when performing the motion estimation

Video compression

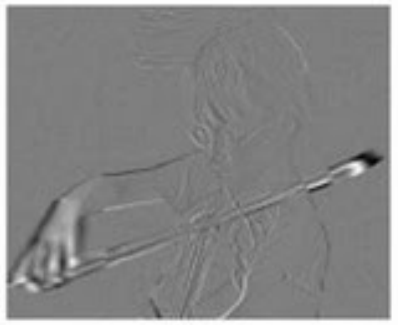
- Video: a sequence of pictures (or frames)
- JPEG algorithm used for intra-frame coding
  - e.g. moving JPEG (MJPEG) exploits only intra-frame coding (*spatial redundancy*)
- High correlation between successive frames (temporal redundancy)
  - motion estimation and motion compensation: inter-frame coding
  - use a combination of actual frame contents and predicted frame contents

Inter-frame and intra-frame coding

- **Inter-frame** compression uses one or more earlier or later frames in a sequence to compress the current frame.
- **Intra-frame** compression uses only the current frame, which is effectively image compression.
- MPEG 1 & 2 use both inter-frame and intra-frame coding

Temporal redundancy

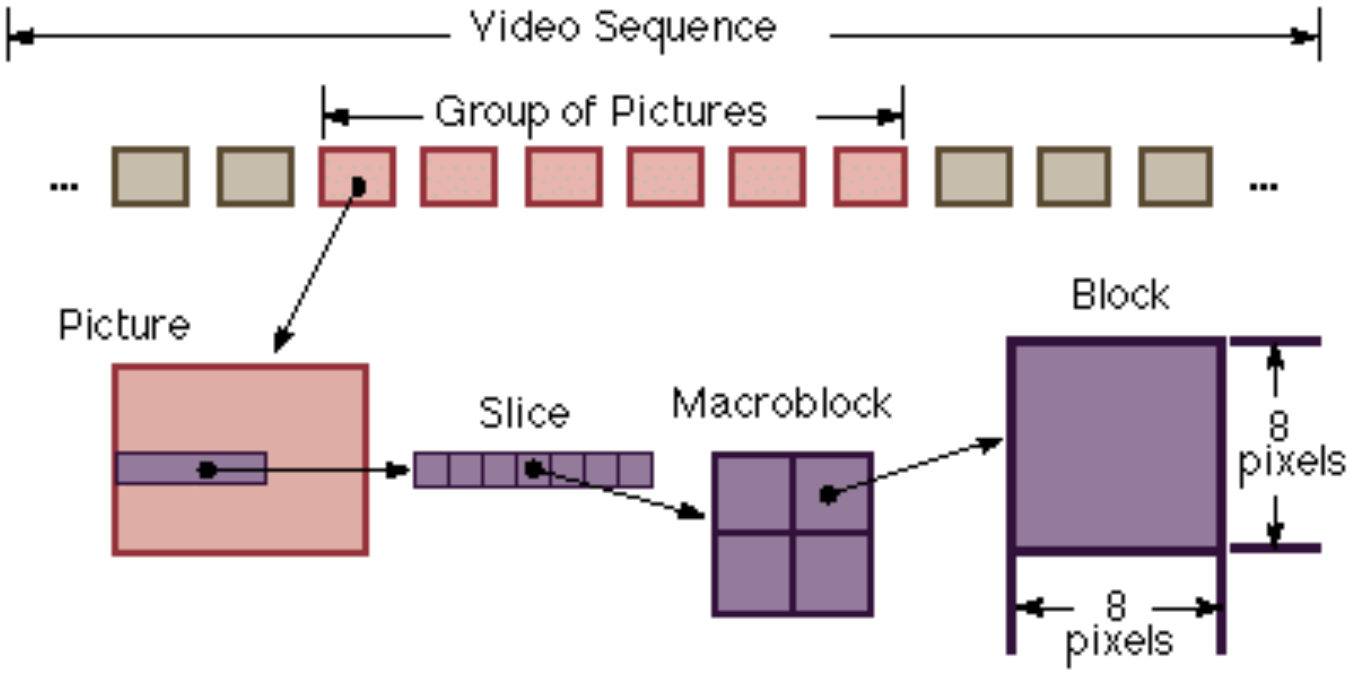
- Temporal redundancy occurs when, in a sequence of images, consecutive images contain similarities.
  - For example, in a sequence of a tennis match, the court does not change; only the players and the ball make significant movements.
- With temporal compression, only the changes from one frame to the next are encoded.



Two main types of compression

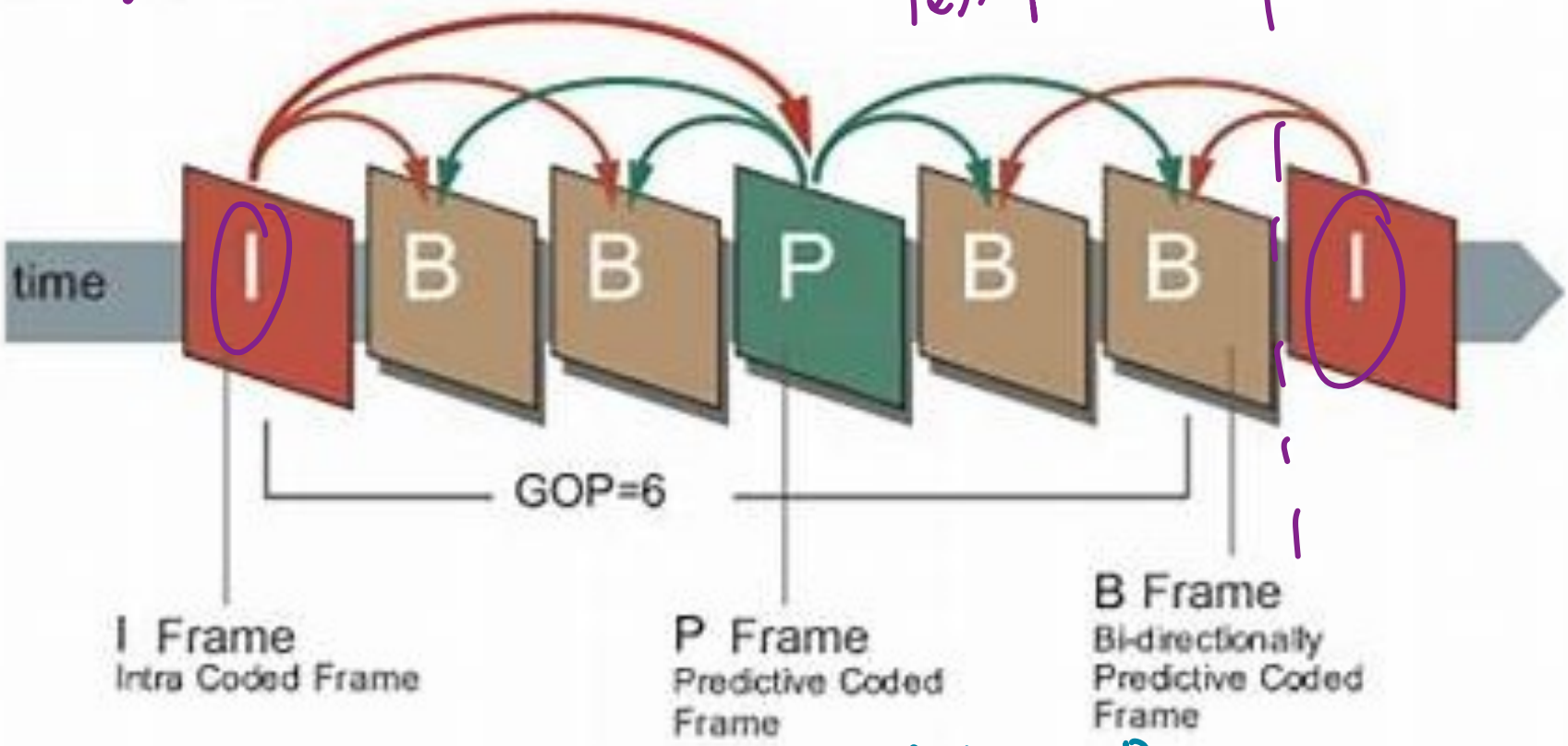
- Spatial compression (intra-frame coding)
  - Allow fast random access
  - Applied to key frames (e.g. every 12th frame, start of new scene, etc)
    - **I-frames** Make no prediction
- Temporal compression (inter-frame coding)
  - Allow high compression
  - Exploit temporal redundancy between frames
  - Applied to frames occurring between key frames
  - Based on motion detection
    - **P- and B- frames**

Structure of Digital Videos



Group of Pictures (GOP)

Shorter GOP. more I frames. lots of rapid change. less predictions. large video files



only 1 frame. but a number of P and B.

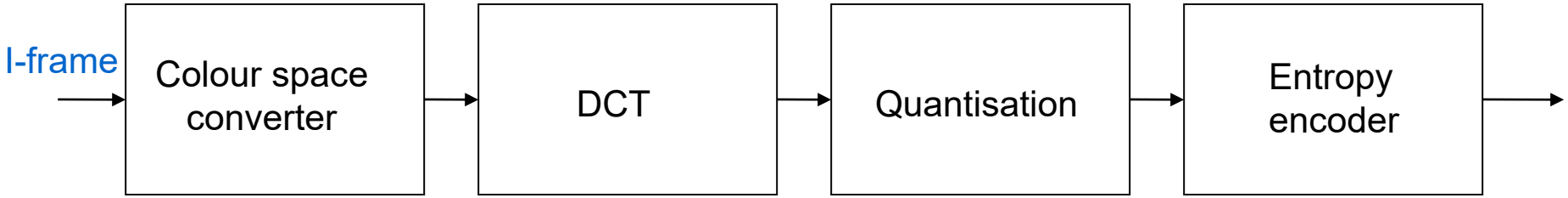
typical one: IPBBPBBPBBP

MPEG video encoding

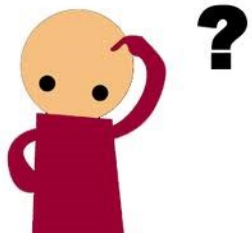
- input frames are preprocessed
  - color space conversion
  - spatial resolution adjustment (chroma sub-sampling)
- frame types (Intra or Inter) are decided for each frame/picture
- each picture is divided into macroblocks of 16 X 16 pixels
- macroblocks
  - are intracoded for I frames
  - are predictive coded or intracoded for P and B frames
  - are divided into six blocks of 8 X 8 pixels
    - 4 luminance and 2 chrominance
    - DCT is applied to each block → transform coefficients
      - » quantized
      - » zig-zag scanned
      - » variable-length coded

I-frames (Intra-coded frames)

- Self-contained (without reference to other frame)
- Treated as a still image and make use of JPEG
- Compression rate is the lowest
- Points for random access in video stream



Question

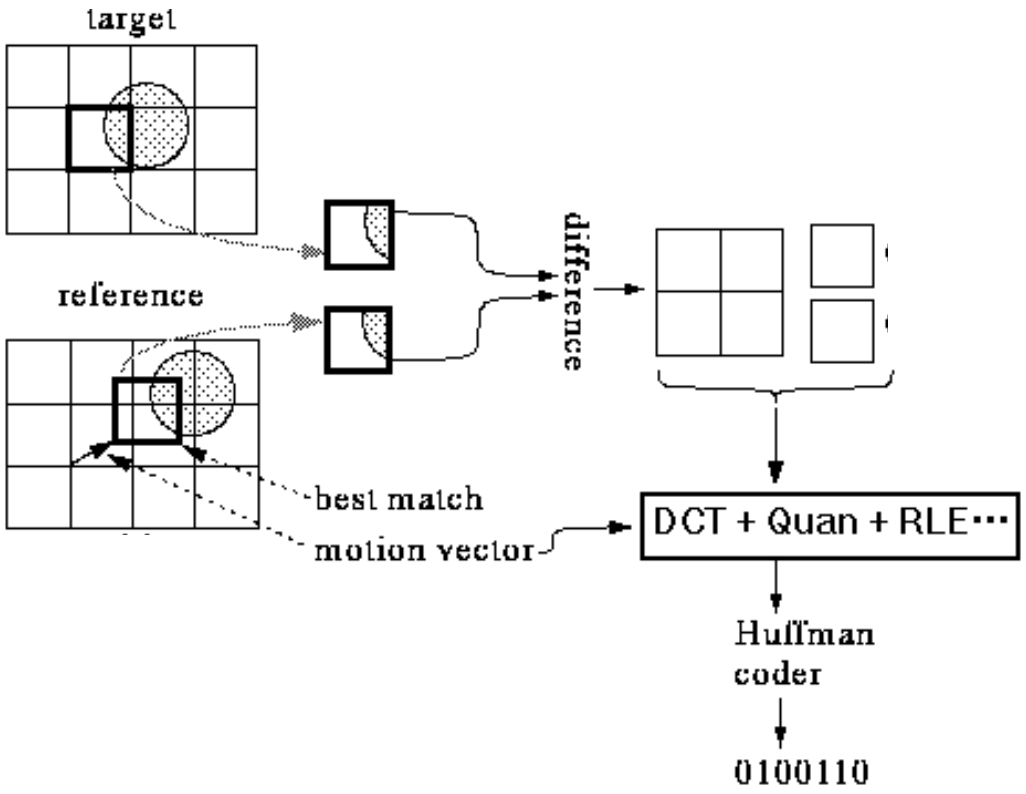


Which of the following statements describes best intra-frame coding? Choose one answer.

- It uses one earlier frame in a sequence to compress the current frame.
- It uses only the current frame.
- It uses both earlier and later frames in a sequence to compress the current frame.
- It uses one later frame in a sequence to compress the current frame.

P-frames (Predictive-coded frames)

- Use a reference frame for motion estimation
- Hold only the changes in the image from a previous frame
- Compression rate is higher



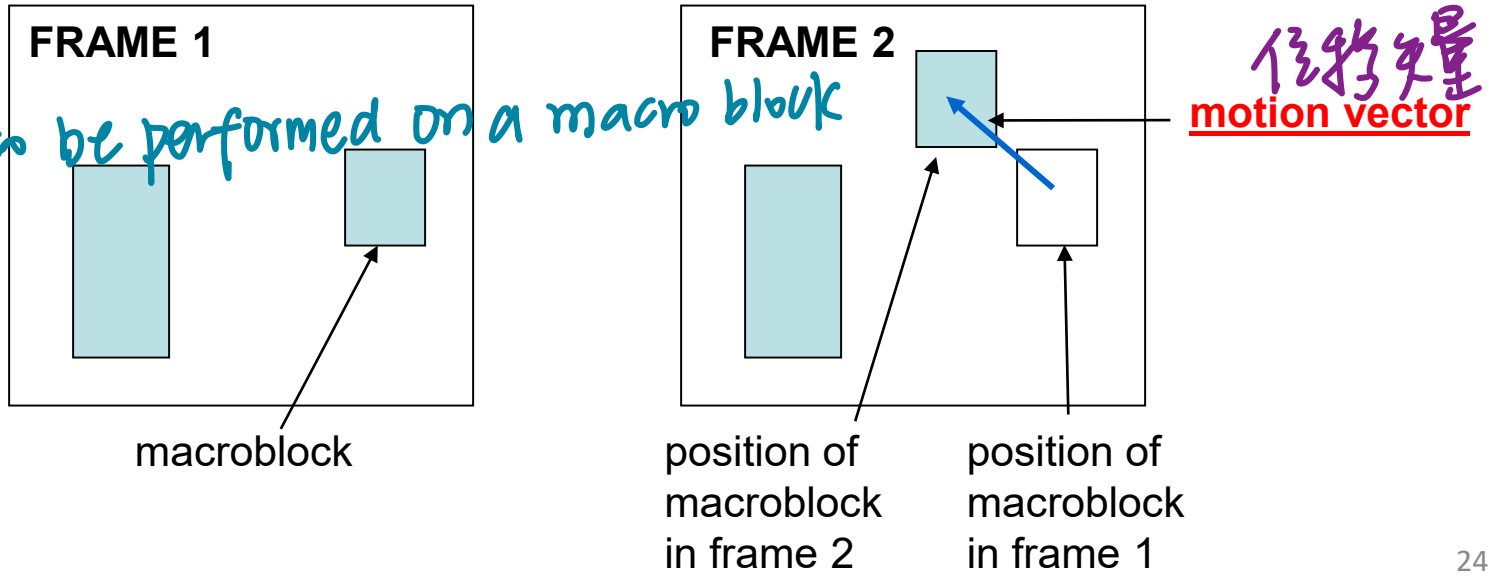
It examines the movement of objects in an image sequence to try to obtain vectors representing the estimated motion.

Motion estimation

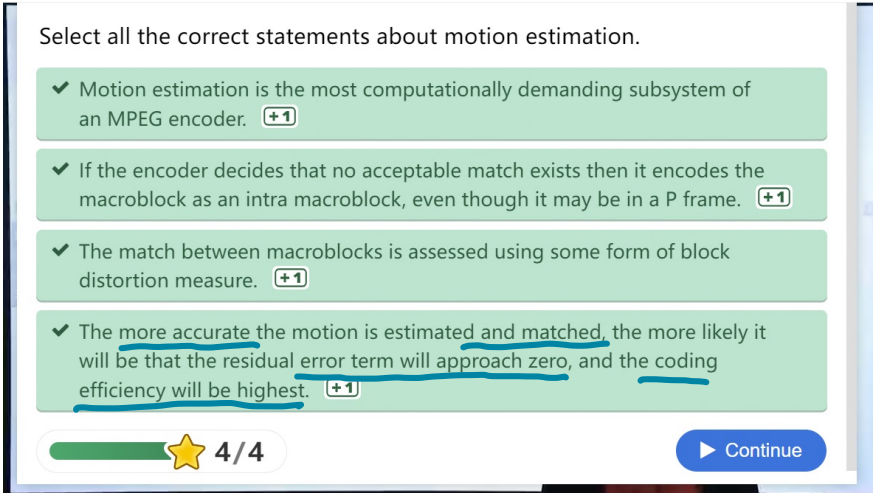
Based on looking for matching macroblocks of data in successive frames, and measuring the difference between matching macroblocks in terms of:

- Position within the frame (expressed as motion vectors)
- Luminance and colour difference (error term)

encoded to be performed on a macro block



The main goal of motion estimation algorithms is the accurate determination of a prediction macroblock for which the absolute difference from the current macroblock is minimum. Such a macroblock roughly approximates the true motion that appears in a video sequence.



What is motion compensation?

运动补偿是一种用于预测视频中的帧的算法技术, won't encode temporal redundancy in successive frames, only encode difference in matching macroblocks. remove the redundancy found through motion estimation.

Encode the motion vector and error terms that we have found in motion estimation. 做法

Motion estimation

- We can't search all possible macro blocks, so we need a smart algorithm
- Matching macroblocks: only if a "close" match can be found
  - Evaluate "closeness" with MSE or other metric
  - If no suitable match found, just encode the macroblock as an I-block
- We don't want too many P or B frames in a row
  - Predictive error will keep propagating until next I frame
  - Delay in decoding

Motion estimation

When comparing macroblocks, three things could happen:

1. One identical macroblock is found: no need to encode anything
2. No matching macroblock is found: use intra-frame encoding
3. One similar macroblock is found: encode difference values between the two macroblocks and a motion vector

error terms

the Block Matching Algorithm (BMA) for motion estimation

A frame is divided into macroblocks of 16x16 pixels. Macroblocks are matched against corresponding blocks in the previous or following frame within the square window of width (16 + 2w) pixels. A function such as the Mean Squared Error (MSE) function (matching criterion) is used to find the best match.

Given a maximum motion displacement of 8 pixels, how many evaluations of the matching criterion are required in the BMA?

the basic principle of fast motion estimation techniques:
the number of search points is reduced by selectively checking only a small number of specific points.

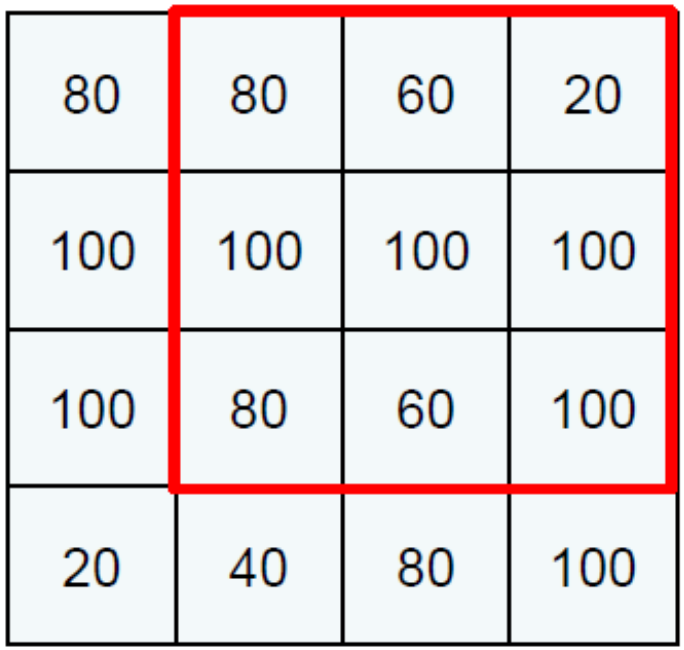
P-frames

encode { error term, motion vector

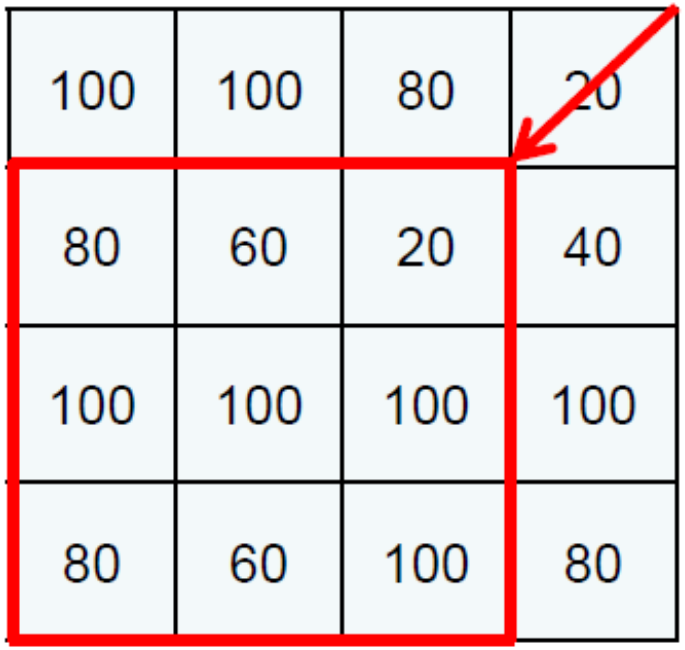
Each macroblock in a P-frame can be encoded either as an I-macroblock or as a P-macroblock.

- An *I-macroblock* is encoded in the same way as a macroblock in an I-frame (i.e. similar to JPEG)
- A *P-macroblock* is encoded as an area of the past reference frame, and an error term. To specify where the macroblock came from in the reference frame, a motion vector is used.
  - If the motion vector is zero, then the macroblock has not moved between frames
  - If the difference values are also zero, then the area does not need to be coded

Motion vector



Reference Picture



Current P-Picture

Error term

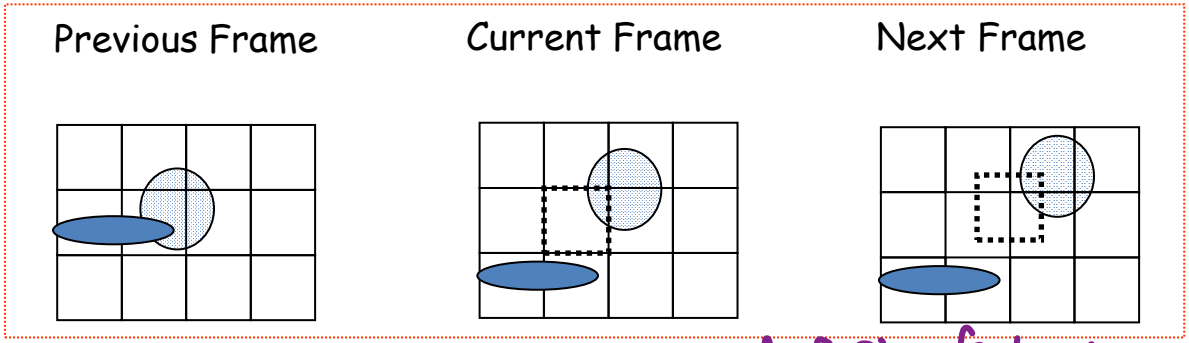
100	100	100
100	80	60
20	40	80

100	100	100
80	80	60
20	20	80

0	0	0
20	0	0
0	20	0

B-frames (Bi-directionally predictive-encoded frames)

- Prediction limit
  - Some macroblocks may need information that is not present in the *previous reference frame*



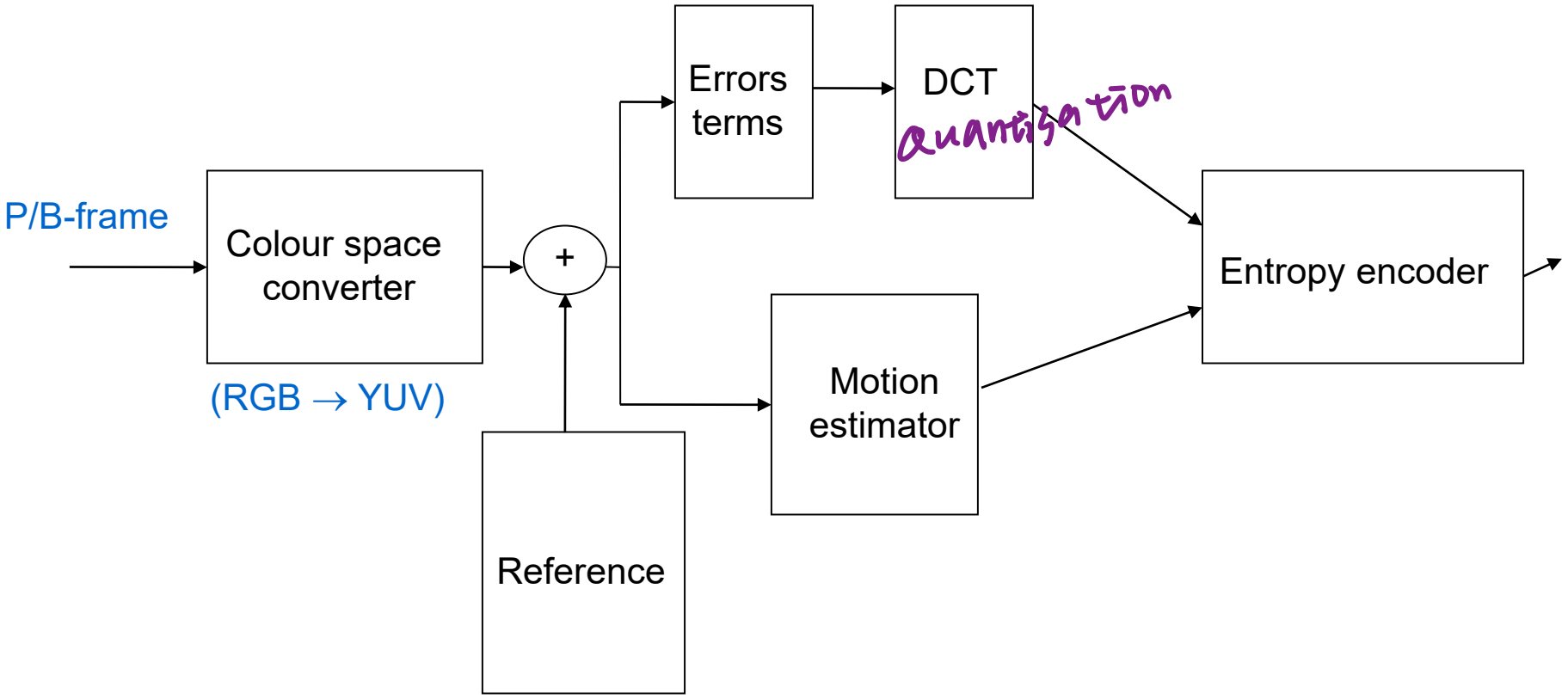
use two references. from past. and from future.

  - Such information might be available in a *subsequent frame*

- MPEG uses a third frame type (B-frame)
  - to form a B-frame: search for matching MBs in both past and future frames
  - typical pattern is IBBPBBPBB IBBPBBPBB IBBPBBPBB
  - actual pattern is up to encoder, and need not be regular

according to the content

P- and B- frames

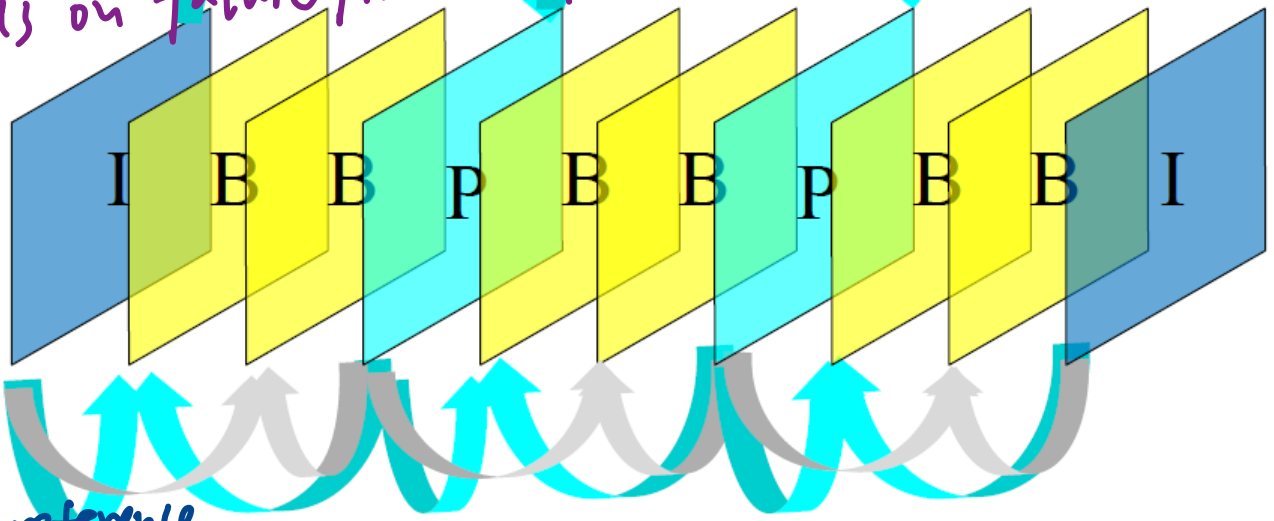


P- and B- frames

- P-frame contains 1/3 data of I-frame
  - motion vectors
  - error macroblock
- B-frame contains 1/2 to 1/5 data of P-frame
  - least data but most computational
  - delay issue:
    - need to transfer future I or P frames before any dependent B-frames can be processed

Group Of Pictures

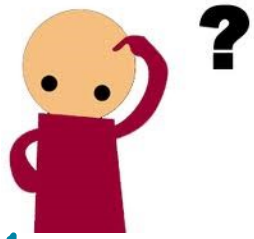
frames should be re-ordered
: B depends on future frames



P : one reference.
B : two references.

- I frames do not rely on other frames — make no prediction.
- P frames rely on previous I and P frames
- B frames rely on previous and future I and P frames — make most prediction
- No frames rely on B frames — because they are highly predicted and prediction errors would be propagated to other frames.

Question



(∴ 快取多)

What can you say about the content of an MPEG video that contains very few I frames?

means it has a lot of P and B frames. → the content is easily predictable, i.e. It contains very little change from one frame to the next. The video doesn't contain much action, it could be the video of an interview.

Select all the correct statements.

- ✔ With ABR (Average Bit Rate) you keep a nominal bitrate, like with CBR (Constant Bit Rate), but you maximise quality by allowing variations in the bitrate, like VBR (Variable Bit Rate). +1
- ✔ VBR (Variable Bit Rate) is intended to provide constant quality. +1
- ✔ ABR (Average Bit Rate) is for the case where you want to optimise for quality but have somewhat flexible bandwidth constraints. +1
- ✔ CBR (Constant Bit Rate) only makes sense for those scenarios where you absolutely need a constant bitrate, like streaming. +1

ate Allocation

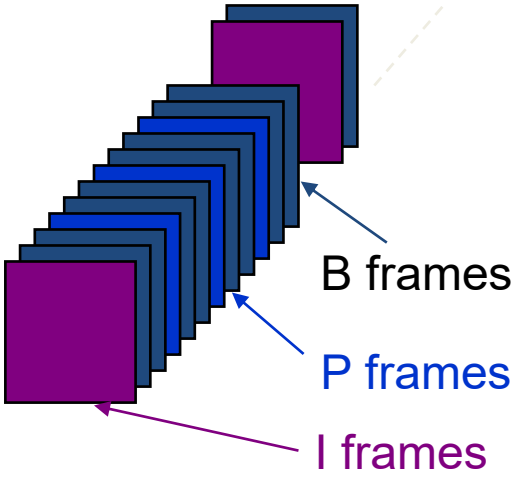
- CBR – Constant BitRate
  - Streaming media uses this
  - Easier to implement
- VBR – Variable BitRate
  - DVD's use this
  - Usually requires 2-pass coding
  - Allocate more bits for complex scenes
  - This is worth it, because you assume that you encode once, decode many times

Fixed size
quicker encoding
useful for streaming
various quality

good for storage
Better quality to space ratio
Smaller files

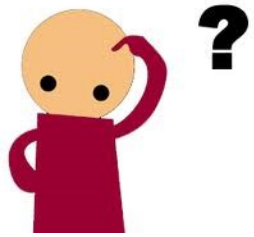
MPEG 1&2 summary

- Images coded INTRA (I):
  - Random access
  - Error resiliance
- Images coded INTER (P):
  - Prediction from previous decoded image (I, P)



- Images coded BI-INTER (B):
  - Prediction from previous and / or future decoded image (I, P)
  - allow effective prediction of uncovered background (areas of the current picture that were not visible in the past and visible in the future)

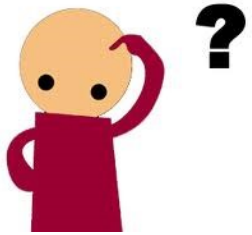
Question



Explain how the choice of GOP structure affects the size of the MPEG video file.

When GOP structures are long (many P & B) the total number of I frames in the video will be small. Since I frames are the biggest, the video files containing a small number of them will tend to be smaller in size.

Question



The frames of any GOP structure need to be re-ordered before transmission. Why?

Solution: The B frames contained in a GOP need two reference frames in order to be reconstructured. One of these references is a "future" frame, ie. it normally comes later in the GOP structure. To make it available for decoding the B frame, it has to be sent before, hence the need to re-order the data stream.

MPEG4

- MPEG 4(1998) provides standardised ways of representing units of aural, visual or audio-visual content, as discrete “media objects”
- These can be of natural of synthetic origin
- For example, they could be recorded with a camera or microphone, or generated with a computer

MPEG4

- MPEG-4 can be used in numerous applications: streaming video; delivering 2D still images; controlling animated 3D models; handling two-way video conferences, etc.
- Coding of audio-visual objects:

– Objects can be coded and decoded *independently*

– Object can be flexibly *composed* to create different scenes

– *Natural and synthetic* object are treated in the same way

– Excellent error resilience

Part 10 and H.264 AVC

- Approx. half the bit rate of MPEG2/MPEG4 Part 2
- Has many new features that allow it to compress video much more effectively, e.g.:

– Up to 16 reference frames can be used, unlike previous limit of 2 for B-frames

– Variable block-size motion compensation (VBSMC) with block size from 16 x 16 to 4 x 4, enabling precise segmentation of moving regions.

H.264 AVC applications

- The AVC encoder is designed for use in a variety of applications: mobile video streaming; video conferencing; internet video; DSL (Digital Subscriber Line) video streaming; Digital HD TV, etc.
- It is a generic standard supporting a wide range of bit rates and a wide range of resolutions.

QuickTime

- A multimedia framework developed by Apple Computer, with 2 main components:

- The QuickTime framework providing a common set of APIs for encoding and decoding audio and video

- The QuickTime Moving (.mov) file format, an open source media container

- Its open architecture supports many file formats and codecs, including Motion JPEG, MPEG, H.264 and is extensible to support future codecs.
- In February 1998, the ISO standards body gave QuickTime a boost by deciding to use it as the basis for parts of the new MPEG-4 standard.

AVI

- Audio Video Interleave
- Video for Windows multimedia container format
- A special case of the RIFF (Resources Interchange File Format), defined by Microsoft
- Video for Windows supports several data compression techniques, including RLE, Indeo (Intel), MPEG4 and DivX

WMV

- Windows Media Video
- Compressed video file format incorporating several proprietary Microsoft codecs
- WMV 9 codec is a competitor to H.264, DivX, etc.
- Adopted as format for Blu-ray discs

Flash Video

- Container file format used to deliver video over the Internet : FLV or F4V
- Can use multiple codecs: Sorenson Spark, VP6, H.264 video

HTML5 Video

- A HTML 5 tag introduced for the purpose of playing videos or movies on web
- Intended to become the new standard way to show video on the web.
- Current draft specification does not specify which video formats browsers should support

EBU5303

Multimedia Fundamentals

Perceptual Encoding

Agenda

- Compression of audio is rarely lossless
- A-law encoding is a nonlinear companding method
- Psychoacoustics is the study of how the human ears and brain perceive sound
- MP3 and AAC compression methods use perceptual encoding

Audio Compression

- Uncompressed 3 minutes song in stereo = 25 MB
- Lossless compression does not work well (complex and unpredictable nature of sound waveforms)
- Obvious compression technique: **silence compression**
  - Detect silence = samples falling below a threshold
  - Treat them as zero and compress using RLE (Run Length Encoding)
  - Silence is rarely absolute $\Rightarrow$ not strictly lossless

Lossless Audio Compression

- Used for editing or further compression, for archival storage, or as master copies.
- Compression ratios are typically around 50–60% of original size.
- Lossless audio compression formats: Free Lossless Audio Codec (FLAC), Apple's Apple Lossless (ALAC), MPEG-4 ALS, Microsoft's Windows Media Audio 9 Lossless (WMA Lossless), Monkey's Audio, TTA, and WavPack

Disadvantage: require more storage space than lossily encoded ones.

Advantage: give an exact copy of the original audio

- FLAC uses linear prediction to convert the audio samples.
- There are two steps: the predictor and the error coding.
- The difference between the predictor and the actual sample data is calculated and is known as the residual.
- The residual is stored efficiently using Golomb-Rice coding (a type of entropy encoding)
- FLAC also uses run-length encoding for blocks of identical samples, such as silent passages.

FLAC uses linear prediction. What characterises linear prediction and why does it make FLAC lossless? Select all the correct statements.

- ✔ Linear prediction is a mathematical operation where future values of a digital signal are estimated as a linear function of previous samples. +1
- ✔ For doing linear prediction, FLAC exploits the fact that audio data typically has a degree of sample-to-sample correlation. +1
- ✔ The digital sample is passed through a prediction stage to find a mathematical description of the signal. This description is typically much smaller than the raw signal itself, hence the lossless compression. +1
- ✔ In linear prediction, if the predictor does not describe the signal exactly, the difference between the original signal and the predicted signal (called the error or residual signal) must be coded losslessy. +1

Companding 压扩

- Companding = compressing and expanding
- Companding = reducing the bit depth
- When reducing the bit depth with a linear method, rounding down has a greater impact on low amplitude samples than on high amplitude ones.

  E.g.: when reducing the bit depth from 16 to 8 bits, all amplitudes between 0 and 255 become 0 (i.e. they are lost!).

↑ decrease the number of bits used to record the strongest signals.
improve SNR at reduced bit rates

Exercise

-32768 to 32767

÷256 =127

Let's assume that converting from 16 bits to 8 bits is done by dividing the sample value by 256 and rounding down.

remainder: 255

error: $\frac{255}{32768}$ <1%

Consider the following two 16 bits samples' amplitudes: 32767 and 255.

→ round to 0 remainder: 255

error: 100%

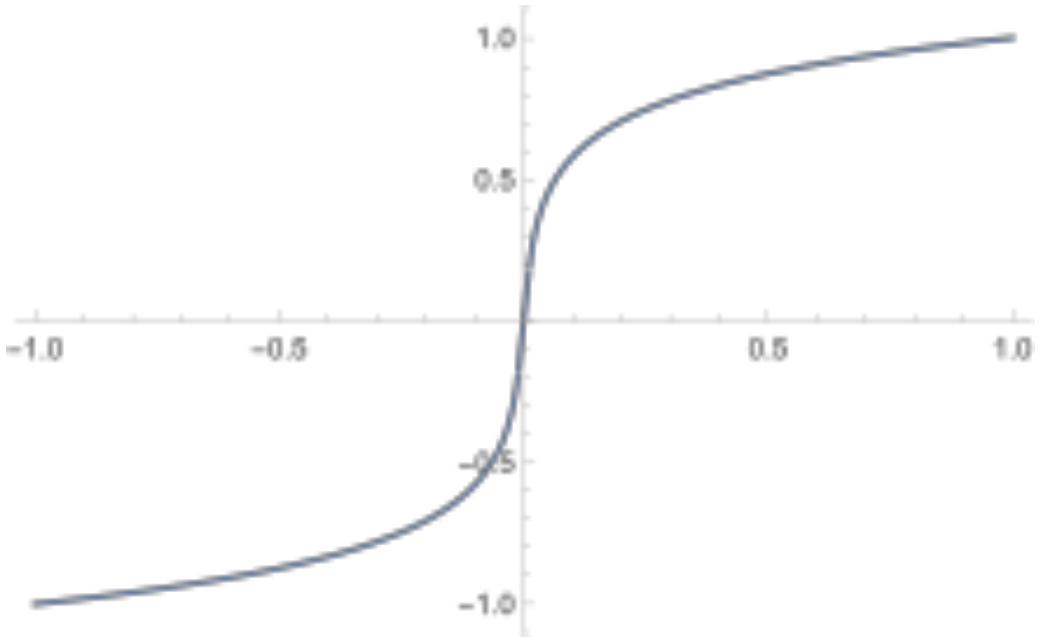
- Convert the two samples to 8 bits.
- Compare the errors from rounding these two samples.

A-law

- A-law: a nonlinear companding method.
- With A-law encoding, not all samples are encoded in the same number of bits.
- The human auditory system is believed to be a logarithmic process in which high amplitude sounds do not require the same resolution as low amplitude sounds: the human ear is more sensitive to quantisation noise in small signals than large signals.
- A-law coding applies a logarithmic quantisation function to adjust the data resolution in proportion to the level of the input signal: smaller signals are represented with greater precision – more data bits – than larger signals.

A-law

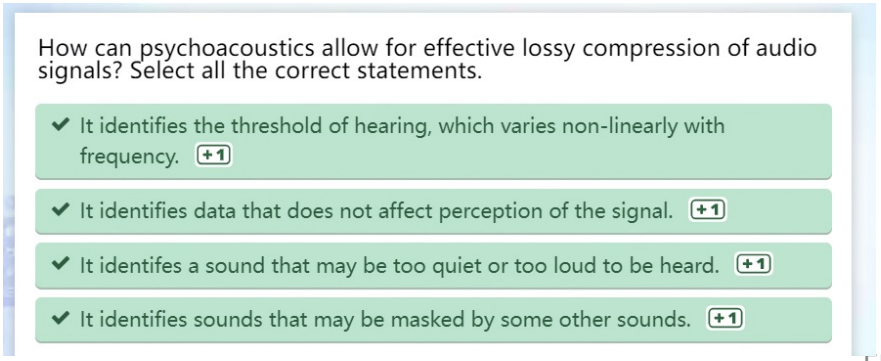
The A-law function has the effect of “spreading out” the quantisation intervals more at lower amplitudes.



Plot of F(x) for A = 87.6

Psychoacoustics

- A branch of science studying the psychological and physiological responses associated with sound (including noise, speech and music), i.e. the study of how the human ears and brain perceive sound.
- Psychoacoustical experiments have shown that human hearing is nonlinear in a number of ways, including perception of octaves, perception of loudness, and frequency resolution.



Hearing is not a purely mechanical phenomenon of wave propagation, but is also a sensory and perceptual event: psychoacoustics is the branch of science studying the psychological responses associated with sound (including noise, speech, and music). It is an interdisciplinary field including areas such as psychology, acoustics, electronic engineering, physics, biology, physiology, and computer science.

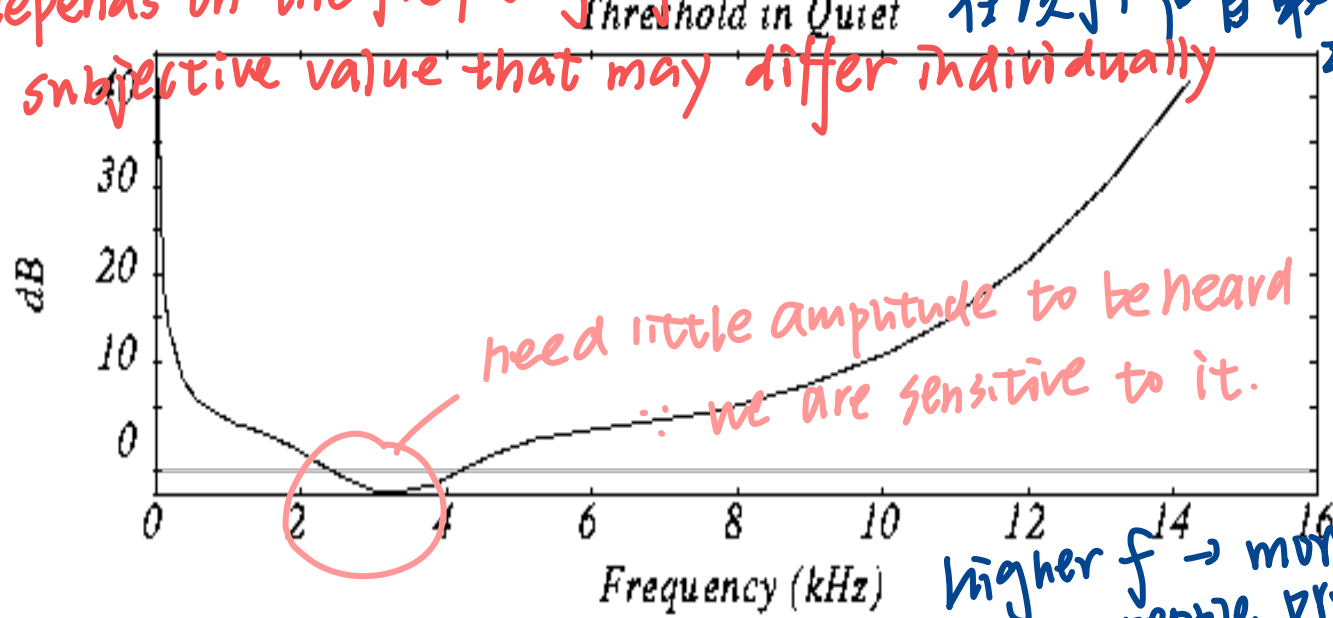
Threshold of Hearing

- Humans hear best (i.e., have the most sensitivity to amplitude) in the range of about 1000 to 5000 Hz, which is close to the range of the human voice.
- Threshold of hearing = minimal level at which sound can be heard

intensity

+ it depends on the frequency of the sound
+ is a subjective value that may differ individually

在該f，声音最低多少dB可以被听见

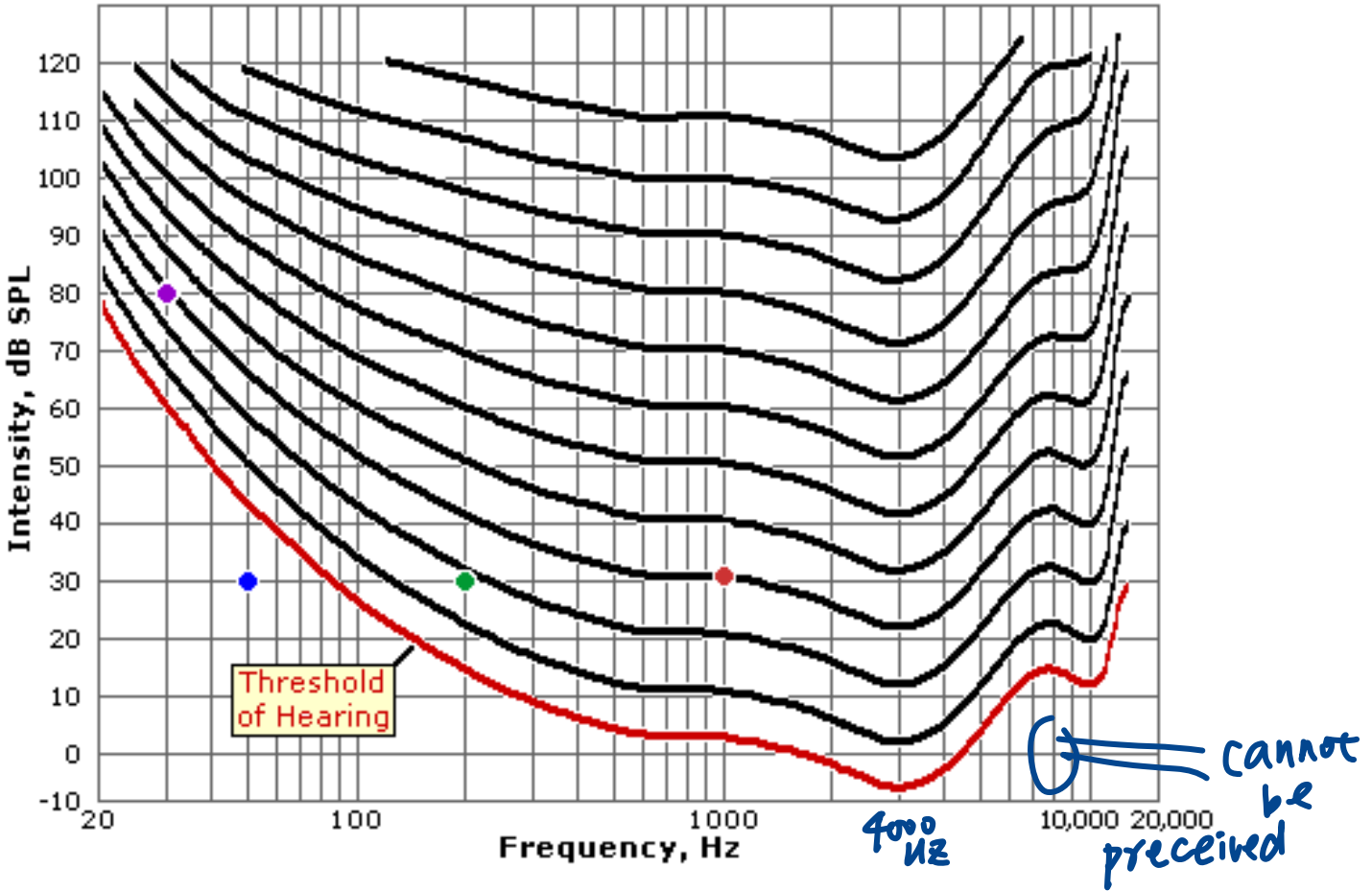


need little amplitude to be heard
∴ we are sensitive to it.

higher f → more amplitude to let people precerve

Threshold of Hearing

The graph shows how loud a sound at a certain frequency needs to be, for it to be perceived as a certain loudness.



Critical Bands

- Human ability to distinguish between frequencies decreases nonlinearly from low to high frequencies: at the very lowest audible frequencies, we can tell the difference between pitches that are only a few Hz apart, while at high frequencies the pitches must be separated by more than 100 Hz before we notice a difference.
- This difference in frequency sensitivity arises from the fact that the inner ear is divided into **critical bands**.
- Each band is tuned to a range of frequencies (in a similar manner to bandpass filters), and there are 24 critical bands in the human hearing range.
- Critical bands for low frequencies are narrower than those for high ones.

reason

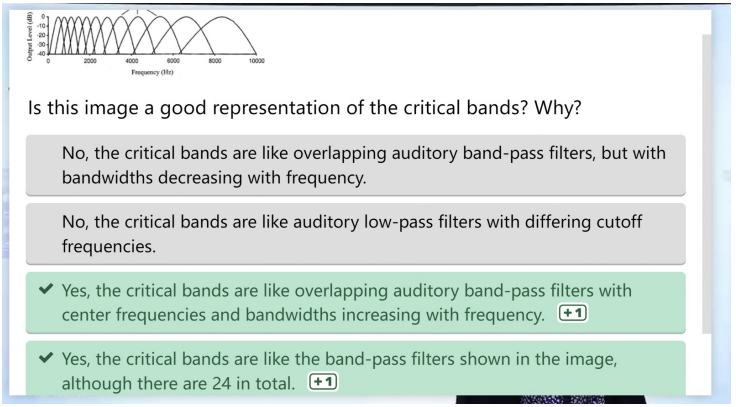
definition: critical band is a range of frequency in which human cannot distinguish the difference.

Critical Bands

Critical Band (Bark)	Center Frequency (Hz)	Bandwidth (Hz)
1	50	100
2	150	100
3	250	100
4	350	100
5	450	110
6	570	120
7	700	140
8	840	150
9	1000	160
10	1170	190
11	1370	210
12	1600	240
13	1850	280
14	2150	320
15	2500	380
16	2900	450
17	3400	550
18	4000	700
19	4800	900
20	5800	1100
21	7000	1300
22	8500	1800
23	10500	2500
24	13500	3500

Critical Bands

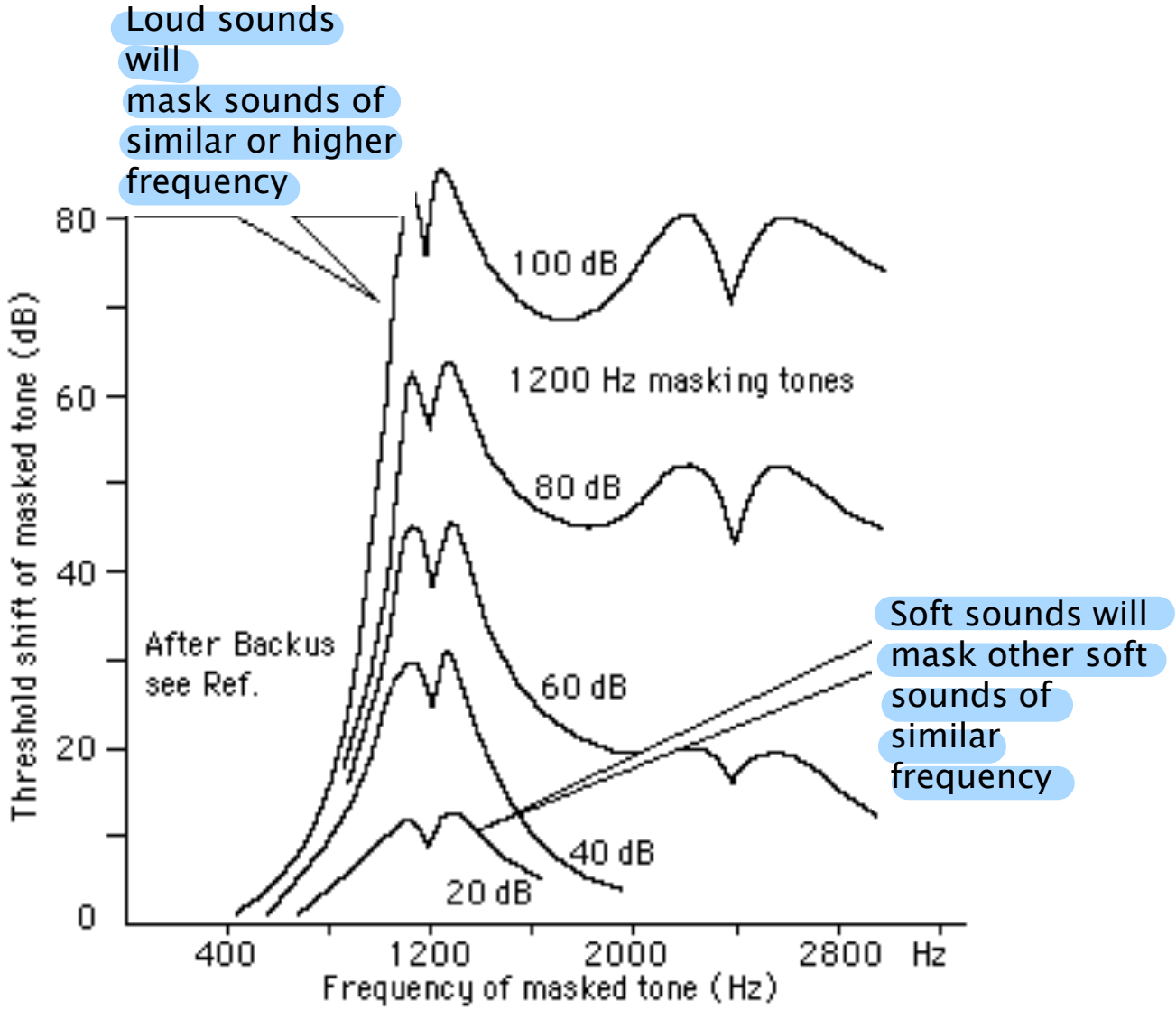
- If two tones are in the same critical band, they are not easily distinguishable as separate, distinct tones.
- When the tones are about 4 Hertz apart, the ear hears a single tone with a low frequency modulation or beating.
- When the tones are about 70 Hertz apart, the ear hears a rapid modulation or beating.
- With a separation of 350 Hertz, the two tones are in different critical bands, and the ear can distinguish them from each other.



Frequency Masking

- **Frequency masking**: A loud tone may mask a softer tone of similar or higher frequency.
- Masking occurs when two frequencies are received by a critical band at about the same moment in time, one of the frequencies being significantly louder than the first such that it makes it inaudible.
- The loud frequency is called the **masking tone**, and the quiet one is the **masked frequency**.

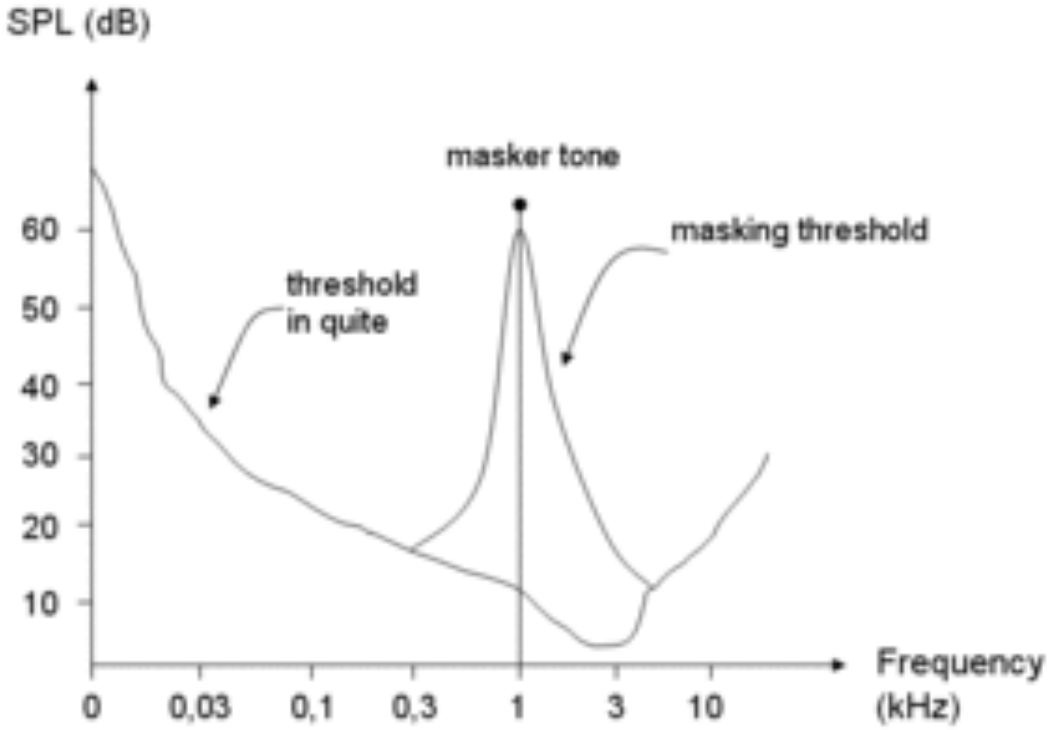
Frequency Masking



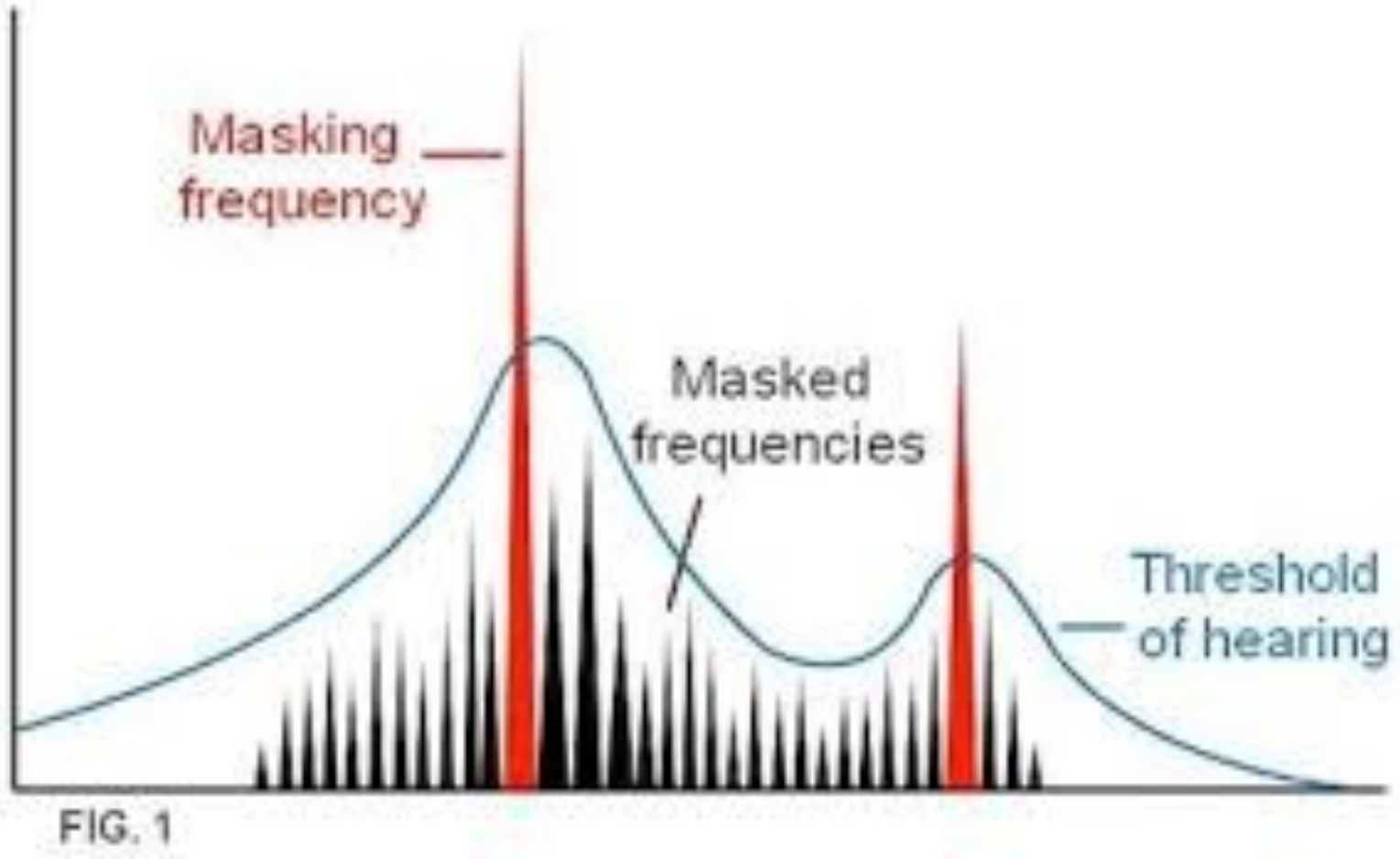
Frequency masking

the effects on threshold of hearing.

- Masking causes the threshold of hearing to be raised within a critical band in the presence of a masking tone.
- The new threshold of hearing is called the **masking threshold**.

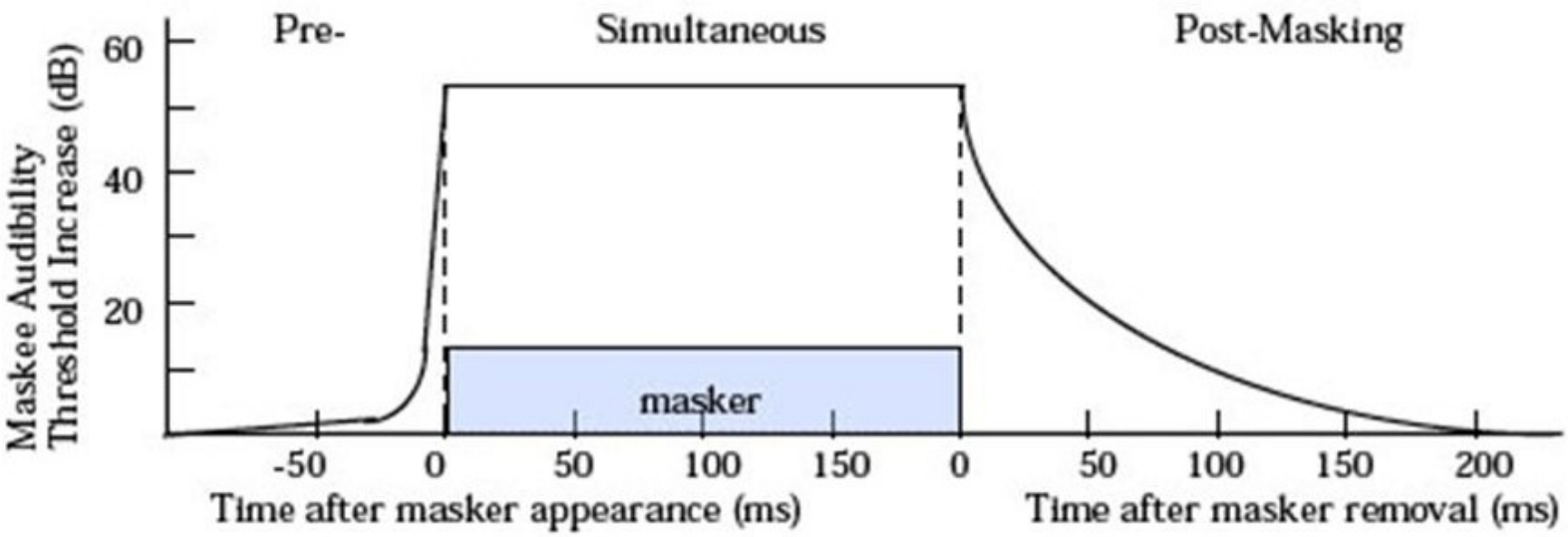


Frequency Masking



Temporal Masking

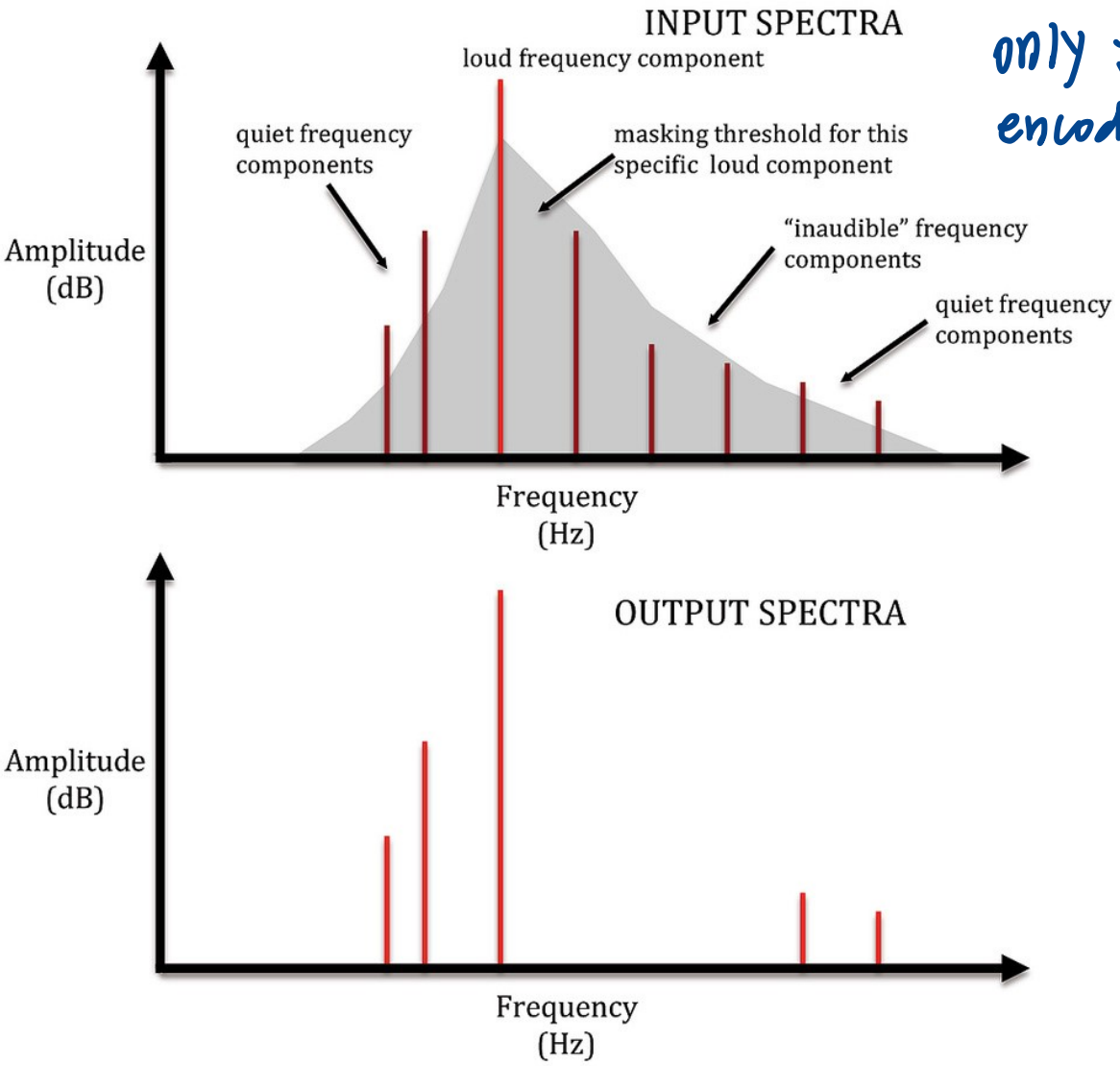
- **Temporal masking**: After a loud sound stops, there is a small delay before we can hear a softer tone.
- The duration of masking depends on the duration of the masker, its amplitude and its frequency.



Perceptual Encoding

- Based on psychoacoustics
- The goal of applying psychoacoustics to compression methods is to determine the components of sounds that human ears don't perceive very well, if at all.
- These are the parts that can be discarded, thereby decreasing the amount of data that must be stored in digitised sound.

Frequency Masking and Encoding



only f above threshold encoding is encoded

Perceptual Audio Coder

1. Use convolution filters to divide the audio signal into frequency subbands --> *subband filtering*.
2. Determine amount of masking for each band caused by nearby band using a *psychoacoustic model*
3. If the power in a band is below the masking threshold, don't encode it.
4. Otherwise, determine number of bits needed to represent the coefficient.
5. Format bitstream

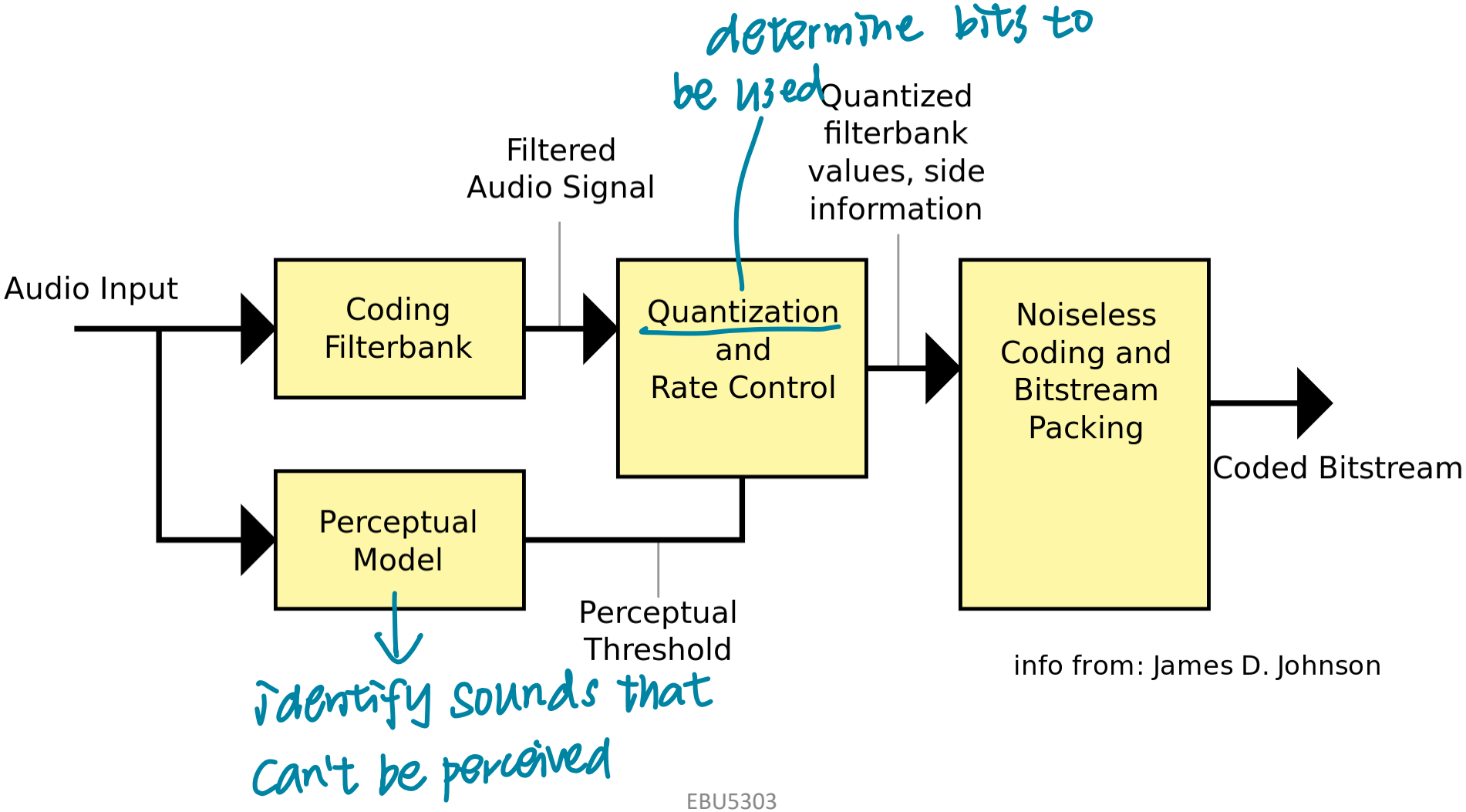
In perceptual encoding, what is the purpose of subband filtering?

Using a psychoacoustic model, subband filtering allows to determine the amount of masking for each band.
Using a psychoacoustic model, subband filtering allows to eliminate entire bands if they are above the threshold of hearing.
Using a psychoacoustic model, subband filtering allows to separate loud tones from softer tones.
Using a psychoacoustic model, subband filtering allows to move from the time domain to the frequency domain

How can psychoacoustics allow for effective lossy compression of audio signals?

Effective lossy compression identifies data that doesn't matter (i.e. that does not affect perception of the signal and throws it away.
Two phenomena in particular cause some sounds not to be heard, despite being physically present: - a sound may be too quiet or too loud to be heard (threshold of hearing)- or it may be obscured by some other sound (masking)
A weak sound emitted soon after the end of a louder sound is masked by the louder sound (temporal masking)
If two sounds occur simultaneously, a sound close in frequency to the louder sound is more easily masked than if it is far apart in frequency (frequency masking in critical bands)
Audio compression algorithms look for places where masking occurs in all of the critical bands over each consecutive window of time for the duration of the audio signal.

Perceptual Audio Coder



info from: James D. Johnson

Illustrative example

Band	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
Level (dB)	0	8	12	10	6	2	10	60	35	20	15	2	3	5	3	1

- If the level of the 8th band is 60dB, it gives a masking of 12dB in the 7th band, 15dB in the 9th (perceptual model).
- Level in 7th band is 10 dB (< 12 dB), so ignore it.
- Level in 9th band is 35 dB (> 15 dB), only the amount above the masking level needs to be sent.

MPEG Audio

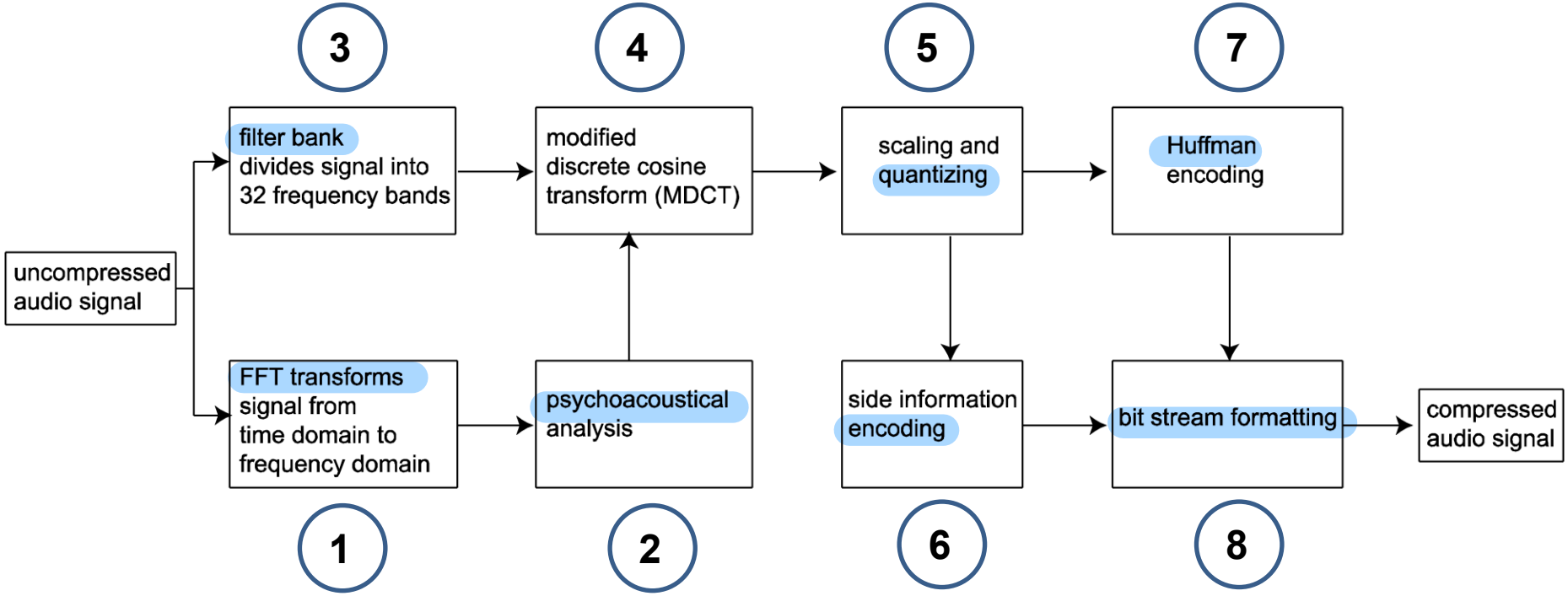
- **MP3**: formally MPEG-1 (and MPEG-2) Audio Layer III
  - 16 bits
  - Sampling rate: 32, 44.1, or 48 kHz
  - Bitrate: 32 to 320 kbps
- **AAC:** Advanced Audio Coding; part of the MPEG-2 and MPEG-4 specifications
  - More sample frequencies (8 kHz to 96 kHz)
  - Higher coding efficiency and simpler filter bank
  - 96 kbps AAC sounds better than 128 kbps MP3

MP3

常用：bit rate：128 kb/s

- Compared to CD-quality digital audio (i.e. 2 channel signed 16-bit sampled at 44,100 Hz), MP3 compression can commonly achieve a 75 to 95% reduction in size.
- The MPEG standard does not include a precise specification for an MP3 encoder, but does provide example psychoacoustic models: implementers devise their own algorithms suitable for removing parts of the information from the audio input.
- The quality of MP3 encoded sound depends on the quality of the encoder algorithm as well as the complexity of the signal being encoded.

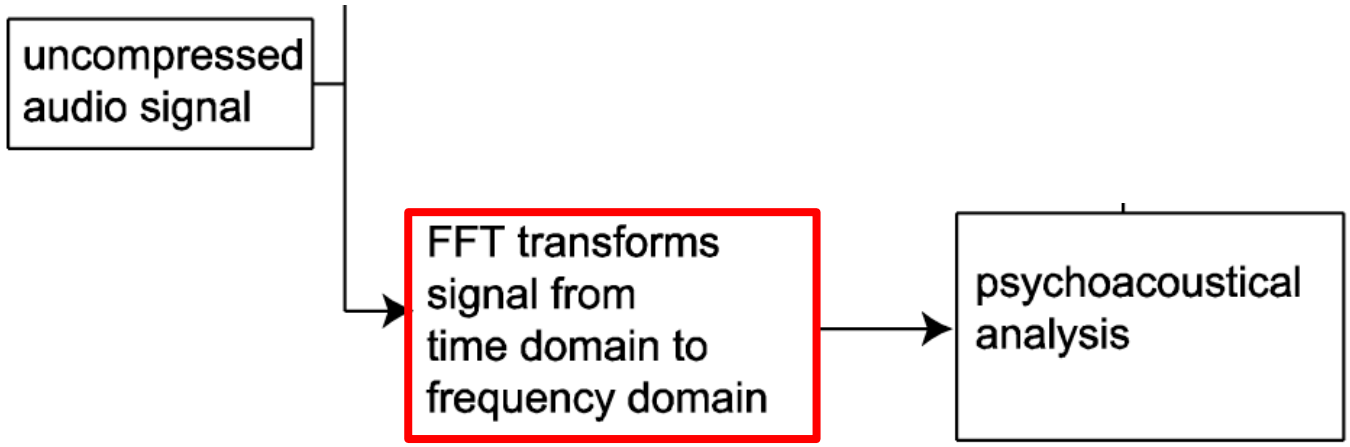
MP3 algorithm



1

FFT

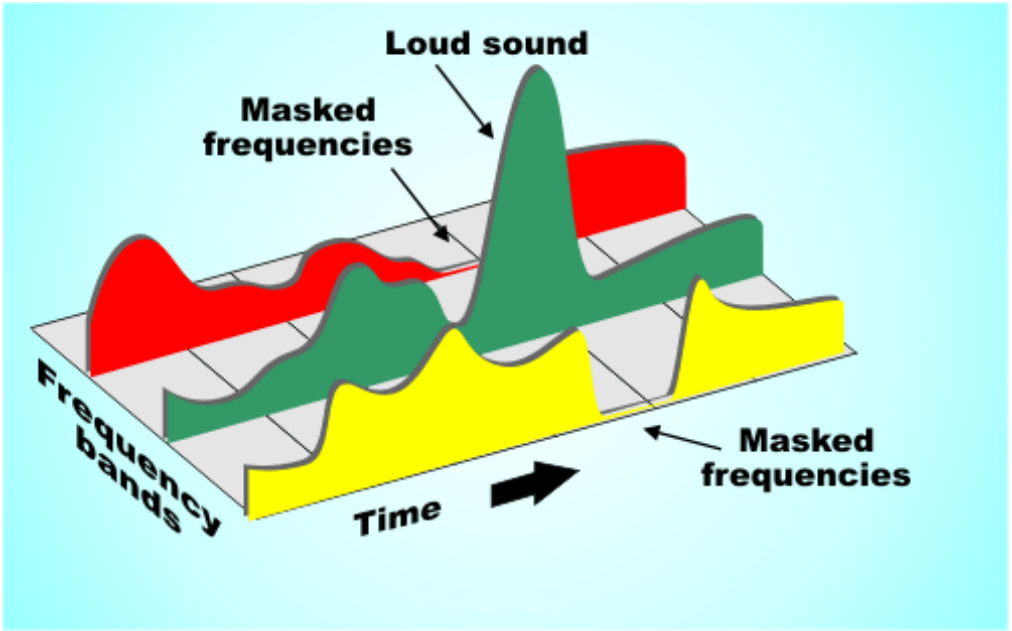
Divide the audio signal in frames of 1152 samples, and use the Fourier transform to transform the time domain data to the frequency domain, sending the results to the psychoacoustical analyser.



2 Psychoacoustic analyser

The psychoacoustic analyser identifies masking tones and masked frequencies in a local neighbourhood of frequencies over a small window of time.

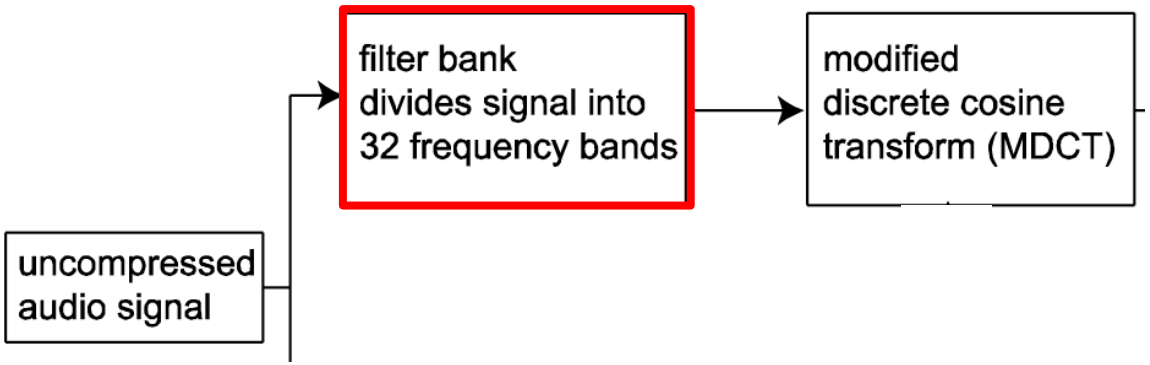
- It outputs a set of **signal-to-mask ratios** (**SMRs**).
- The SMR is the ratio between the amplitude of a masking tone and the amplitude of the minimum masked frequency in the chosen vicinity.

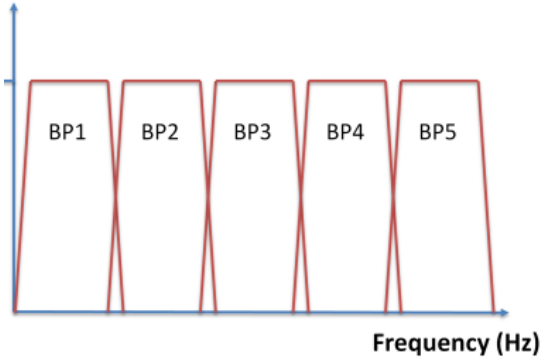


Filter bank

Divide each frame into 32 frequency bands between 0 and 22.05 kHz, using filter banks (bandpass filters).

- The 32 resulting bands are still in the time domain.
- That is, there are 32 sets of 1152 time-domain samples, each holding just the frequencies in its band.





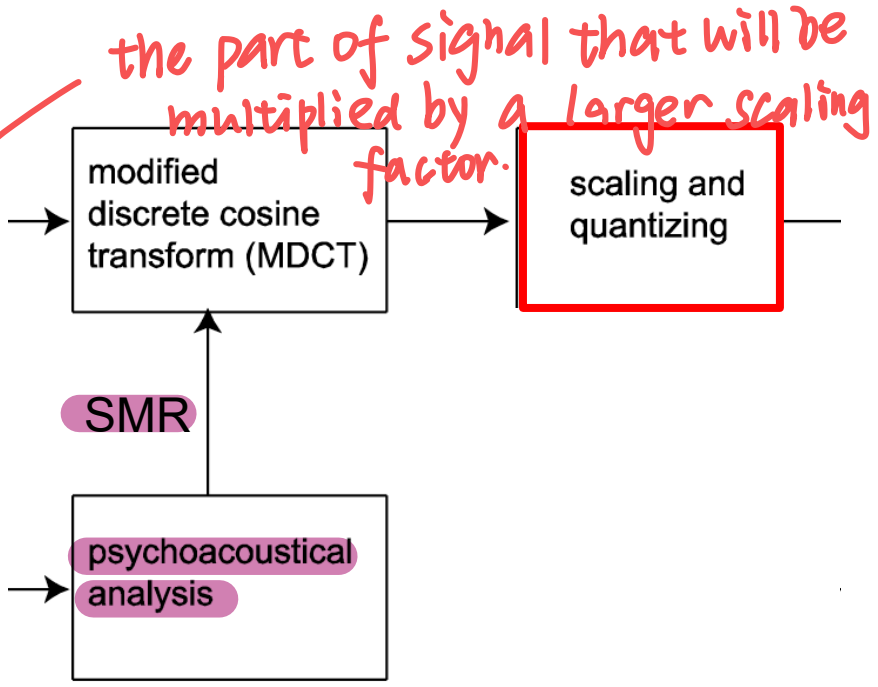
MDCT

Use the MDCT (Modified Discrete Cosine Transform) to divide each of the 32 frequency bands into 18 subbands for a total of 576 frequency subbands.

Scaling and Quantising

Sort the subbands into 22 groups, called scale factor bands.

- Based on the SMR, the scale factor bands cover several MDCT coefficients and more closely match the critical bands of the human ear.
- Use nonuniform quantisation combined with scaling factors: bands that have a lower SMR are multiplied by larger scaling factors because the quantisation error for these bands has less impact, falling below the masking threshold.



Scaling and Quantising example

quantized value: 39

Say that an uncompressed band value is 20,000 and values from all bands are quantised by dividing by 128 and rounding down. Thus the quantised value would be 156. When the value is restored by multiplying by 128, it is 19,968, for an error of 32/20,000 = 0.0016 (or 0.16%).

Now suppose the psychoacoustical analyser reveals that this band requires less precision because of a strong masking tone. Thus, it determines that the band should be scaled by a factor of 0.1. Now we have 20,000*0.1/128= 15. Restoring the original value we get 15*128/0.1 = 19200, for an error of 800/20,000 = 0.04 (or 4%).

$5000 \div 128 = 39$

$5000 \times 0.2 \div 128$
$= 7.8125$
≈ 7
$= 4480$

Scaling and Quantising

An appropriate psychoacoustical analysis provides scaling factors that increase the quantisation error where it doesn't matter, in the presence of masking tones.

Scale factor bands effectively allow less precision (i.e., fewer bits) to store values if the resulting quantisation error falls below the audible level.

This is one way to reduce the amount of data in the compressed signal.

Say that an uncompressed band value is 5,000 and values from all bands are quantised by dividing by 128 and rounding down.

- What is the quantised value?

5000/128 = 39 取整 (round down)

- What is the quantisation error?

39*128 = 4992

(5000-4992)/5000 = 0.0016

Now suppose the psychoacoustical analyser reveals that this band requires less precision because of a strong masking tone. Thus, it determines that the band should be scaled by a factor of 0.2.

- What is the new quantised value?

(5000*0.2)/128 = 7

scaling factor

- What is the new quantisation error?

7*128/0.2 = 4480

(5000-4480)/5000 = 0.104

Side information and Huffman

6

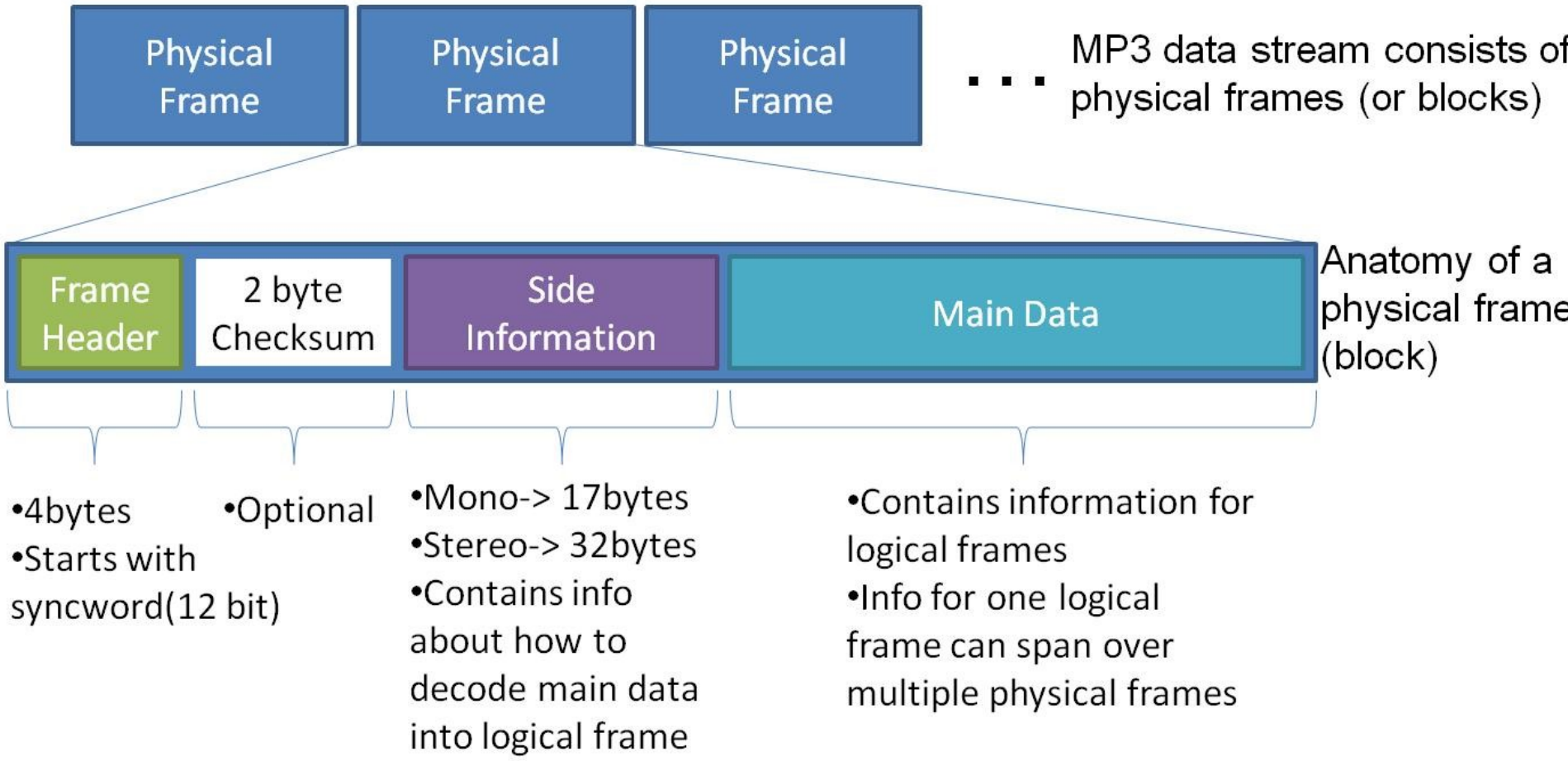
Encode side information: side information is the information needed to decode the rest of the data, including where the main data begins, where scale factors and Huffman encodings begin, the Huffman table to use, the quantisation step, and so forth.

7

Use Huffman encoding on the resulting 576 quantised MDCT coefficients.

Bit stream

Put the encoded data into a properly formatted frame in the bit stream.



MP3 algorithm

/*Input: An audio signal in the time domain
Output: The same audio signal, compressed
/*
Process the audio signal in frames of 1152 samples

For each frame {
Use the Fourier transform to transform the time domain data to the frequency domain, sending the results to the
Psychoacoustical analyser {
    Based on masking tones and masked frequencies, determine the signal-to-masking noise ratios (SMR) in areas across the frequency spectrum
    } // end of Psychoacoustical analyser
Divide the frame into 32 frequency bands

For each frequency band {

Use the modified discrete cosine transform (MDCT) to divide each of the 32 frequency bands into 18 subbands, for a total of 576 frequency subbands

} // end of For each frequency band

Sort the subbands into 22 groups, called scale factor bands, and based on the SMR, determine a scaling factor for each scale factor band

Use nonuniform quantisation combined with scaling factors to quantise

Encode side information

Use Huffman encoding on the resulting 576 quantised MDCT coefficients

Put the encoded data into a properly formatted frame in the bit stream

} // end of For each frame

MP3 bitrates

- CD audio is 44100 samples per second, stereo and 16 bits per channel: the bitrate of uncompressed CD digital audio is 44100x32 = 1411 kbit/s.
- MP3 was designed to encode data at 320 kbit/s or less.
- A bit rate of 128 kbit/s is commonly used.
- Bitrates 128, 160 and 192 kbit/s represent compression ratios of approximately 11:1, 9:1 and 7:1 respectively.
- With some advanced MP3 encoders, it is possible to specify a given quality, and the encoder will adjust the bit rate accordingly (i.e. Variable Bit Rate or VBR).
- Average Bit Rate (ABR) is a type of VBR where the bitrate is allowed to vary for more consistent quality, but is controlled to remain near an average value chosen by the user, for predictable file sizes.

AAC

- AAC compression, the successor to MP3, uses similar encoding techniques but improves on MP3 by offering more sampling rates (8 to 96 kHz), more channels (up to 48), and arbitrary bit rates.
- Filtering is done solely with the MDCT, with improved frequency resolution for signals without transients and improved time resolution for signals with transients.
- Frequencies over 16 kHz are better preserved.
- The overall result is that many listeners find AAC files to have better sound quality than MP3 for files compressed at the same bit rate.

EBU5303

Multimedia Fundamentals

Digital Broadcasting

Agenda

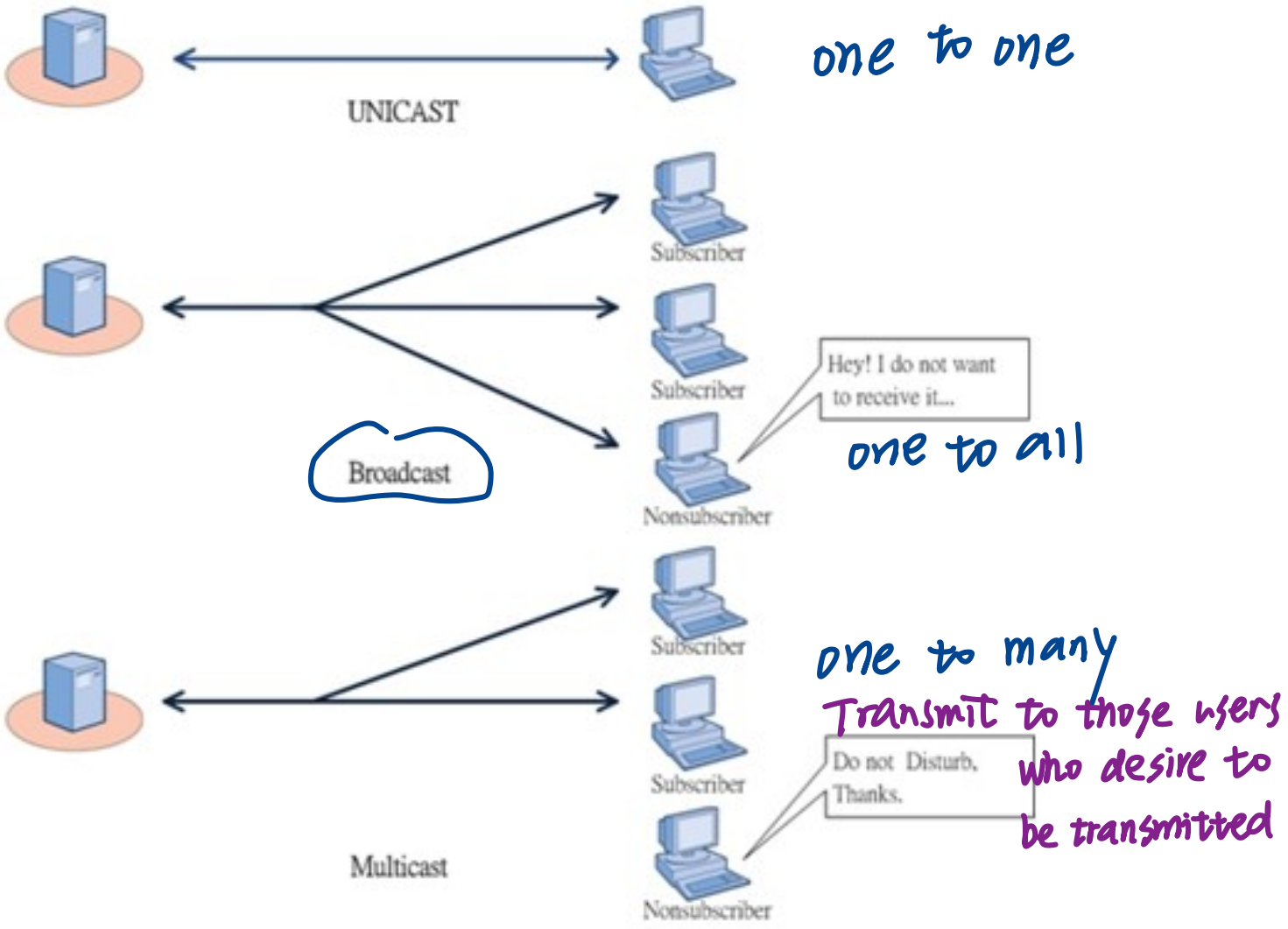
- What is broadcasting?
- Building blocks of a digital broadcasting system
- Major standards for digital broadcasting
- Digital Video Broadcasting - Satellite (DVB-S)

Digital broadcasting is the practice of using digital signals rather than analogue signals for broadcasting over radio frequency bands (to multiple devices and users). Medium can be satellite and can be also cable territorial combination of these.

What is Broadcasting?

- In Telecommunications, broadcasting refers to a method of transferring a message to all recipients simultaneously.
- Traditionally Point-to-multipoint communication
  - Simpler transmission scheme than point-to-point
  - High transmission power ← drawback
- In principle: one-way communication
  - In practice: increasingly interactive, thanks to digital technology
    - View on demand
    - Two-way (movies, shopping, …)

What is Broadcasting?



Why Digital Broadcasting?

Advantages

- Digital signals are more robust
- Better quality: lossless data transmission
- More reliable
- Less expensive
- More flexible
- Time- vs. Frequency-Domain Multiplexing
- Additional devices requiring digital data
- Commercial reasons: better spectral efficiency, more channels, more services (gaming, shopping, internet), mobile reception, data transmission

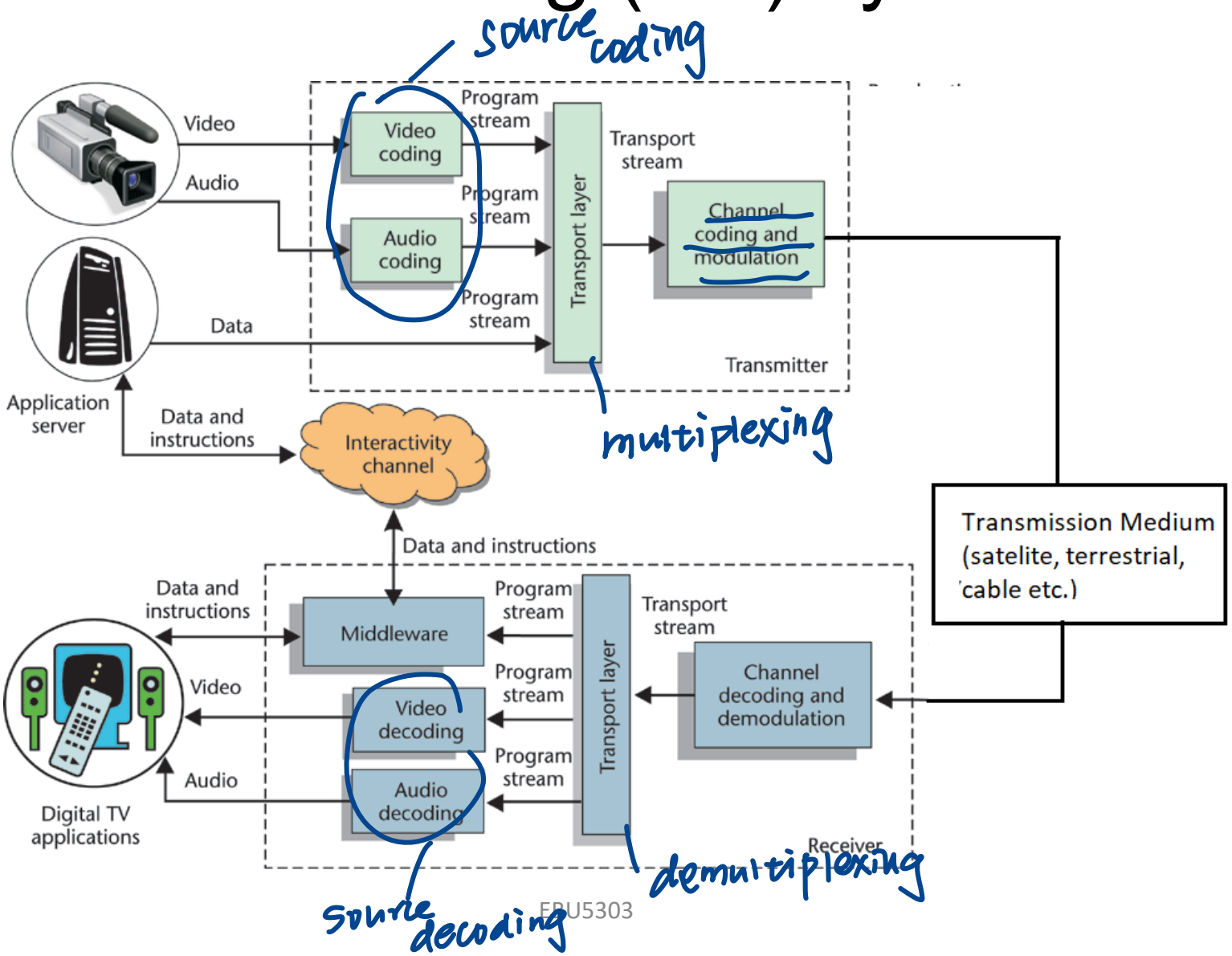
Benefits of Digital Switchover

- Potential benefits to consumers:
  - A greater choice of services
  - Extra information on programmes and interactive features
  - Easier tuning and new functions
  - Less interference to pictures or sound
- Potential benefits for the company:
  - Less cost due to no more need of simultaneous analogue/digital transmission
  - Requires less spectrum and so saves huge capital expenditure
  - Possibility of diversifying devices, services and applications
- Potential Benefit to Government/regulatory body
  - Wider coverage [reducing digital inequality]
  - Freeing up spectrum which can be sold/offered for other services
  - Better management/regulation

Aspects of Digital Broadcasting

- Essential stages:
  - Channel coding: error protection of bits
  - Modulation: for transmitting signal onto carrier
- But also…
  - Source coding: data compression
  - Multiplexing: combining into single data stream
    - Signal processing
- Involves video/audio and data

Building blocks of a Digital Broadcasting (DB) system



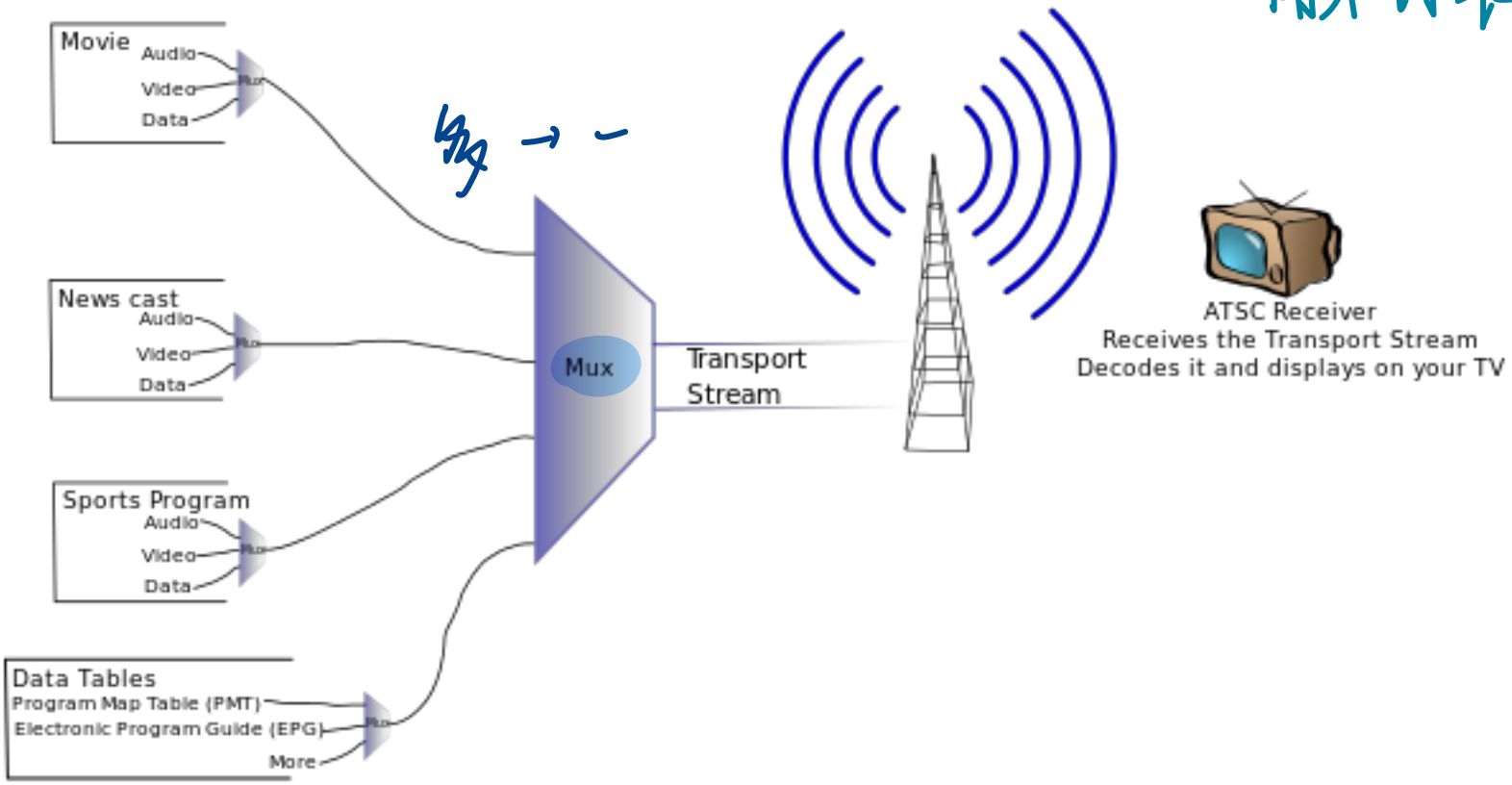
Fundamental Components of DB

- Audio,Visual (Still/Video), and/or Data
- Transmitter:
  - Compression (***source coding***), e.g. using MPEG2
  - Multiplexing information to single ***Transport Stream*** (TS)
  - ***Forward error correction*** (***Channel Coding***), e.g., Reed-Solomon
  - ***Modulation*** (e.g. OFDM: Orthogonal Frequency-Division Multiplexing)
  - ***Transmission*** (antenna (Yagi-Uda, dish) or optical fibre)
- Reverse process at receiver

Transport Stream (TS)

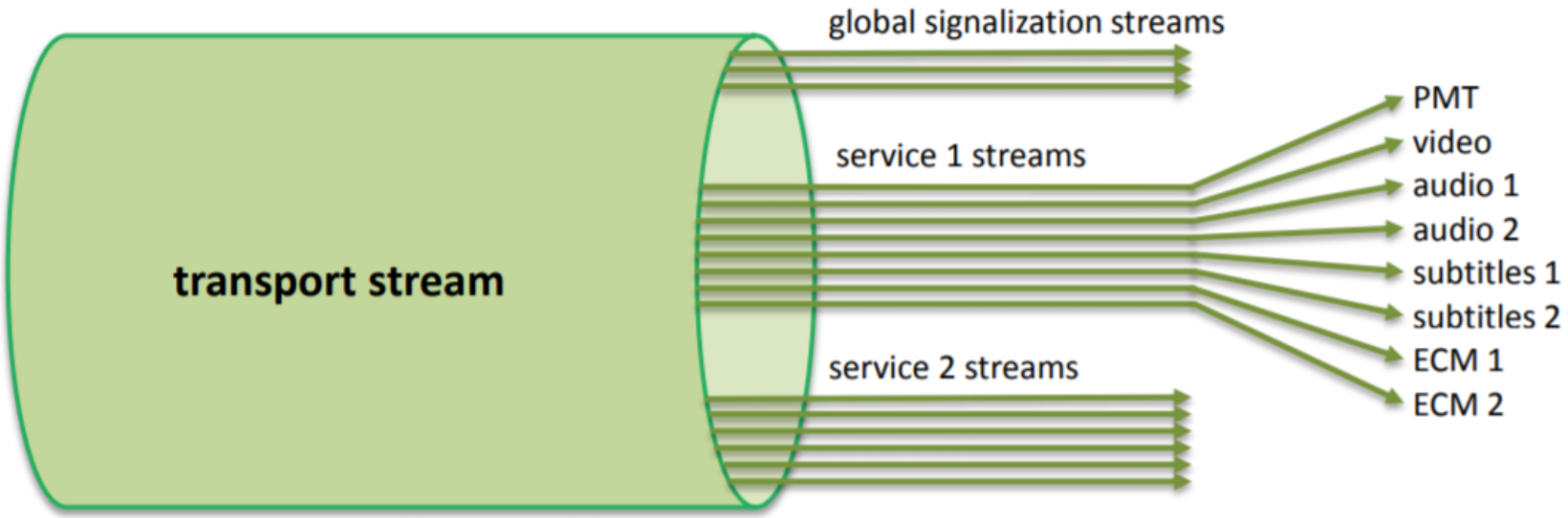
- *Transport stream* specifies a container format encapsulating packetised Elementary Streams (ES).

用来封装基本流



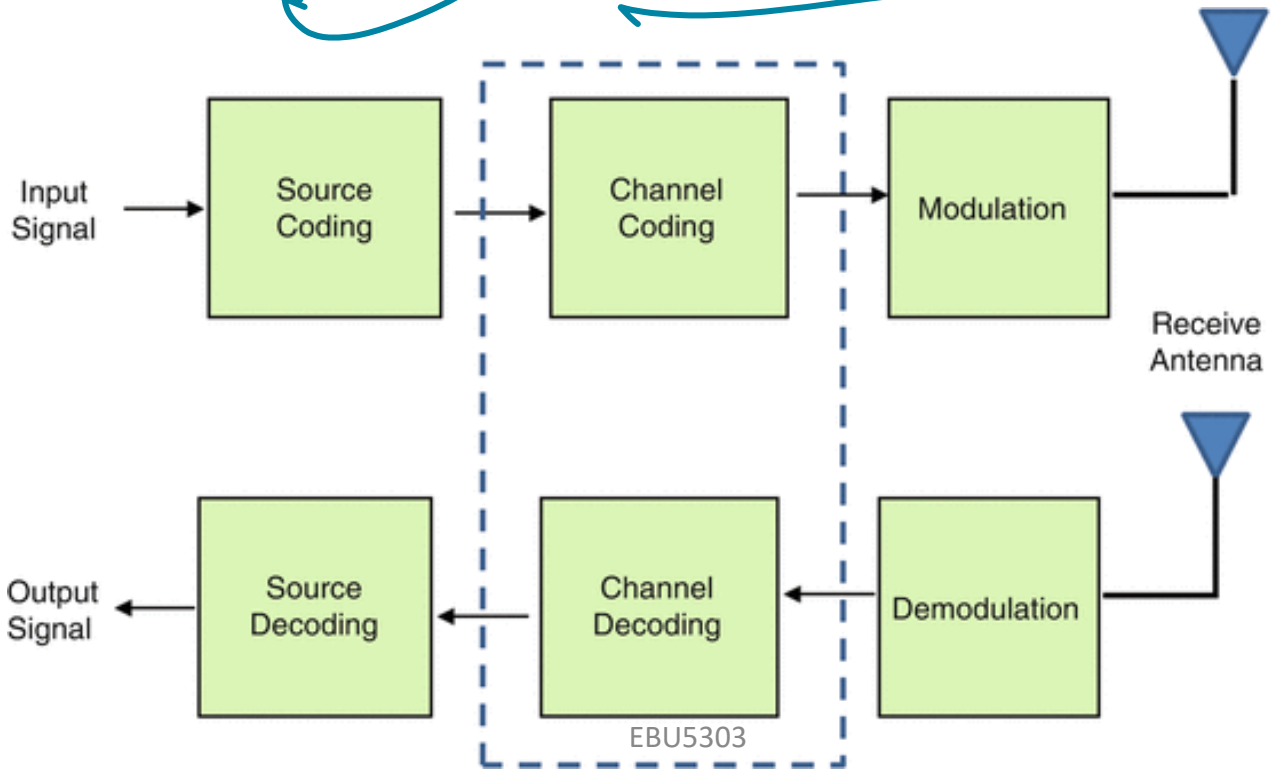
Transport Stream (TS)

- A transport stream is a multiplex of elementary streams
  - elementary stream = sequence of TS packets with same PID value in header
  - one set of elementary streams for global signalization
    - describe the TS, the network, the operator, the services, the events, EMM's, etc.
  - one set of elementary streams per service
    - a service is typically a TV channel



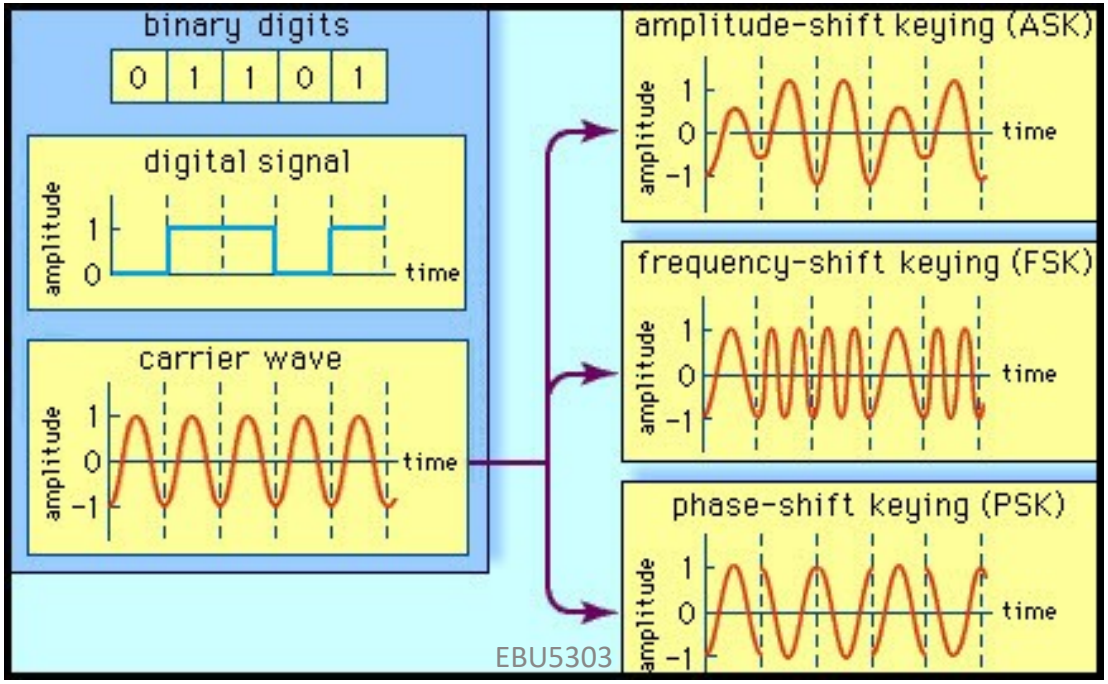
Channel Coding

- Channel coding, also known as *forward error control coding* (FECC), is a process of detecting and correcting bit errors in digital communication systems.
- It is performed both at the transmitter and at the receiver.



Modulation

- *Digital Modulation* uses discrete signals to modulate a carrier wave.
- The three main types of digital modulation are Amplitude Shift Keying (ASK), Frequency Shift Keying (FSK) and Phase Shift Keying (PSK).



Why do we need standards?

- **Safety and reliability**
  - Users perceive standardised products and services as more dependable
  - Increases sales and the take-up of new technologies
- **Support of government policies and legislation**

  Standards are frequently referenced by regulators and legislators for protecting user and business interests
- **Interoperability**

  The ability of devices to work together relies on products and services complying with standards

Major Standards for Digital Brodcasting

- Digital Video Broadcasting (**DVB**)
  - Europe, Singapore, Australia and New Zealand.
- Advanced Television System Committee (**ATSC**)
  - United States, Canada, Mexico, South Korea, Dominican Republic and Honduras.
- Integrated Services Digital Broadcasting (**ISDB**)
  - Japan and the Philippines.
- Digital Terrestrial Multimedia Broadcasting (**DTMB**)
  - Peoples Republic of China.

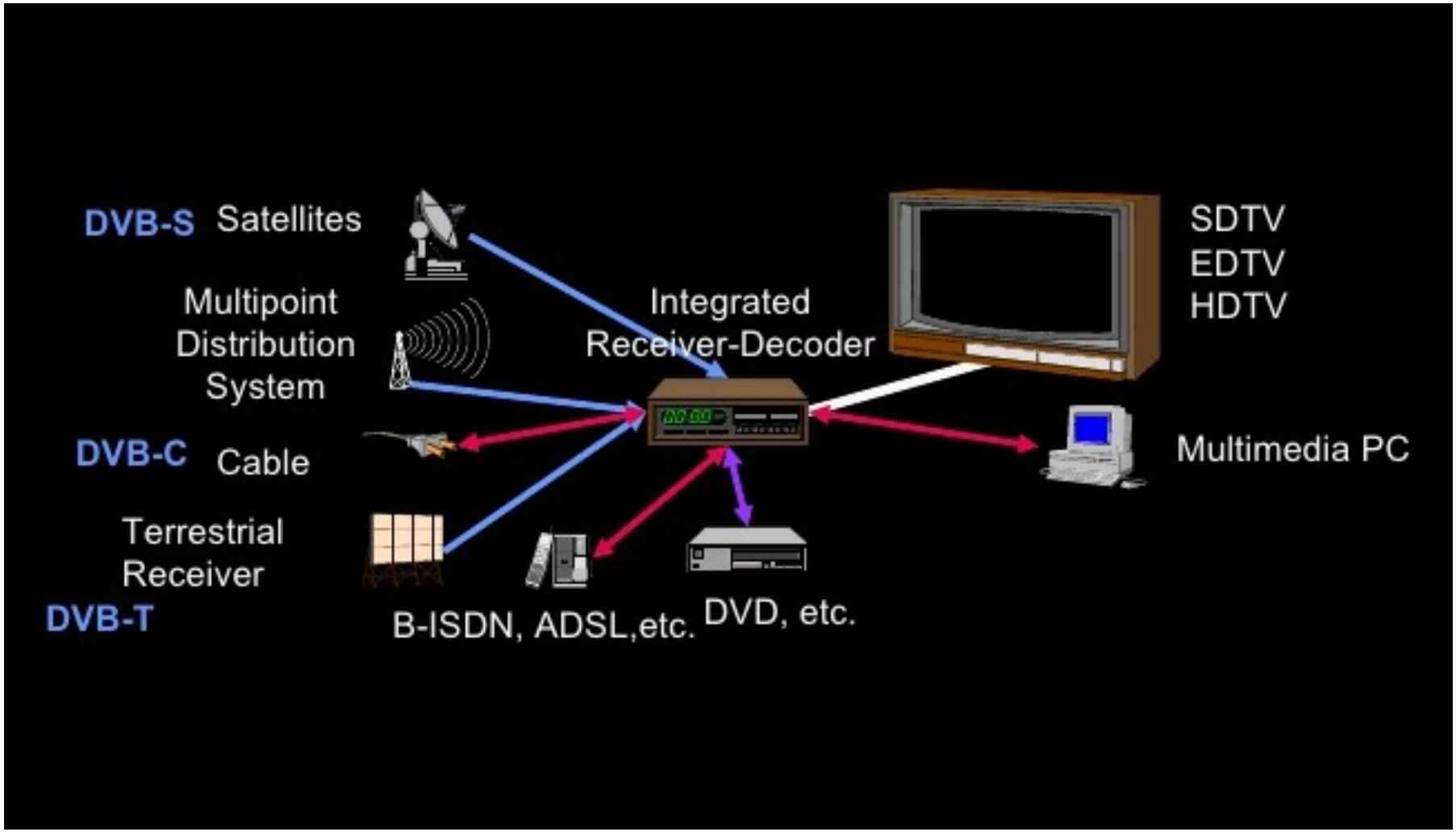
DVB (Digital Video Broadcasting)

- Set of standards that defines digital broadcasting using existing satellite, cable, and terrestrial infrastructures
- Most widely used transmission standard in the world
- Focus of digital television development
- Based on MPEG2 source coding
- DVB-S (1993), -C (1994), -T (1995), -SH, …
  - Different coding and modulation w.r.t. channel
  - e.g. QPSK in DVB-S, QAM in DVB-C, COFDM in DVB-T
- Spawned ISDB-T, -C, -S (Japan) and A/53-T (US)

DVB family of standards

- Every DVB standard defines the ***channel coding*** and ***modulation***, since every channel has a different set of characteristics.
- But for all of them, the system input and output signals are ***MPEG-2 Transport Streams***.
  - DVB-S and DVB-S2 for satellite broadcasting
  - DVB-C for cable systems
  - DVB-T and DVB-T2 for terrestrial broadcasting
  - DVB-H for handheld systems
  - DVB-SH for satellite-to-handheld systems
  - DVB-IPDC for internet protocol datacastover
  - DVB-CPCM for content protection & copy management

DVB family of standards



ATSC (Advanced Television Systems Committee)

- ATSC depends on numerous interwoven standards
  - ATSC (Terrestrial, Cable, Satellite)
  - ATSC-M/H (Mobile/Handheld)
- Established 1990s
- Original specification for HDTV (High Definition Television)
- Uses Dolby, not MPEG for audio
- Mobile reception difficult/impossible until 2008
  - New ATSC-M/H since 2009

ISDB (Integrated Services Digital Broadcasting)

- Covers digital television (DTV) and digital radio
- Maintained by Association of Radio Industries and Businesses (ARIB), Japan
- Main differences compared to DVB:
  - ISDB-S uses 8-PSK (Phase Shift Keying) and Trellis coding instead of QPSK in DVB-S (modulation)
  - In ISDB-T, single 6 MHz TV channel can be split into 13 x 432 kHz subchannels for adaptive use
    - digital audio (1 subchannel),
    - SDTV (multiple subchannels),
    - HDTV (all 13 subchannels).

ISDB family of standards

- ISDB-S for satellite television
- ISDB-T for terrestrial television (mainly used in South America)
- ISDB-Tsb for terrestrial sound
- ISDB-C for cable television
- ISDB-1seg for cell phones, laptops, vehicles

DTMB (Digital Terrestrial Multimedia Broadcast)

- Chinese GB20600-2006 standard
- Initially called DMB-T/H
- DMB-T, T-DMB (South Korea)
- Established 2006
- China
- CMMB: Chinese Mobile Multimedia Broadcasting
  - Mobile and stationary
  - Satellite and terrestrial
  - Standard GY/T 220.1

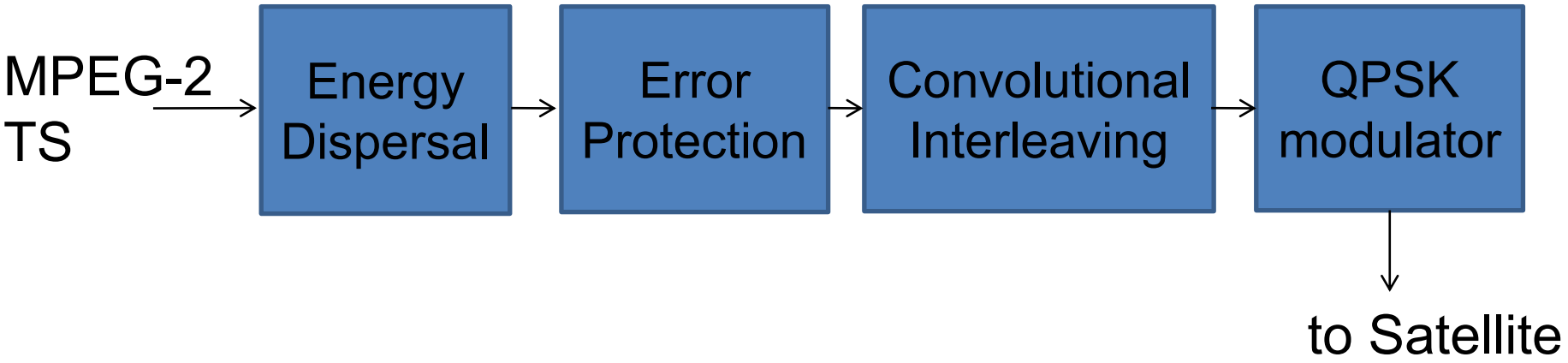
Digital Video Broadcasting – Satellite (DVB-S)

- DVB-S: standard for Direct-to-home Broadcasting via Satellite (DBS)
- Defined between 1993 and 1997 by European Standard EN 300 421.

DVB-S Encoding

After the data has been coded following the MPEG-2 standard, it needs to go through the next steps before being transmitted to the satellite:

- Multiplexing and randomisation for energy dispersion
- Reed-Solomon Encoder (Error Protection)
- Convolutional Interleaving → further error protection
- QPSK modulation

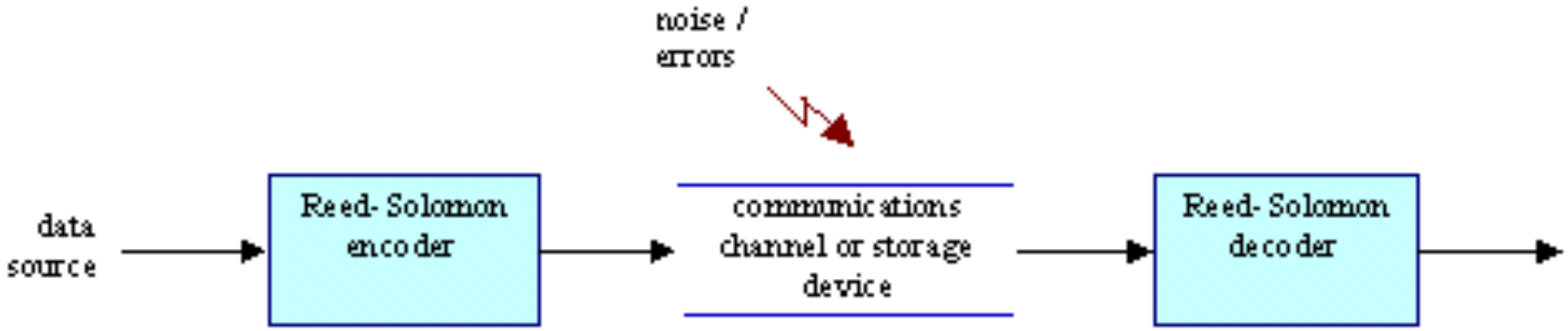


Energy Dispersal and Error Protection

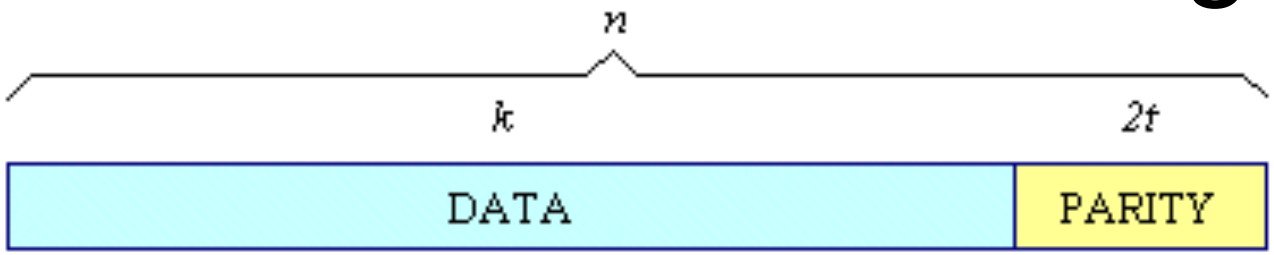
- ***Energy Dispersal*** is carried out at the encoding end by scrambling with a pseudo random sequence
  - Energy dispersion is the randomising of the input signal in order to achieve a power-density spectrum of the modulated signal that is as even as possible.
- ***Error Protection*** scheme permitting various code rate
  - ***Reed-Solomon coding*** RS(204, 188, t=8) is used, where 16 parity bits are introduced in each transport packet. With this the decoder is able to correct up to 8 error bytes in each packet of 204 received bytes.

Reed-Solomon coding

- The Reed-Solomon encoder takes a block of digital data and adds extra "redundant" bits.
- The number and type of errors that can be corrected depends on the characteristics of the Reed-Solomon code.
- A Reed-Solomon code is specified as RS(*n*,*k*) with *s*-bit symbols.



Reed-Solomon coding



- The encoder takes *k* data symbols of *s* bits each and adds parity symbols to make an *n* symbol codeword.
- A Reed-Solomon decoder can correct up to *t* symbols that contain errors in a codeword, where *2t = n-k.*
- Example: RS(204,188) with 8-bit symbols. Each codeword contains 204 code word bytes, of which 188 bytes are data and 16 bytes are parity. For this code: n = 204, k = 188, s = 8, 2t = 16, t = 8

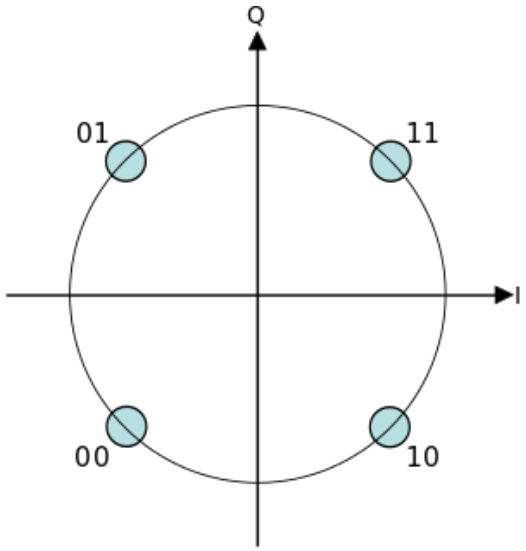
  2t = 204 - 188

  2t = 24 t = 12

  Errors in up to 8 bytes anywhere in the codeword can be automatically corrected.

Interleaving and Modulation

- ***Interleaving***
  - In order to avoid errors in consecutive packets, the packets are interleaved.
- ***Modulation***: QPSK (***Quadrature Phase Shift Keying***) is used for modulation
  - With four phases, QPSK can encode two bits per symbol



Data Rate Calculation (example)

- Assuming the symbol rate is 27.5 MS/s.
- QPSK offers 2 bits/Symbol , a code rate : $\frac{3}{4}$
- gross_data_rate = 2 bits/symbol X 27.5 Megasymbols/s
  = 55 Mbit/s;
- QPSK-modulated signal must first be provided with error protection before being fed into the actual modulator.
- Reed-Solomon code with rate (204, 188) is used.
- net_data_rate Reed-Solomon
  = gross_data_rate x 188/204 (data percent)
  = 55 Mbit/s x 188/204
  = 50.69 Mbit/s;

net data rate = code rate x U = 3/4 x 50.69 = 38.02 Mbit/s

Data Rate Calculation (example)

add redundancy

- further error protection in the form of convolutional coding is inserted after the Reed Solomon forward error correction.
- This further expands the data stream.
- $CodeRate = \frac{Input\ Data\ Rate}{Output\ Data\ Rate}$
- In DVB-S, the code rate can be selected within the range of 1/2, 3/4, 2/3,...7/8.
- The net data rate in DVB-S with a code rate of 3/4, after convolutional coding, is then: $Net_data_rate_{3/4} = code_rate \times net_data_rate_{reed\ solomon}$

$= 3/4 \bullet 50.69\ Mbit/s = 38.01\ Mbit/s;$

≠ redundancy

Observation on data rate

- If the code rate is 1/2, the data stream is expanded by a factor of 2. The error protection is now maximum and the net data rate has dropped to a minimum.
- A code rate of 7/8 provides only a minimum overhead but also only a minimum of error protection. The available net data rate is then at a maximum.
- The code rate can then be used to control the error protection and thus, as a reciprocal of this, also the net data rate.

Adding error protection means adding redundancy to the data, hence reduce the net data rate.

DVB-S2

- Improved version of DVB-S standard, ratified in March 2005.
- It has been designed for:
  - Broadcast Services for standard definition TV and HDTV.
  - Interactive Services including Internet Access for consumer applications
  - Professional applications, Data Content distribution, etc..
- Supports recent improvements in channel coding
- Supports recent improvements in channel modulation
- Typically offers 30% data rate increase under the same condition compared to DVB-S

DVB-S vs. DVB-S2

Satellite EIRP (dBW)	51		53.7	
System	DVB-S	DVB-S2	DVB-S	DVB-S2
Modulation & coding	QPSK 2/3	QPSK 3/4	QPSK 7/8	8PSK 2/3
Symbol rate (Mbaud)	27.5 (α =0.35)	30.9 (α =0.0)	27.5 (α =0.35)	29.7 (α =0.25)
C/N at 27. 5 MHz (dB)	5.1	5.1	7.8	7.8
Useful bitrate (Mbit/s)	33.8	46 (gain = 36%)	44.4	58.8 (gain = 32%)
Number of SDTV programmes	7 MPEG-2 15 AVC	10 MPEG-2 21 AVC	10 MPEG-2 20 AVC	13 MPEG-2 26 AVC
Number of HDTV programmes	1-2 MPEG-2 3-4 AVC	2 MPEG-2 5 AVC	2 MPEG-2 5 AVC	3 MPEG-2 6 AVC